

# Inflation Inequality Across the Euro Area<sup>1</sup>

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## Abstract

This article documents inflation inequality across household categories in the euro area using harmonized HICP prices and household budget surveys. We construct household- and group-level inflation indicators and assess recent fiscal interventions using counterfactual price indices. During the 2021–2023 inflation episode, the annual inflation gap between the first and fifth income quintiles peaks at 1.2 percentage points in 2022 and is close to zero in 2021 and 2023. Counterfactual indices indicate that household price policies, including tariff shields, VAT and charge cuts, fuel rebates, and price caps, lower annual inflation by about 0.7 percentage point on average over 2021–2023 and reduce the annual income-quintile inflation gap by 0.4 percentage point on average.

**Keywords:** Inflation, Distribution of Income, Unconventional Fiscal Policy

**JEL codes:** E31, D30, C43

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<sup>1</sup>This paper presents the authors' views and should not be interpreted as reflecting the views of the Banque de France or the Eurosystem. This paper primarily uses HBS household microdata, supplemented by published tabulations from Eurostat. The responsibility for all conclusions drawn from the data lies entirely with the authors. We thank Pradana Aumars and Lyna El Kamel for excellent research assistance.

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## Introduction

This paper examines the effect of the recent inflation surge on euro-area households. Inflation rates in the euro area rose sharply from mid-2021 onward, displaying considerable dispersion across countries and socio-demographic groups. Aggregate inflation masks differences in household exposure because expenditure shares vary across income, age, and residence groups. High inflation disproportionately affects households whose consumption baskets are more exposed to rapidly rising prices, in particular households with higher expenditure shares on energy and food. The 2021–2023 episode was a large relative-price shock, initially driven by imported energy and commodity prices.

Inflation inequality requires both relative-price changes and heterogeneous expenditure shares. If all prices rose proportionally, fixed-basket inflation would be identical across households. Inequality emerges when relative-price movements interact with differences in household consumption baskets. We examine how recent energy-price shocks and the fiscal response shaped household inflation exposure across the euro area. Specifically, we construct monthly price indices by income, age, and residence area for each euro-area country and aggregate them using Eurostat’s HICP methodology. The consumption baskets are individual 2020 HBS expenditure records, from which we construct both household-group baskets and household-specific indices. Published group tabulations and the Italian national approximation are used only where the required microdata are unavailable. We use the household indices to measure dispersion within and across groups and construct counterfactual price indices that remove the main price-based fiscal interventions of 2021–2023. This counterfactual is a direct consumer-price accounting exercise: it isolates the contribution of the covered measures to household inflation, holding expenditure weights fixed between scenarios and abstracting from behavioral responses and general-equilibrium effects.

Our main results are the following. Inflation inequality varies substantially across euro-area countries and years. In 2022, the annual Quintile 1–Quintile 5 gap exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. In the euro area, the annual income gap is 0.1 point in 2021, 1.2 points in 2022, and 0.0 point in 2023 after rounding. In regressions pooling 2021–2023, the average annual age and rural–city gaps are about 0.6 and 0.4 points. The inflation burden is 2.5, 8.7, and 4.8 percent of net income for the first quintile in 2021, 2022, and 2023, compared with 1.5, 4.7, and 3.0 percent for the fifth quintile. Older and rural households experience higher inflation during the rise in energy prices. In household-level regressions, income, age, and residence are associated with exposure during the energy shock. The rural–urban gap is large in 2021–2022 and reverses sign in 2023, whereas the age gradient

remains visible in 2022–2023; the association with income varies over time. Finally, household price policies lower annual inflation by about 0.7 percentage point on average over 2021–2023 and reduce the annual income-quintile gap by about 0.4 percentage point on average. A complementary input–output simulation shows that the observed extra-EU energy import-price shock generates 6.5 points of inflation exposure and widens the Q1–Q5 gap by 1.1 points. Household energy contributes 1.7 points to this gap, partly offset by vehicle operation and other products. Thus, households remain more exposed to the energy shock at the bottom of the income distribution, while the fiscal response primarily cushions the household-energy channel.

The literature measures household inflation by combining price changes with differences in expenditure patterns. Early contributions reweight aggregate price indices for demographic groups ([Hobijn and Lagakos, 2005](#)) in the United States. More recently, [Jaravel \(2024\)](#) constructs long-run distributional consumer price indices for the United States, closely following the BLS methodology. Using household-level expenditure or scanner data, [Argente and Lee \(2021\)](#), [Kaplan and Schulhofer-Wohl \(2017\)](#) document substantial dispersion within those groups. In the euro area, [Ampudia et al. \(2024\)](#) show how household shopping behavior shapes the effect of monetary policy on inflation heterogeneity along the income distribution. [Pallotti et al. \(2024\)](#) combine consumption, income, and balance-sheet data to measure welfare losses across euro-area households. Building on [Charalampakis et al. \(2022\)](#), we construct monthly indices by income, age, and residence area for all 20 euro-area countries. Unlike existing cross-country indicators, our RAS-calibrated baskets aggregate to official HICP item weights. Because we assign a common price index to all households within each COICOP category, our measures capture expenditure-share heterogeneity but not differences in prices paid, product quality, or within-category substitution ([Sangani, 2026](#)).

A related literature evaluates the fiscal response to the recent inflation surge. Using household microsimulation models, [Pallotti et al. \(2024\)](#) and [Amores et al. \(2025\)](#) show that fiscal interventions reduced welfare losses and inequality, although broad price measures were less targeted than income support. Using detailed UK household data, [Levell et al. \(2026\)](#) highlight a trade-off between targeting and efficiency: energy price subsidies better protect households with high exposure to energy-price shocks, including within income groups, but distort consumption choices, while transfers preserve price signals but may be less well targeted. For Germany, [Krebs and Weber \(2024\)](#) examine conditions under which energy price controls can be optimal, while [Dertwinkel-Kalt and Wey \(2025\)](#) show that the design of the energy price brakes can induce suppliers to raise prices while reinforcing households’ incentives to save energy. [Bonnet et al. \(2025\)](#) show, for France, that income-targeted transfers are more effective than fuel price subsidies at limiting welfare losses relative to household income. We instead reconstruct monthly counterfactual price paths at the

COICOP-component level for all 20 euro-area countries. This approach identifies which products drove the distributional effect of price-based policies: lower electricity and gas prices reduce the Quintile 1–Quintile 5 inflation gap, while fuel measures partly offset that reduction because higher-income households spend more on private transport.

Our contribution is to combine harmonized household inflation measures with a product-level decomposition of price-policy effects across 20 euro-area countries.

The remainder of the paper is organized as follows. Section 1 describes the data and methodology. Section 2 examines developments in inflation inequality in the euro area. Section 3 presents simulations of the heterogeneous effects of unconventional fiscal policies. Section 4 turns to exposure to import-price shocks, and the inflation burden.

## 1 Data and Methodology

Our approach constructs price indices for households and household groups using the same accounting logic as the Harmonised Index of Consumer Prices (HICP). We describe the data and harmonization challenges, then calibrate expenditure weights to official HICP weights.

### 1.1 Calculating Household Inflation Rates

Headline HICP inflation measures price changes for the aggregate household consumption basket. We construct corresponding indices for income groups, age groups, residence areas, and individual households. Where microdata are available, household groups aggregate individual expenditure records using HBS survey weights; their consumption baskets therefore reflect the expenditure of their constituent households. Prices are common within each national product category, as in the official HICP, while expenditure weights vary across groups or households. We first construct individual household indices. Household-group indices provide an intermediate level of aggregation; the final step summarizes inflation across the euro area population and distinguishes the average household rate from inflation in aggregate expenditure.

This analysis combines public HICP data with restricted HBS microdata and national HBS inputs. The HICP is constructed by national statistical institutes in collaboration with Eurostat. The HICP offers a standardized measure of inflation, ensuring comparability across Member States. Each country’s HICP is based on price changes for a fixed basket of goods and services, weighted according to household expenditure from country-specific sources, including national accounts,

household budget surveys, and other national sources. The national indices are then aggregated into the euro-area HICP using country-specific weights that reflect their share of total household consumption expenditure.<sup>1</sup>

To ensure the HICP reflects recent consumption patterns, Eurostat updates its weighting structure annually. The updated weights are implemented with the January release of the HICP and remain fixed throughout the calendar year, which is the basis for the within-year Laspeyres aggregation. Appendix A examines how the measured Q1–Q5 inflation gap changes across COICOP digit depths 2 to 5, with a 3-digit benchmark. Let  $\mathcal{J}_y$  denote the set of product categories with official HICP weights in year  $y$ . Let  $p_{j,y,m}$  denote the price index of product category  $j$  in month  $m$  of year  $y$ , and let  $p_{j,y,0}$  denote the price reference for the HICP weight year, corresponding to the previous December. The official within-year Laspeyres index is obtained by aggregating unchained product indices with the annual HICP product weights:

$$P_{y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,y}^{\text{HICP}} \cdot \frac{p_{j,y,m}}{p_{j,y,0}}. \quad (1)$$

The chained HICP index is then obtained by linking this within-year index to the previous December:

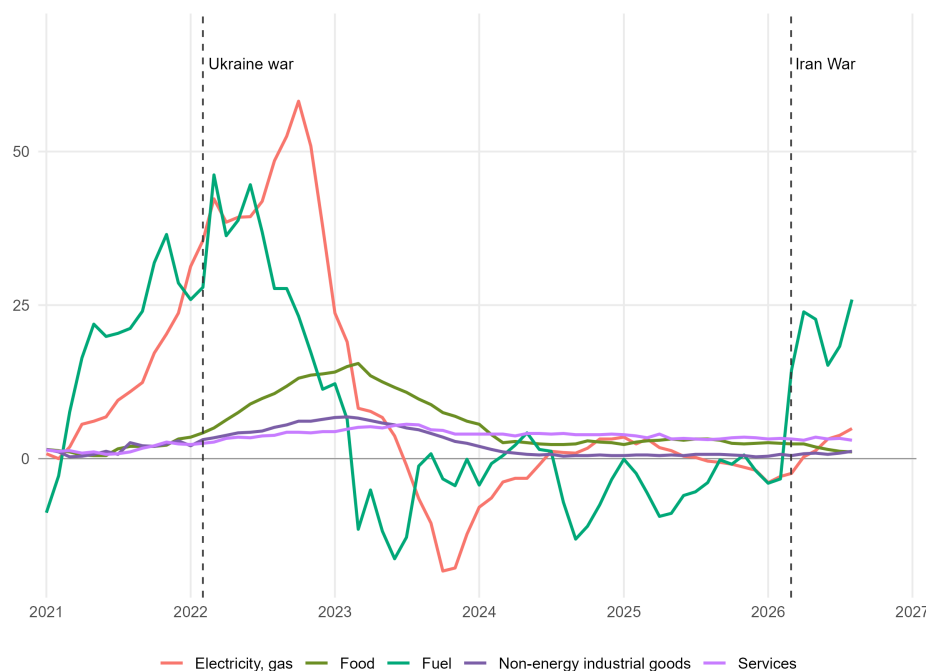
$$I_{y,m} = I_{y-1,12} \cdot P_{y,m}. \quad (2)$$

Figure 1 shows inflation developments across the main HICP components in the euro area since 2014. The series show marked volatility in energy and fuel prices. The sharp rise starting around mid-2021 and peaking in 2022 reflects the post-COVID recovery, supply-chain disruptions, and geopolitical tensions. After the 2022 peaks, inflation declined across most components. Energy prices dropped sharply, while food prices and other components stabilized more gradually.

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<sup>1</sup>The calculation ensures comparability by adhering to Regulation (EU) 2016/792 and consistent data collection practices. The HICP is compiled according to the European Classification of Individual Consumption according to Purpose (ECOICOP). Historical series are mapped to the classification used in the HBS.

Figure 1: Inflation developments in the euro area (year on year, in %)



Sources: Eurostat HICP euro-area aggregate; authors' calculations.

## Consumption Data and Harmonization Challenges

Our primary consumption source is the 2020 HBS household microdata. We harmonize household expenditure records for 19 euro-area countries, excluding Italy. These records contain expenditure by product, survey weights, total net household income, and household characteristics. They serve two purposes: survey-weighted aggregation produces the baskets used for income, age, and residence groups, while the underlying records identify household-specific inflation and its dispersion. Group baskets are therefore constructed from the individual observations wherever the required detail is available, rather than taken from published Eurostat HBS tabulations.

The available inputs support different country samples. Group-level indicators cover all 20 euro-area countries through the explicitly documented national and tabulation-based exceptions below. The household distributions cover 152,613 households in 18 countries: Italy is excluded because the retained national input supplies group baskets rather than household records, and Portugal because its individual expenditure divisions cannot separate rents. These households are held fixed across price years; the monthly series is not a panel of newly observed consumption choices. Figure 3 follows this sample from January 2021 to December 2025, and the paired policy-effect

distribution uses 2021–2023.

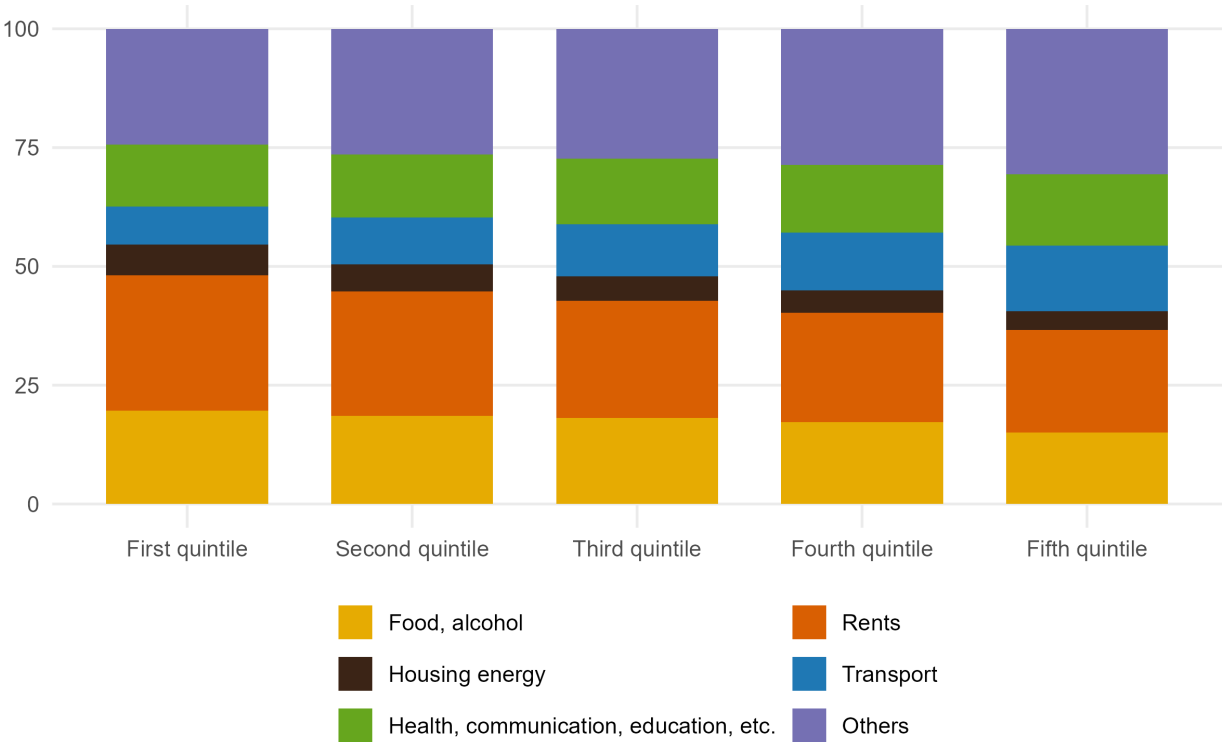
HBS expenditure records combine survey information collected over country-specific reference periods and classified by COICOP product category. The harmonized wave used here is labelled 2020; fieldwork dates and national survey frequency need not coincide across countries. The fixed wave provides a cross-section of consumption heterogeneity rather than annual observations on changing household baskets.<sup>2</sup>

National statistical institutes in Italy and Spain also provide higher-frequency HBS data that can be used for country-specific extensions. Repeated HBS waves available for Spain suggest that, while consumption patterns can change substantially over a few years, differences in the expenditure shares of the main consumption categories between low- and high-income households remain relatively stable (see Appendix C). Changes in overall consumption patterns may therefore have a limited effect on measured inflation inequality when consumption shares move in the same direction across income groups.

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<sup>2</sup>See [Accardo et al. \(2023\)](#) on estimating developments in consumption inequality between survey waves.

Figure 2: Consumption baskets by income quintiles in the euro area (in % of total expenditure)



Notes: Country budget shares are averaged using 2020 HICP country weights, normalized across the 20 countries and held common across quintiles. These are national income quintiles, not euro-area income ranks. Rents comprise actual and imputed rents; the inflation indices exclude both. Microdata supply the baskets except for the documented national and tabulation-based fallbacks.

Sources: HBS microdata, Eurostat, Istat, authors' calculations.

For Figure 2, common 2020 HICP country weights provide a consistent geographic reference for comparing national quintile baskets. They do not measure each country's share of total expenditure within a euro-area quintile. The Italian custom baskets are described in Appendix E. Portugal's unavailable 2020 housing breakdown uses its 2015 within-housing quintile shares, rescaled to the 2020 housing total. Elsewhere, missing imputed rents are supplied by Eurostat, preserving relative shares among observed non-rent expenditures.

While HBS expenditure shares are observed only at low frequency, our weighting approach combines household-specific HBS expenditure structures with monthly HICP inflation rates. Thus, the aggregate weighting structure changes with annual HICP margins, while household expenditure profiles remain fixed between survey observations.

Figure 2 presents consumption baskets by income quintile. Lower-income households allocate a larger share of expenditure to essentials such as food and energy, increasing their exposure to price shocks in these categories. Housing and food account for a substantially larger share of expenditure among lower-income households. Energy-related expenditure for housing and private transport is also much higher in rural areas than in cities, reflecting differences in heating needs and mobility constraints.

HICP and HBS are not directly harmonized. HICP item weights are updated every year and may rely on national accounts or other national sources, while HBS expenditure structures are observed at lower frequency. Depending on the level of data available, HICP and HBS are therefore merged at the three-digit COICOP in each country. In some cases, data published directly by national statistical institutes can be used instead of relying only on the Eurostat dataset. This approach is particularly valuable when the nationally published data provide greater granularity, recency, or frequency.

The HBS contains many household characteristics that can be used to segment inflation exposure (income, education, age of the household reference person, and area of residence).<sup>3</sup> Our microdata income groups rank households by total net household income, without adjustment for household size. This choice differs from ranking households by equivalised income; it describes income at the household level rather than living standards per consumption unit. Published fallback tabulations retain their source classifications, and the Italian income groups are estimated as described below. Our baseline excludes actual and imputed rents and the corresponding HICP rent components before matching, reaggregating parent categories and normalizing weights. The remaining HBS baskets and HICP product shares are normalized over the matched non-rent products. This common coverage applies to household- and group-level indices and policy counterfactuals. It measures non-rent consumption-price exposure rather than an all-items cost-of-living index. Appendix F describes the calibration; Table F.1 retains alternative rent conventions as a separate sensitivity exercise.

The main 2021–2023 analysis and the short-run figures use fixed 2020 HBS baskets. For each national group, we sum individual product expenditure using survey weights and divide by the group’s weighted household population. The RAS expenditure margins use the same households

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<sup>3</sup>In recent HBS waves, Eurostat generally requests the Degree of Urbanisation (DEGURBA) variable, built from a common European methodology based on a 1 km<sup>2</sup> population grid. It classifies residence areas into cities (densely populated areas), towns and suburbs (intermediate density areas), and rural areas (thinly populated areas), using comparable thresholds across EU countries: urban centers of at least 50,000 inhabitants and 1,500 inhabitants per km<sup>2</sup>, urban clusters of at least 5,000 inhabitants and 300 inhabitants per km<sup>2</sup>, and all remaining areas.

and expenditure coverage. Income quintiles use survey-weighted national thresholds of total net household income, without equivalisation. Missing income excludes a household from income baskets, but not from age, residence, or unconditional analyses when the required information is available.

Missing or negative expenditure is treated as zero. Rents are excluded before parent categories are rebuilt. Products with no observed expenditure are not treated as observed; the group-basket implementation retains its numerical positivity floor for calibration.

There are three exceptions to the microdata construction. Italy retains the national group baskets described in Appendix E. Portugal uses published group tabulations for non-rent inflation because its individual expenditure divisions cannot isolate rents. Malta retains published residence-area baskets and relative-expenditure weighting because the required groups and RAS margins are unavailable. Historical and validation exercises retain their separately stated HBS waves.

These data constraints motivate the weighting methodology below. A simple use of HBS expenditure shares would generally not aggregate back to the official HICP, because the HBS all-household basket does not necessarily coincide with the annually updated HICP item weights. We therefore start from the official HICP product weights and use HBS to distribute each product's expenditure across households and, at the intermediate aggregation level, household groups. The Row and Column Adjustment Scaling (RAS) calibration preserves the relative HBS expenditure profiles while imposing the row and column margins required by the HICP construction.

## Household-Level Price Indices

For household  $i$  in country  $c$ , let  $a_{ic}$  denote its survey weight and  $E_{ic}$  its expenditure on the matched non-rent product set  $\mathcal{J}_{c,y}$ . Let  $\omega_{c,j,y}^{\text{HICP}}$  be the official weight of product  $j$  in year  $y$ . We normalize the household expenditure margins and HICP product margins to sum to one:

$$s_{ic,y} = \frac{a_{ic}E_{ic}}{\sum_h a_{hc}E_{hc}}, \quad \bar{\omega}_{c,j,y}^{\text{HICP}} = \frac{\omega_{c,j,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_{c,y}} \omega_{c,k,y}^{\text{HICP}}},$$

where  $h$  indexes households in country  $c$ . RAS rescales the household-product expenditure matrix to match both margins. Its calibrated cells  $M_{j,ic,y}$  satisfy

$$\sum_{j \in \mathcal{J}_{c,y}} M_{j,ic,y} = s_{ic,y}, \quad \sum_i M_{j,ic,y} = \bar{\omega}_{c,j,y}^{\text{HICP}}.$$

Normalizing each household's calibrated expenditure gives its product weights:

$$\omega_{j,ic,y} = \frac{M_{j,ic,y}}{s_{ic,y}}, \quad \sum_{j \in \mathcal{J}_{c,y}} \omega_{j,ic,y} = 1. \quad (3)$$

Thus, expenditure-weighted household baskets reproduce the HICP product shares on matched support. Appendix F details the RAS algorithm.

Product indices are first expressed relative to the preceding December, aggregated with annual household weights, and then chained to the previous December’s household index.<sup>4</sup> Let  $p_{c,j,y,m}$  denote the product price index in month  $m$  of year  $y$ .

The indices are rebased to an annual average of 100 in 2020. Monthly year-on-year household inflation is  $100(I_{ic,t}/I_{ic,t-12} - 1)$ . Figure 3 reports country-weighted averages of national means and empirical percentiles. The household-level application uses fixed 2020 HBS baskets at COICOP digit 3. Annual HICP margins update aggregate product composition while household expenditure profiles remain fixed; historical exercises using successive HBS waves retain their stated matching convention. Native 3-digit expenditure is preferred; finer codes are aggregated where available. Distributional analyses retain households with positive retained expenditure, and finite inflation rates; they require no income filter. Income quintiles are defined within each country using total net household income, without equivalisation. Income-based baskets retain households with finite income. The regression and burden samples additionally require a positive reported equivalence scale, without equivalising income.

For Figure 3, source price series must be complete over the full calculation period before recoding to ECOICOP v1 at digit 3. This keeps the set of detailed products within each aggregate constant over time. Price coverage is separate from the treatment of missing expenditure cells.

## 1.2 Aggregating Household Inflation Rates

### Household-Group Price Indices

For each income, age, or residence group  $g$ , we aggregate expenditure records using survey weights before applying RAS. Group indices therefore need not equal averages of separately calibrated household indices. Country subscripts are suppressed below.

**Group-Specific Weights.** The calibration follows the household procedure, replacing individual expenditure margins with the group’s share  $s_{g,y}$  of total HBS expenditure. If  $M_{j,g,y}$  is the calibrated product–group mass, the group basket is  $\omega_{j,g,y} = M_{j,g,y}/s_{g,y}$ . It sums to one, and its expenditure-weighted average across groups reproduces the normalized HICP product shares. Appendix F

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<sup>4</sup>See Eurostat’s `hicp` R package, which implements the Eurostat HICP aggregation.

gives the group margins and their matching to HBS waves. The baseline uses the 2020 wave throughout.

**Group-Level Consumer Price Indices.** Using the same December-relative aggregation and chaining as for households, we obtain

$$P_{g,y,m} = \sum_{j \in \mathcal{J}_y} \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y-1,12}}, \quad I_{g,y,m} = I_{g,y-1,12} P_{g,y,m}, \quad (4)$$

where  $\mathcal{J}_y$  is the matched non-rent product set,  $P_{g,y,m}$  the within-year price link, and  $I_{g,y,m}$  the chained group index. Group calculations can use parent-category prices when detailed series are missing; household calculations retain usable prices and weights at the required depth. Calibration targets the matched non-rent basket; the all-items reference exercises are reported in Appendices F and G.

## Mean Household Inflation and Aggregate Inflation

Figure 3 uses a two-stage aggregation. Within each country, we calculate the mean and the 10th, 20th, 80th and 90th empirical percentiles of household year-on-year inflation using HBS survey weights. We then average each national statistic across countries using Eurostat’s annual HICP country expenditure weights, normalized over the 18 countries in the sample. For a national statistic  $s_{c,y,m}$ , the reported series is

$$s_{y,m} = \sum_c \omega_{c,y} s_{c,y,m}, \quad \omega_{c,y} = \frac{W_{c,y}}{\sum_{k \in \mathcal{C}} W_{k,y}},$$

where  $W_{c,y}$  is the published all-items HICP country weight and  $\mathcal{C}$  is the fixed country sample. The bands therefore average national percentile endpoints; they are not percentiles of a pooled euro-area distribution. These weights are not additionally adjusted for rents. Averaging these statistics is distinct from constructing aggregate HICP inflation.

To obtain an HICP-comparable total, let  $L_{ic,y,m}$  denote the household price link relative to the preceding December. Let  $q_{c,y}$  be the annual HICP country expenditure weight, adjusted to exclude rents and normalized over the countries included. We aggregate links before chaining:

$$L_{c,y,m} = \frac{\sum_i a_{ic} E_{ic} L_{ic,y,m}}{\sum_i a_{ic} E_{ic}}, \quad (5)$$

$$L_{y,m} = \sum_c q_{c,y} L_{c,y,m}, \quad I_{y,m} = I_{y-1,12} L_{y,m}. \quad (6)$$

Averaging household inflation rates directly does not implement this annual chain aggregation.

All baseline inflation results exclude actual and imputed rents. Official headline series and the explicitly identified rent-sensitivity and validation exercises retain their stated comparison coverage.

## **2 Main Results**

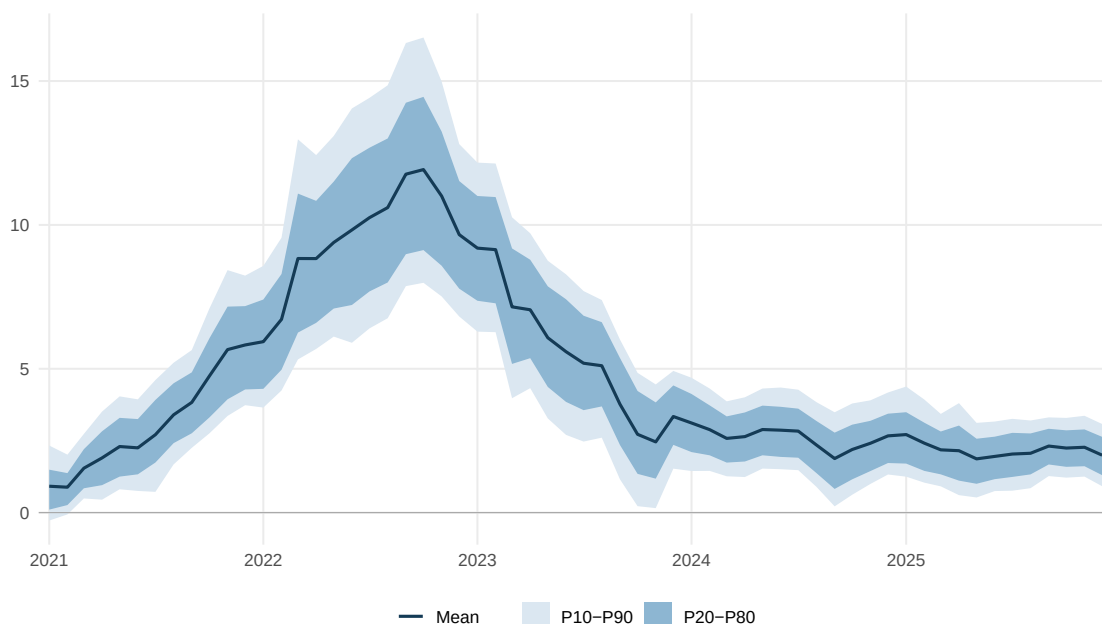
This section presents our main results. We focus on inflation inequality in the wake of the 2021–2023 energy crisis. We further document cross-country and cross product heterogeneity in inflation inequality.

### **2.1 Inflation Inequality in the Euro Area**

#### **2.1.1 Short-Run Inflation Inequality**

Figure 3 traces country-weighted national means and percentile ranges from January 2021 to December 2025. The mean rises from 0.9% in January 2021 to a peak of 11.9% in October 2022. The weighted average of national P90–P10 ranges increases from 2.6 percentage points in January 2021 to a maximum of 8.5 points in October 2022. In October 2022, the weighted national P10 and P90 endpoints are 8.0% and 16.5%, respectively. The inflation surge therefore coincides with greater dispersion across household baskets within countries.

Figure 3: Household inflation: country-weighted national means and percentiles, 2021–2025 (year on year, in %)



Notes: Monthly year-on-year inflation for 152,613 households in 18 euro-area countries, using fixed 2020 HBS microdata baskets. Italy and Portugal are excluded because household baskets or the detail needed to remove rents are unavailable. Actual and imputed rents (041 and 042) are excluded. Household indices use annual RAS calibration with expenditure margins at the three-digit COICOP level. National means and empirical percentiles first use HBS survey weights, without an income filter. Each national statistic is then averaged using annual Eurostat HICP country weights normalized over these 18 countries. The solid line shows the mean; the darker and lighter bands show averages of national P20–P80 and P10–P90 endpoints. These are neither pooled euro-area percentiles nor confidence intervals.

Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

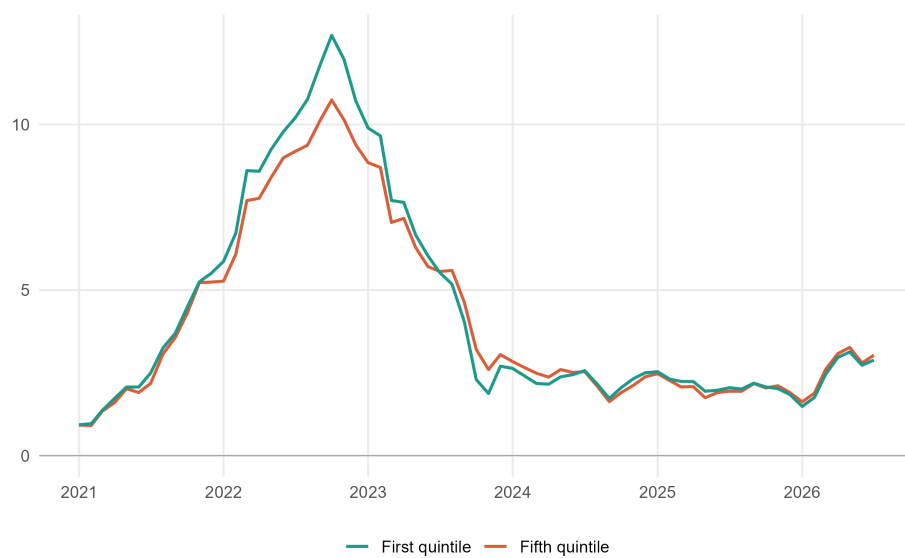
The band widths summarize within-country differences across household baskets, averaged with annual country expenditure weights. They do not measure the dispersion of national inflation levels around the euro-area mean or statistical uncertainty. Unlike the transaction data used by [Kaplan and Schulhofer-Wohl \(2017\)](#), our data do not identify differences in prices paid by households for the same product within a country. The households at each national percentile can change from month to month, although the underlying sample is fixed. The country-weighted mean of national household inflation rates need not coincide with inflation calculated from an expenditure-weighted, chained aggregate HICP index.

Figure 4 shows moderate income-based inflation inequality at the euro-area level during the post-COVID inflation surge. This aggregate result masks substantial cross-country heterogeneity,

including reversals in the sign of the Quintile 1–Quintile 5 gap across countries and years. Inflation rates for the first and fifth income quintiles remain close in the EA20 aggregate throughout 2021–2023 because national consumption structures and country-level price dynamics partly offset one another. The small aggregate gap should therefore not be interpreted as evidence that distributional exposure was uniformly moderate within member countries.

This pattern is broadly consistent with recent evidence from [Bobasu et al. \(2023\)](#) and [Claeys et al. \(2022\)](#), although small differences remain because of our weighting scheme or exclusion of actual and imputed rents.

Figure 4: Short-run inflation inequality in the euro area, by income quintile (year on year, in %)



Notes: This figure reports year-on-year inflation rates for the first and fifth income quintiles in the euro area from January 2020 to July 2026. Group baskets are constructed from fixed 2020 HBS microdata, using total net household income without equivalisation, with the Italian approximation and Portuguese tabulation fallback described in Section 1. The country series are aggregated using annual HICP country weights.

Sources: Eurostat HICP, HBS, authors' calculations

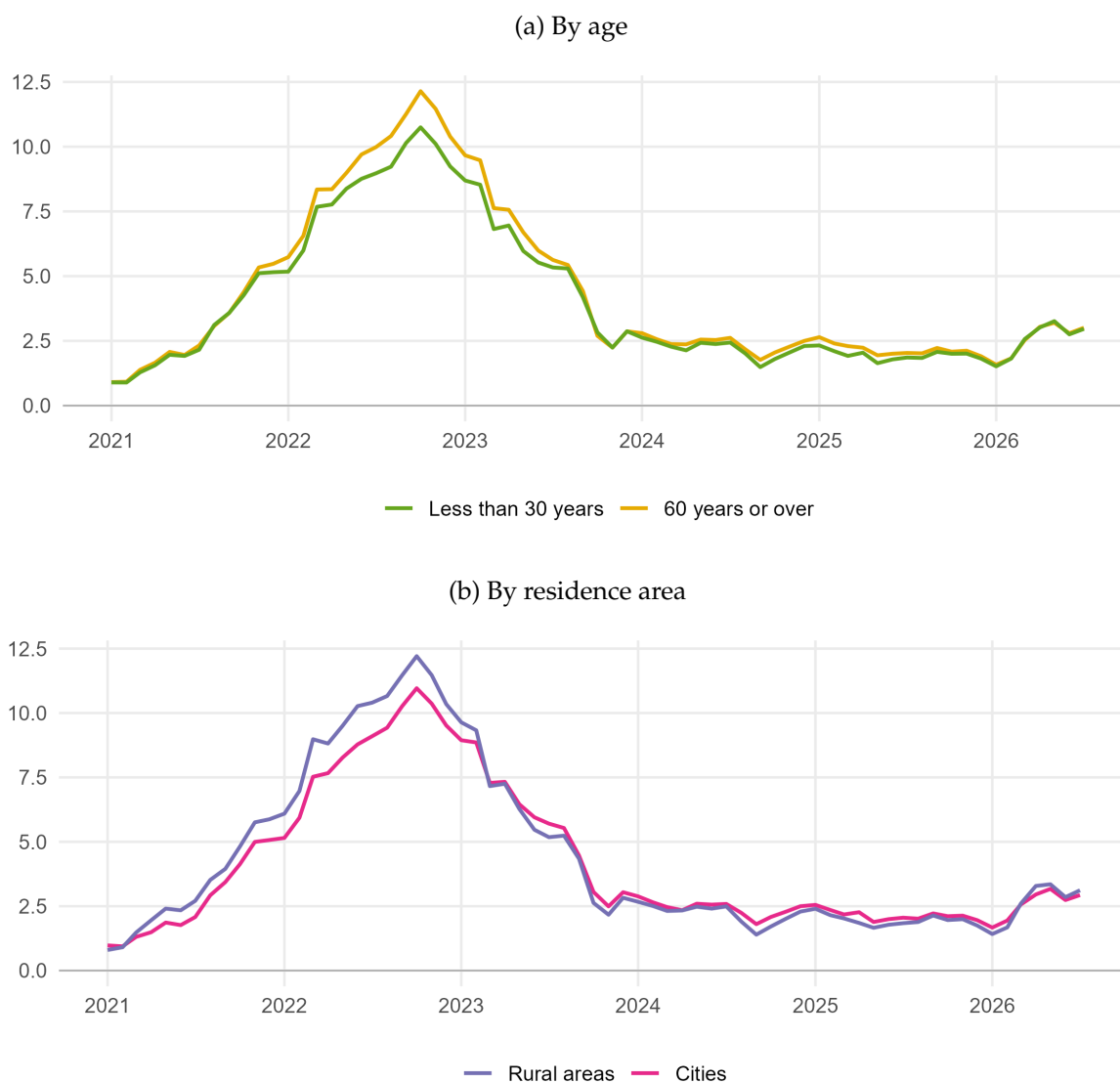
### 2.1.2 Inflation Inequality by Age and Residence Area

Figure 5 presents inflation rates by household category in the euro area, segmented by age and residence area. In the household-level regressions pooling 2021–2023, average annual inflation is 0.6 point higher for households whose reference person is aged 60 or over than for those under 30, and 0.4 point higher in rural areas than in cities. The monthly year-on-year inflation rates in the figure show that both gradients vary over time and temporarily narrow or reverse as energy inflation recedes. [Gautier and Montornes \(2024\)](#) obtain similar age-related results under slightly different assumptions.

The residence-area differentials reflect differences in the structure of energy demand. Rural households tend to spend a larger share of their budget on transport fuels and home energy, partly because commuting distances are longer and the set of available transport substitutes is more limited. This raises their exposure when energy prices accelerate, but also makes their measured inflation fall faster when those price pressures unwind.

Appendix H complements these short-run comparisons with evidence on long-run inflation inequality across income quintiles in the euro area.

Figure 5: Inflation by household categories in the euro area (year on year, in %)



Notes: Year-on-year inflation from fixed 2020 HBS group baskets and annual HICP country weights, excluding rents. National exceptions are documented in Section 1. The conditional annual gaps discussed in the text come from Table 6. Sources: Eurostat HICP and HBS, authors' calculations.

## 2.2 Heterogeneous Effects of the 2022–2023 Energy Shock Across Countries

The 2021–2023 inflation episode was caused by a large energy-price shock, especially in natural gas markets, following Russia’s invasion of Ukraine. Although this shock was common to the euro area, its distributional effects differed across countries because energy mixes, import dependence, household consumption patterns, and fiscal responses were not uniform.

Table 1 documents this cross-country heterogeneity by reporting annual-average inflation and the annual Quintile 1–Quintile 5 inflation gap. The gap is largest in 2022 in most countries: it exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. Several countries subsequently record a negative gap in 2023 as energy-price pressures recede, while the euro-area gap falls from 1.2 points in 2022 to zero after rounding in 2023. This cross-country pattern is qualitatively similar to [Claeys et al. \(2022\)](#), who also find larger inflation gaps between low- and high-income households in countries with higher inflation.

The interaction between country-specific price dynamics, exposures, and fiscal policies and group-specific expenditure shares explains these differences. A price increase in a product category generates inflation inequality only to the extent that the affected category has different expenditure weights across household groups. Hence, similar income-quintile consumption profiles can still produce different inflation gaps when national energy prices, food prices, or fiscal cushioning measures evolve differently. Conversely, large aggregate shocks may generate limited inequality when price increases fall on categories with similar weights across quintiles or when policy measures dampen prices in essential energy components.

Table 1: Annual-average inflation rates and Q1–Q5 inflation inequality in the 20 euro-area countries, 2021–2023

| Country          | Weight<br>(2022, %) | Average inflation |            |            | Inflation inequality |            |            |
|------------------|---------------------|-------------------|------------|------------|----------------------|------------|------------|
|                  |                     | 2021              | 2022       | 2023       | 2021                 | 2022       | 2023       |
| Germany          | 28.1                | 3.5               | 9.7        | 6.5        | -0.4                 | 1.4        | 0.5        |
| France           | 20.4                | 2.3               | 6.6        | 6.1        | 0.4                  | 0.2        | 0.2        |
| Italy            | 16.3                | 2.0               | 9.0        | 6.0        | 0.1                  | 1.6        | 0.5        |
| Spain            | 11.0                | 3.1               | 8.6        | 3.4        | 0.8                  | 0.9        | -0.7       |
| Netherlands      | 5.4                 | 3.2               | 13.2       | 4.0        | 0.0                  | 3.1        | -1.2       |
| Belgium          | 4.0                 | 3.4               | 11.0       | 2.0        | 1.3                  | 2.3        | -2.7       |
| Austria          | 3.3                 | 2.8               | 9.1        | 7.7        | -0.5                 | 0.0        | 0.5        |
| Portugal         | 2.2                 | 0.9               | 8.3        | 5.3        | 0.8                  | 0.2        | -0.9       |
| Greece           | 2.2                 | 0.6               | 9.6        | 4.1        | 0.3                  | 1.2        | -0.6       |
| Finland          | 1.9                 | 2.2               | 7.8        | 4.5        | -0.2                 | -0.9       | 0.4        |
| Ireland          | 1.5                 | 2.4               | 7.9        | 5.0        | 0.5                  | 2.9        | 1.1        |
| Slovakia         | 0.8                 | 2.9               | 12.3       | 11.1       | -0.9                 | 0.1        | 0.5        |
| Croatia          | 0.7                 | 2.7               | 10.8       | 8.5        | 0.0                  | 0.2        | -0.3       |
| Lithuania        | 0.6                 | 4.7               | 18.9       | 8.7        | 0.2                  | 3.3        | 0.4        |
| Slovenia         | 0.4                 | 2.1               | 9.4        | 7.2        | 0.0                  | 0.4        | 1.5        |
| Luxembourg       | 0.3                 | 3.6               | 8.6        | 3.0        | 0.1                  | 0.2        | -0.1       |
| Latvia           | 0.3                 | 3.3               | 17.6       | 9.1        | -0.2                 | 4.5        | 2.5        |
| Estonia          | 0.2                 | 4.5               | 19.4       | 9.4        | 0.6                  | 4.8        | 1.3        |
| Cyprus           | 0.2                 | 2.3               | 8.2        | 3.9        | 0.0                  | 0.8        | 0.2        |
| Malta            | 0.1                 | 0.8               | 6.1        | 5.4        | -0.2                 | 0.3        | 0.6        |
| <b>Euro area</b> | <b>100.0</b>        | <b>2.8</b>        | <b>9.0</b> | <b>5.6</b> | <b>0.1</b>           | <b>1.2</b> | <b>0.0</b> |

Notes. Annual inflation compares annual-average indices with the preceding year. The inflation gap is Q1 minus Q5, in percentage points. Weights are 2022 EA20 HICP shares. Sources: Eurostat, authors' calculations.

Table 2 decomposes euro-area inflation and the Q1–Q5 gap by product group. In 2022, household energy accounts for 3.1 points of average inflation and contributes 1.9 points to the 1.2-point gap; food adds 1.9 points to average inflation and 0.7 point to the gap. These positive distributional contributions are partly offset by personal transport, other transport, and other products, which contribute -0.4, -0.3, and -0.5 point to the gap. In 2023, the positive food contribution is again offset by other product groups, leaving a zero aggregate gap after rounding.

Table 2: Product-group contributions to annual-average euro-area inflation and Q1–Q5 inflation inequality, 2021–2023

| Product group                              | Weight<br>(2022, %) | Average inflation |            |            | Inflation inequality |            |            |
|--------------------------------------------|---------------------|-------------------|------------|------------|----------------------|------------|------------|
|                                            |                     | 2021              | 2022       | 2023       | 2021                 | 2022       | 2023       |
| Food and non-alcoholic beverages           | 17.8                | 0.3               | 1.9        | 2.1        | 0.1                  | 0.7        | 0.7        |
| Alcoholic beverages, tobacco and narcotics | 4.6                 | 0.1               | 0.2        | 0.3        | 0.1                  | 0.1        | 0.2        |
| Clothing and footwear                      | 5.6                 | 0.1               | 0.1        | 0.2        | 0.0                  | 0.0        | -0.1       |
| Housing maintenance and furnishings        | 8.0                 | 0.2               | 0.6        | 0.5        | -0.1                 | -0.3       | -0.2       |
| Water, electricity, gas and other fuels    | 6.8                 | 0.7               | 3.1        | 0.2        | 0.4                  | 1.9        | 0.1        |
| Operation of personal transport equipment  | 6.3                 | 0.8               | 1.4        | 0.1        | -0.3                 | -0.4       | 0.0        |
| Transport (excluding vehicle operation)    | 9.6                 | 0.1               | 0.4        | 0.3        | -0.1                 | -0.3       | -0.2       |
| Other                                      | 41.1                | 0.4               | 1.3        | 1.9        | -0.1                 | -0.5       | -0.6       |
| <b>Total</b>                               | <b>100.0</b>        | <b>2.8</b>        | <b>9.0</b> | <b>5.6</b> | <b>0.1</b>           | <b>1.2</b> | <b>0.0</b> |

Notes. Product rows contribute to annual EA20 average inflation (percent) and the Q1–Q5 gap (percentage points).

Annual rates compare annual-average indices with the preceding year. Weights are 2022 EA20 expenditure shares.

Sources: Eurostat, authors' calculations.

### 3 Effects of Household Fiscal Policies on Inflation Inequality

This section evaluates the distributional effects of unconventional fiscal policies implemented during the 2021–2023 energy crisis that directly changed consumer prices faced by households, including electricity and gas price caps and shields, VAT and energy-tax reductions, cuts in regulated charges, and fuel rebates.

#### 3.1 Constructing Counterfactual Price Indices

In response to the energy crisis, euro-area governments introduced a broad array of measures to mitigate household exposure to rising energy costs (Arregui et al., 2022, Dao et al., 2023, García-Miralles, 2023, Levell et al., 2026). Euro-area countries differed in their choice of instruments, combining tax reductions, price controls, energy-bill rebates, and direct income transfers in different proportions. France froze regulated gas tariffs at their October 2021 level through 2022, then limited

their increase in early 2023, while also limiting increases in regulated electricity tariffs and cutting electricity taxes. Germany’s 2023 household gas and electricity price brakes instead provided bill rebates linked to the positive difference between contractual and guaranteed prices on 80 percent of a predetermined reference quantity, preserving the contractual price at the margin ([Dertwinkel-Kalt and Wey, 2025](#), [Krebs and Weber, 2024](#)). The Netherlands capped household gas and electricity prices in 2023 up to uniform annual consumption limits; when binding, the cap lowered the marginal price within the covered block, whereas the German rebate was based on a predetermined reference quantity ([Haan and Schinkel, 2023](#)). Spain combined energy-tax cuts and fuel rebates with limits on regulated gas-tariff increases and the Iberian mechanism, jointly implemented with Portugal, which compensated eligible fossil-fuel generators to lower wholesale electricity prices, with the compensation financed by a levy on benefiting buyers and congestion revenues. These differences in policy design and coverage matter for both household exposure and the construction of counterfactual prices. We combine the Bruegel datasets of unconventional fiscal policies with national sources to identify price measures, including Spanish and Portuguese food-product VAT cuts. The fiscal accounting assigns a documented cost to each of the 191 measures active in the counterfactual and counts shared envelopes once. Some shared envelopes cannot be split by instrument or targeting status, and some include business beneficiaries, so those decompositions do not isolate amounts received by households.

Table [D.1](#) in the appendix reports the euro-area inventory and identifies the measures simulated in our price counterfactuals. We construct these indices for measures for which the necessary statutory, price, and unit-conversion information is available; Appendix [D](#) reports the country-level series.

We compare observed policy-inclusive retail prices with estimated counterfactual price paths for energy and food products. Because the measures operate through administered prices, tax-inclusive prices, retail price ceilings, or unit rebates, their direct effects are confined to the covered product categories. The counterfactuals are a direct consumer-price accounting exercise. We hold expenditure weights fixed between the observed and counterfactual scenarios and do not model changes in consumption, firms’ pricing responses outside the covered components, or feedback through wages or demand. The resulting differences measure the direct contribution of the covered price measures to inflation and inflation inequality, rather than their welfare effects.

We distinguish direct reconstructions, scalar calibrations, and conditional tariff or bill benchmarks. Observed counterfactual tariffs, statutory tax rates, and statutory unit rebates permit direct reconstruction when coverage and pass-through are also specified. For aggregate-equivalent

calibrations, we choose one measure-specific intensity parameter so that the geometric mean of the monthly counterfactual-to-observed price ratios over the active policy months matches the assumed component-level ratio recorded in the policy inventory, which summarizes coverage and pass-through without reconstructing individual household bills.

Counterfactual price indices are constructed at COICOP digit 3 by replacing each observed HICP component index with its counterfactual counterpart. Let  $p_{j,t}$  denote the observed HICP component index for product  $j$  in month  $t$ , and let  $\tilde{p}_{j,t}$  denote the corresponding counterfactual index. Here  $t$  indexes calendar months, corresponding to the year-month pair  $(y, m)$  used above. Let  $r_{j,t}$  denote the observed retail price per unit and  $\tilde{r}_{j,t}$  its counterfactual counterpart, expressed in the same monetary units. For each simulated measure, we apply the implied price ratio to the observed component index<sup>5</sup>:

$$\tilde{p}_{j,t} = p_{j,t} \frac{\tilde{r}_{j,t}}{r_{j,t}}. \quad (7)$$

**VAT and Excise-Tax Reductions.** For VAT reductions, the observed retail price includes the temporary tax rate. We recover the counterfactual price by removing that rate and applying the standard rate (Gautier and Lalliard, 2013):

$$\tilde{r}_{j,t} = r_{j,t} \frac{1 + \tau_{j,t}^0}{1 + \tau_{j,t}^1}, \quad (8)$$

where  $\tau_{j,t}^0$  is the standard VAT rate and  $\tau_{j,t}^1$  is the temporary rate. Ad valorem excises follow the same ratio logic; specific excise cuts require a unit-price adjustment, as described below. Coverage and pass-through are calibrated where the statutory information is incomplete.

**Administered Prices.** For regulated-price shields and caps, we use an external reference path where available, such as a commodity-price benchmark or a tariff published by the national regulator (for example, the CRE in France):

$$\tilde{r}_{j,t} = r_{j,t}^{\text{ref}}. \quad (9)$$

When the observed tariff and the counterfactual reference tariff are both available, the counterfactual retail price is obtained by rescaling the observed price. If  $T_{j,t}^{\text{obs}}$  denotes the regulated tariff actually applied and  $T_{j,t}^{\text{ref}}$  the tariff that would have applied without the intervention, then:

$$r_{j,t}^{\text{ref}} = r_{j,t} \frac{T_{j,t}^{\text{ref}}}{T_{j,t}^{\text{obs}}}. \quad (10)$$

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<sup>5</sup>When several measures apply to the same product in the same country and month, the R scripts combine their counterfactual-to-observed price ratios by adding their deviations from one to a baseline ratio of one.

For the Netherlands, contract coverage and consumption limits require particular care. [Statistics Netherlands \(CBS\) \(2023\)](#) report tariffs with and without the cap and account for consumption above the statutory limits, but their temporary series covers only newly concluded variable contracts, excluding existing and fixed-term contracts. It is therefore not an all-household price series. Our Dutch reconstruction uses the broader CBS contract tariffs and net component bills documented by [de Bondt et al. \(2024\)](#). For each month of 2023, we add the cap discount back to the published gas and electricity bills at the same reference quantities, retain fixed charges and taxes, and avoid counting the energy-tax refund twice. The ratio of uncapped to capped bills is applied to the corresponding HICP component without fitting a scalar parameter. This removes the cap mechanically at observed contractual tariffs; it does not estimate suppliers' pricing responses. [Haan and Schinkel \(2023\)](#) explain why partial coverage and compensation linked to suppliers' own tariffs can weaken competition and create upward pricing incentives.

**Subsidies and Rebates.** For unit subsidies and fuel rebates, we add the statutory absolute price wedge to the observed monthly price path or, without a retail unit-price series, calibrate it in HICP component-index units. Evidence from [Drolsbach et al. \(2023\)](#) suggests that temporary fuel-tax reductions and discounts were largely passed through to consumer prices in 2022 in several euro-area countries, although pass-through varied over time and by fuel type. This supports treating these measures as retail-price wedges in the counterfactual construction. Under full pass-through, adding the statutory support amount  $s_{j,t}$  to the observed unit price gives:

$$\tilde{r}_{j,t} = r_{j,t} + s_{j,t}. \quad (11)$$

For transport-fuel rebates,  $s_{j,t}$  is recovered from the statutory pump discount, expressed in euros per litre, and applied to a monthly retail-fuel price  $r_{j,t}^M$ , where the upright superscript M denotes the observed market price per litre corresponding to HICP component  $j$ . Following [Bonnet et al. \(2025\)](#), we refer to these interventions as pump-price rebates or subsidies because they reduced the tax-inclusive price paid by consumers, although their technical implementation sometimes operated through fuel excise taxation rather than through a direct transfer to households. The no-rebate fuel index is obtained by rescaling the observed HICP transport-fuel component:

$$\tilde{p}_{j,t} = p_{j,t} \frac{r_{j,t}^M + s_{j,t}}{r_{j,t}^M}.$$

These transport-fuel measures are applied only to fuels and lubricants for personal transport equipment, not to the whole vehicle-operation group. The product decompositions nevertheless report the broader vehicle-operation aggregate; its remaining components retain their observed

price paths unless they are affected by a separately identified measure. More generally, when the Bruegel dataset identifies a subsidy but does not supply a monthly unit price, we set  $s_{k,t} = \delta_k \bar{p}_j$  in component-index units. This yields the time-varying ratio  $1 + \delta_k \bar{p}_j / p_{j,t}$ . The single parameter  $\delta_k$  is calibrated.

For a small set of price-relevant interventions, the policy inventories identify the measure but do not provide the monthly counterfactual tariff schedule, the distribution of household consumption across tariff blocks, or the eligible-household weights needed for a direct statutory reconstruction. We therefore add these cases as separate calibrated layers, identified from [Sgaravatti et al. \(2021\)](#) and [Arregui et al. \(2022\)](#), rather than merging them with the VAT or unit-subsidy calculations. This applies to Cyprus’s electricity-bill subsidy, Ireland’s electricity credits and Public Service Obligation levy reduction, Italy’s system-charge reductions and social energy bonuses, and to Belgium’s extension of the social energy tariff and basic energy package. The calibrated ratios are applied only to the relevant COICOP energy components and policy months, and the corresponding assumption tables are reported with the counterfactual outputs. These layers should be interpreted as aggregate-equivalent price effects, not as statutory tariff reconstructions.

Germany’s price brakes instead use a representative-bill model anchored on Bundesnetzagentur’s April 2023 prices and the statutory rebate on 80 percent of reference consumption, with monthly HICP movements used to infer the uncapped bill under fixed quantity assumptions; this is a conditional model, not an observed counterfactual tariff series. For Slovenia’s electricity and gas caps, we replace constant ratios with a published-contract benchmark. We add the positive difference between the GEN-I July 2022 new-customer supply tariff and the statutory household cap, including the VAT applicable that month, to a full-bill price anchored on SURS fourth-quarter 2022 averages and scaled by the monthly product HICP. Network charges and excise are held fixed in this cap calculation. The legal cap window runs from September 2022 through December 2023; the benchmark gas tariff falls below the cap in November 2023, so its cap effect is then zero. This calculation assumes full exposure to the reference contract. National contract shares are unavailable, and the resulting effect remains a conditional benchmark, with alternative contract and exposure scenarios reported separately. The cap calculation retains the contemporaneous VAT rate; it does not itself reverse the separate VAT reduction. The Slovenian inventory is not exhaustive. Enterprise-only support is excluded from the household inventory. The counterfactual assumptions table reports the executed measures and their information basis.

Table 3 summarizes the policy decomposition. Fiscal costs sum documented budget envelopes linked to measures active in the counterfactual, counting each shared envelope once. All 191 active

measures have a documented cost, yielding an EA20 total of 1.58 percent of 2022 GDP. Envelopes without an identified instrument split are retained in the “Other” category rather than allocated mechanically, and Bruegel national totals serve only as external comparisons. Source envelopes can cover different periods and beneficiaries and should not be interpreted uniformly as expenditure attributable exclusively to households.

Administered-price measures reduce the Q1–Q5 inflation gap by 0.4 point in both 2022 and 2023. Targeted measures reduce it by 0.3 point in 2022 and 0.1 point in 2023; untargeted measures reduce it by 0.3 and 0.5 points, respectively. These separate effects need not sum to the joint effect because the measures interact. The targeting classification describes policy packages, not household-specific eligibility identified in the HBS.

The tax category includes VAT and excise reductions, including transport-fuel excises, and the other tax and charge relief assigned to this fiscal category. Instrument effects are computed from their separate additive price wedges, so a tax cut is not absorbed by an overlapping subsidy or price cap. Joint fiscal envelopes retain the documented budget allocations. The other calibrated layer mainly comprises Italian social energy bonuses and electricity-system charge relief. Fiscal costs and annual inflation effects are different objects: the latter depend on the affected products, policy timing and price assumptions, and are not calibrated to the fiscal amounts.

Table 3: Annual effects of household price-policy categories on average inflation and inflation inequality in the euro area, 2021–2023

| Policy category                         | Fiscal cost<br>(% of<br>2022 GDP) | Effect of price policies on<br>average inflation |             |             | Effect of price policies on<br>Q1–Q5 inflation inequality |             |             |
|-----------------------------------------|-----------------------------------|--------------------------------------------------|-------------|-------------|-----------------------------------------------------------|-------------|-------------|
|                                         |                                   | 2021                                             | 2022        | 2023        | 2021                                                      | 2022        | 2023        |
| <i>Instrument classification</i>        |                                   |                                                  |             |             |                                                           |             |             |
| VAT and excise cuts                     | 0.55                              | -0.1                                             | -0.6        | 0.0         | 0.0                                                       | -0.2        | -0.1        |
| Administered prices                     | 0.81                              | 0.0                                              | -0.7        | -0.7        | 0.0                                                       | -0.4        | -0.4        |
| Fuel subsidies and rebates              | 0.08                              | 0.0                                              | -0.2        | 0.1         | 0.0                                                       | 0.0         | 0.0         |
| Other calibrated layer                  | 0.14                              | 0.0                                              | -0.1        | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| <b>Total</b>                            | <b>1.58</b>                       | <b>-0.1</b>                                      | <b>-1.5</b> | <b>-0.6</b> | <b>-0.1</b>                                               | <b>-0.6</b> | <b>-0.5</b> |
| <i>Targeted and untargeted measures</i> |                                   |                                                  |             |             |                                                           |             |             |
| Untargeted price measures               | –                                 | -0.1                                             | -1.0        | -0.4        | 0.0                                                       | -0.3        | -0.5        |
| Targeted price measures                 | –                                 | 0.0                                              | -0.5        | -0.2        | 0.0                                                       | -0.3        | -0.1        |
| <b>Total</b>                            | <b>1.58</b>                       | <b>-0.1</b>                                      | <b>-1.5</b> | <b>-0.6</b> | <b>-0.1</b>                                               | <b>-0.6</b> | <b>-0.5</b> |

Notes. Observed minus counterfactual annual inflation or Q1–Q5 gap, in percentage points; negative values indicate a reduction. Category effects need not sum to Total because policies interact. Targeting classifies measures, not individual eligibility. Fiscal costs are recorded policy envelopes divided by nominal 2022 GDP for the same 20 countries; budget periods may span several years. Fiscal costs refer to entire schemes affecting household prices, not amounts received exclusively by households. A recorded zero denotes a regulatory measure with no identified direct fiscal cost. Sources: Eurostat, national sources, authors' calculations.

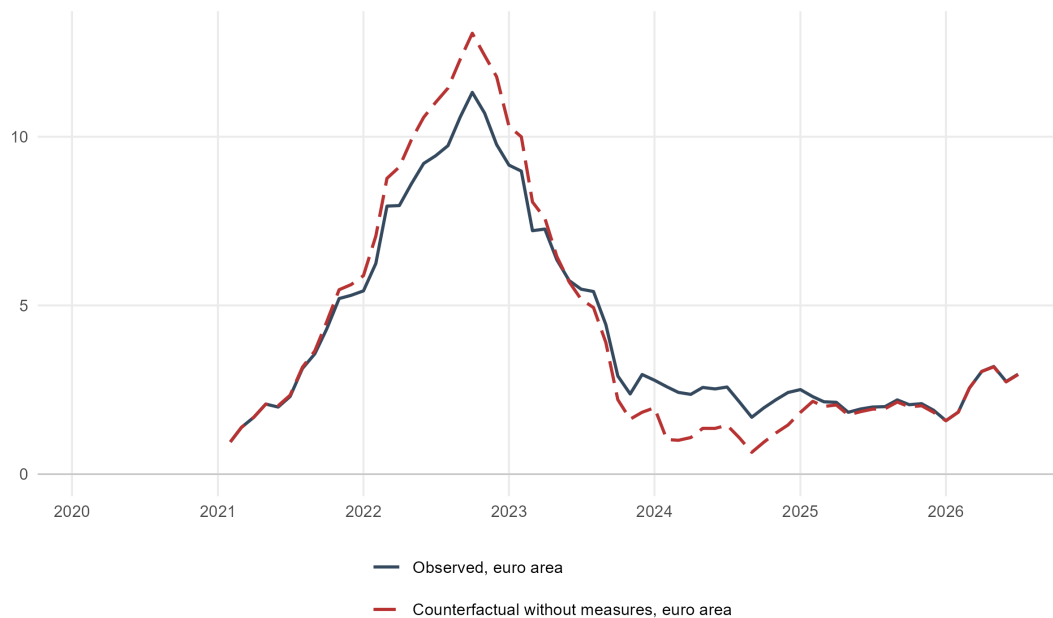
### 3.2 Effects of Unconventional Fiscal Policies

The price-based measures changed both the level and the timing of inflation during the energy crisis. We compare observed inflation with counterfactual paths that remove fiscal policies. This comparison isolates their direct effect on consumer prices.

In annual-average terms, observed minus counterfactual euro-area inflation is –1.5 percentage points in 2022 (Table 4). This estimate is broadly consistent in magnitude with [Dao et al. \(2023\)](#), who find that unconventional fiscal policy measures reduced euro-area inflation by 1–2 percentage points in 2022 under different modeling assumptions. Their estimate includes indirect effects through the pass-through to core inflation, whereas our accounting exercise isolates the direct effect on consumer prices; the comparison therefore concerns the order of magnitude rather than identical effects. Figure 6 shows the effect of unconventional policies on average euro-area inflation.

Counterfactual inflation lies above observed inflation during the main phase of the energy shock, with the largest difference around the inflation peak in late 2022. VAT and excise reductions, fuel rebates, price shields, and regulated-price caps therefore compressed the peak rather than preventing the underlying rise in inflation. The ordering reverses in late 2023 and throughout 2024, when observed inflation exceeds the no-policy counterfactual. The withdrawal of measures and the comparison with policy-depressed prices twelve months earlier can raise observed year-on-year inflation even though the measures lowered prices while they were in place. The two paths largely converge during 2025.

Figure 6: Household price policies: observed and counterfactual euro-area inflation (year on year, in %)



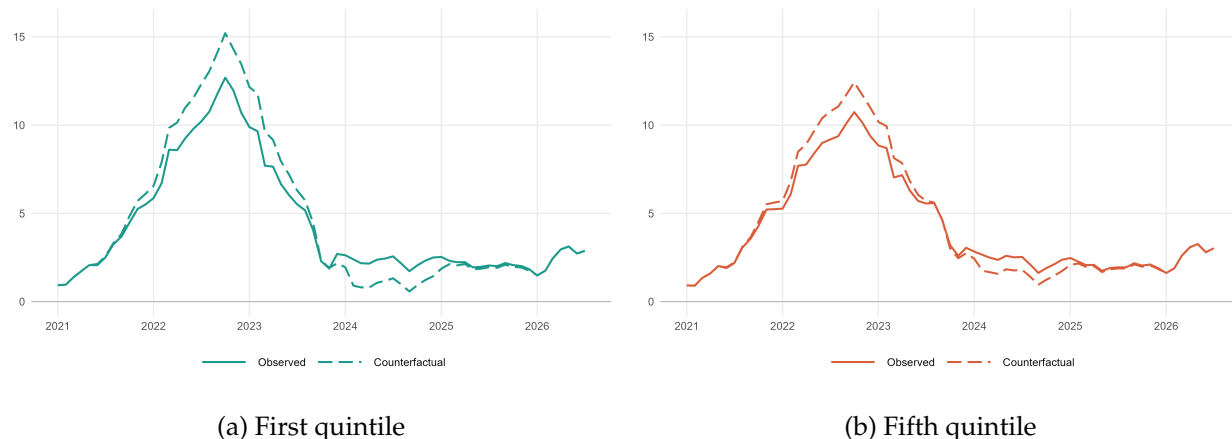
Notes: The counterfactual series removes fiscal interventions that directly affected consumer prices. The figure reports year-on-year inflation for all 20 euro-area countries, aggregated with annual HICP country weights, from January 2021 to July 2026. Policy effects are defined as observed minus counterfactual inflation.

Sources: Eurostat HICP; Bruegel energy-policy dataset; authors' calculations.

Figure 7 compares observed and counterfactual inflation separately for the first and fifth income quintiles. The reduction around the 2022 peak is larger for the first quintile, so the policies narrow the Q1–Q5 gap during the energy shock. In both panels, observed inflation exceeds counterfactual inflation during much of 2024, as the earlier policy wedges enter the year-on-year comparison base. This reversal concerns the policy effect within each quintile and is distinct from a reversal of the Q1–Q5 gap. The two scenarios largely converge in 2025. This pattern indicates that the

measures primarily protected lower-income households when energy prices were rising fastest. This finding is consistent with microsimulation evidence that the European fiscal response reduced distributional losses, although broad price measures were less targeted than income support (Amores et al., 2025, Pallotti et al., 2024).

Figure 7: Household price policies: observed and counterfactual euro-area inflation by income quintile (year on year, in %)



Notes: Solid lines show observed year-on-year inflation and dashed lines show inflation in the absence of household price policies for the first and fifth income quintiles, from January 2021 to July 2026. Income transfers and firm-only measures are outside the counterfactual. Both scenarios use the same group baskets constructed primarily from individual 2020 HBS records, with the Italian approximation and Portuguese tabulation fallback. All 20 countries use RAS expenditure calibration for these income groups and are aggregated using annual HICP country weights.

Sources: Eurostat HICP and country weights; household budget surveys; Bruegel energy-policy dataset; authors' calculations.

### 3.3 Heterogeneity Across Countries and Products

We first compare countries and then decompose the euro-area effect by product group.

#### 3.3.1 Differences Across Countries

Table 4 reports the budgetary cost of schemes affecting household prices and their annual effects on average inflation and the Quintile 1–Quintile 5 inflation gap. These costs describe the scale of the schemes, including business beneficiaries where indicated by an asterisk; they are not inputs into the price-index counterfactual.

Table 4: Household price policies: fiscal cost and effects on annual-average inflation and Q1–Q5 inflation inequality in the 20 euro-area countries, 2021–2023

| Country          | Weight<br>(2022, %) | Fiscal cost<br>(% of 2022 GDP) | Effect of price policies on<br>average inflation |             |             | Effect of price policies on<br>Q1–Q5 inflation inequality |             |             |
|------------------|---------------------|--------------------------------|--------------------------------------------------|-------------|-------------|-----------------------------------------------------------|-------------|-------------|
|                  |                     |                                | 2021                                             | 2022        | 2023        | 2021                                                      | 2022        | 2023        |
| Germany          | 28.1                | 1.2*                           | 0.0                                              | -0.7        | -1.5        | 0.0                                                       | -0.3        | -1.2        |
| France           | 20.4                | 2.9                            | -0.1                                             | -2.7        | -0.6        | 0.0                                                       | -1.1        | -0.5        |
| Italy            | 16.3                | 1.5                            | -0.2                                             | -1.4        | 0.7         | -0.1                                                      | -0.2        | -0.1        |
| Spain            | 11.0                | 2.0*                           | -0.5                                             | -2.1        | -0.3        | -0.2                                                      | -0.8        | -0.3        |
| Netherlands      | 5.4                 | 1.0*                           | 0.0                                              | -2.5        | -0.5        | 0.0                                                       | -1.2        | -0.3        |
| Belgium          | 4.0                 | 0.7*                           | 0.0                                              | -2.4        | 0.5         | 0.0                                                       | -1.0        | 0.1         |
| Austria          | 3.3                 | 0.9*                           | 0.0                                              | -0.2        | -1.0        | 0.0                                                       | -0.1        | -0.5        |
| Portugal         | 2.2                 | 1.5                            | 0.0                                              | -1.0        | -1.8        | 0.0                                                       | -0.4        | -1.1        |
| Greece           | 2.2                 | 2.4*                           | -0.5                                             | -1.6        | 0.3         | -0.4                                                      | -0.8        | -0.1        |
| Finland          | 1.9                 | 0.2                            | 0.0                                              | -0.1        | -0.3        | 0.0                                                       | 0.1         | 0.0         |
| Ireland          | 1.5                 | 0.4*                           | 0.0                                              | -0.8        | 0.4         | 0.0                                                       | -0.5        | 0.1         |
| Slovakia         | 0.8                 | 1.7                            | 0.0                                              | 0.0         | -3.1        | 0.0                                                       | 0.0         | -1.7        |
| Croatia          | 0.7                 | 0.9                            | 0.0                                              | -1.1        | 0.3         | 0.0                                                       | -0.9        | 0.4         |
| Lithuania        | 0.6                 | 0.8                            | 0.0                                              | -0.6        | -0.1        | 0.0                                                       | -0.5        | -0.1        |
| Slovenia         | 0.4                 | 0.3                            | 0.0                                              | -0.8        | 0.0         | 0.0                                                       | -0.1        | -0.5        |
| Luxembourg       | 0.3                 | 0.8*                           | 0.0                                              | -0.4        | -0.6        | 0.0                                                       | -0.1        | -0.2        |
| Latvia           | 0.3                 | 1.1                            | 0.0                                              | -1.9        | 0.3         | 0.0                                                       | -0.9        | 0.0         |
| Estonia          | 0.2                 | 0.5*                           | -0.2                                             | -0.8        | 0.3         | -0.1                                                      | -0.6        | 0.3         |
| Cyprus           | 0.2                 | 0.7                            | -0.1                                             | -0.7        | 0.6         | 0.0                                                       | -0.3        | 0.2         |
| Malta            | 0.1                 | 2.6*                           | 0.0                                              | -1.4        | 0.0         | 0.0                                                       | -0.3        | 0.0         |
| <b>Euro area</b> | <b>100.0</b>        | <b>1.6</b>                     | <b>-0.1</b>                                      | <b>-1.5</b> | <b>-0.6</b> | <b>-0.1</b>                                               | <b>-0.6</b> | <b>-0.5</b> |

Notes. Effects are observed minus counterfactual annual inflation, in percentage points; the gap is Q1 minus Q5. Costs are the best available euro estimates for every simulated measure, divided by 2022 GDP and counting shared envelopes once. An asterisk marks costs including business beneficiaries; the household share is not identified. A recorded zero denotes a regulatory measure for which no direct government transfer or tax expenditure is identified. Sources: Eurostat, Bruegel, national sources, authors' calculations.

The distributional effects are heterogeneous across countries and years. In 2022, the largest reductions in the Q1–Q5 gap occur in the Netherlands (1.2 points), France (1.1), Belgium (1.0), and Croatia and Latvia (0.9 each). In 2023, Slovakia records the largest reduction (1.7 points), followed

by Germany (1.2) and Portugal (1.1). France and Spain record reductions of 0.5 and 0.3 point. Policies increase the gap by 0.4 point in Croatia, 0.3 in Estonia, and 0.2 in Cyprus in 2023. Italy's gap effects are  $-0.1$ ,  $-0.2$ , and  $-0.1$  points in 2021–2023, while its effect on average inflation turns positive, at 0.7 point, in 2023.

For Germany, the reduction in annual-average inflation rises from 0.7 point in 2022 to 1.5 points in 2023, while the reduction in the Q1–Q5 gap rises from 0.3 to 1.2 points. This direct effect aligns with [Krebs and Weber \(2024\)](#), who emphasize lower measured inflation and protection of household purchasing power. [Dertwinkel-Kalt and Wey \(2025\)](#) examine suppliers' pricing and energy-saving incentives. Our accounting counterfactual does not estimate these endogenous responses. For the Netherlands, quintile effects capture different basket exposure to common component prices, not household eligibility for the cap or the distribution of cap transfers.

Estimates close to zero arise either where no policy layer from Appendix D enters the country-year cell or where temporary counterfactual price wedges have little effect on the annual comparison.

France's documented active-measure costs amount to 2.9 percent of 2022 GDP; its Q1–Q5 gap reductions are 1.1 points in 2022 and 0.5 point in 2023. Slovakia's recorded gross expenditure attributed to 2023 amounts to 1.7 percent of 2022 GDP. Differences in budget periods and gross/net accounting prevent these figures from supporting a fiscal-efficiency ranking. More generally, inflation-gap effects are percentage-point changes, not monetary household gains. A cost-effectiveness comparison would require eligibility, quantities consumed, tax and transfer rules, and counterfactual expenditures on a common monetary and beneficiary basis.

### 3.3.2 Differences Across Products

For the euro area, household price policies lower annual-average inflation by 0.1 point in 2021, 1.5 points in 2022, and 0.6 point in 2023. Their effects on the annual Q1–Q5 gap are  $-0.1$ ,  $-0.6$ , and  $-0.5$  points. The three-year average reductions are about 0.7 point for inflation and 0.4 point for the gap.

Table 5 reproduces the product-group decomposition of Table 2 using the counterfactual paths. It reports each product group's contribution to the annual policy effects on average inflation and the Quintile 1–Quintile 5 gap.

Table 5: Product-group contributions of household price policies to annual-average euro-area inflation and Q1–Q5 inflation inequality, 2021–2023

| Product group                              | Weight<br>(2022, %) | Effect of price policies on<br>average inflation |             |             | Effect of price policies on<br>Q1–Q5 inflation inequality |             |             |
|--------------------------------------------|---------------------|--------------------------------------------------|-------------|-------------|-----------------------------------------------------------|-------------|-------------|
|                                            |                     | 2021                                             | 2022        | 2023        | 2021                                                      | 2022        | 2023        |
| Food and non-alcoholic beverages           | 17.8                | 0.0                                              | 0.0         | -0.1        | 0.0                                                       | 0.0         | 0.0         |
| Alcoholic beverages, tobacco and narcotics | 4.6                 | 0.0                                              | 0.0         | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| Clothing and footwear                      | 5.6                 | 0.0                                              | 0.0         | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| Housing maintenance and furnishings        | 8.0                 | 0.0                                              | 0.0         | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| Water, electricity, gas and other fuels    | 6.8                 | -0.1                                             | -1.2        | -0.8        | -0.1                                                      | -0.6        | -0.5        |
| Operation of personal transport equipment  | 6.3                 | 0.0                                              | -0.4        | 0.3         | 0.0                                                       | 0.1         | 0.0         |
| Transport (excluding vehicle operation)    | 9.6                 | 0.0                                              | 0.0         | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| Other                                      | 41.1                | 0.0                                              | 0.0         | 0.0         | 0.0                                                       | 0.0         | 0.0         |
| <b>Total</b>                               | <b>100.0</b>        | <b>-0.1</b>                                      | <b>-1.5</b> | <b>-0.6</b> | <b>-0.1</b>                                               | <b>-0.6</b> | <b>-0.5</b> |

Notes. Policy effects are observed minus counterfactual product contributions, in percentage points; the gap is Q1 minus Q5. Annual columns compare annual-average indices. Weights are 2022 EA20 expenditure shares. Sources: Eurostat, national statistical institutes, authors' calculations.

Household energy is the main contributor to the reduction in the income gap: its contribution is  $-0.1$  point in 2021,  $-0.6$  point in 2022, and  $-0.5$  point in 2023. Vehicle-operation expenditure partly offsets this effect in 2022, contributing 0.1 point to the gap. Within this group, the policy wedge is applied only to fuels and lubricants. Its effect on average inflation reverses from  $-0.4$  point in 2022 to 0.3 point in 2023, partly offsetting the 0.8-point reduction from household energy. Food policies contribute a further 0.1-point reduction in average inflation in 2023. The total annual effects on the Q1–Q5 gap are  $-0.1$ ,  $-0.6$ , and  $-0.5$  points. Product contributions are rounded separately, so displayed rows need not sum exactly to the displayed total.

Household-energy and motor-fuel policies have offsetting distributional effects. Electricity, gas, and other household fuels receive larger budget weights among lower-income households, so policies that reduce their retail prices narrow the Q1–Q5 inflation gap. Relief on motor fuels can move the gap in the opposite direction because higher-income households devote more expenditure to private transport. Measures with similar effects on average inflation need not have similar effects on inflation inequality.

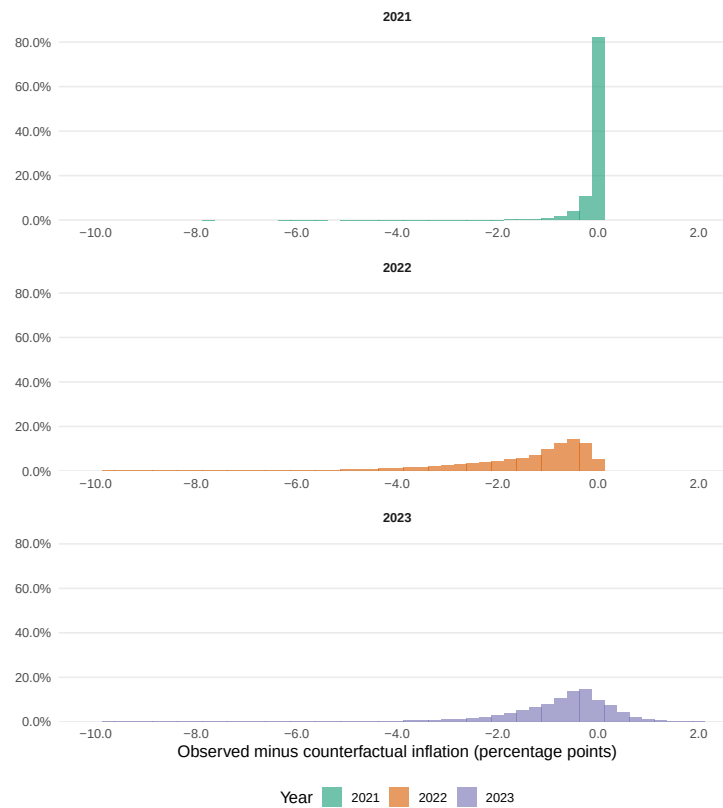
## 4 Household Inflation Exposure: Determinants and Burden

In this section, we examine the distribution of household policy effects and the association between inflation and observable household characteristics. We then assess direct and indirect exposure to extra-EU import-price shocks and measure the inflation burden relative to household income.

### 4.1 Household Characteristics and Inflation Exposure

Figure 8 reports the distribution of the household-level difference between observed and counterfactual annual inflation. Each difference pairs the same household in the same year. Negative values indicate lower inflation under the observed policies. Both rates receive the same observed-scenario expenditure weight when constructing this distribution. The weighted mean effects are  $-0.1$ ,  $-1.5$ , and  $-0.8$  point in 2021–2023; the corresponding medians are  $0.0$ ,  $-1.0$ , and  $-0.6$  point. These household-sample estimates differ from the EA20 group aggregates in coverage and aggregation weights.

Figure 8: Distribution of household inflation differences: observed minus counterfactual, 2021–2023



Notes: Each observation is a household’s observed annual inflation rate minus its no-policy counterfactual, in percentage points. The sample contains the same 152,613 households in each year, covering 18 countries excluding Italy and Portugal. Both scenarios use fixed 2020 HBS microdata baskets, RAS calibration at COICOP digit 3, identical annual HICP product margins, and the same exclusion of actual and imputed rents. Counterfactual product prices remove the household price measures described in Section 3. Bar heights are shares of revalued expenditure, using the observed scenario’s survey weight times retained expenditure times previous-year annual-average household index. Bins have width one quarter of a percentage point. The pooled outer 0.1 percent tails are omitted from display without renormalizing bin shares. Differences smaller than  $10^{-10}$  point in absolute value are treated as numerical zero. Positive inflation differences can reflect base effects and do not imply higher price levels under the policies.

Sources: Eurostat HICP and HBS microdata; authors’ calculations.

We pool individual household records from the euro area to examine whether differences in consumption baskets translate into systematic differences in inflation exposure. We examine how income quintile, age, household size, education, and degree of urbanization are associated with household-specific inflation.

The regression tables use the household indices underlying the distributions, with the income and complete-case restrictions described here. Table 6 reports unweighted ordinary least squares

estimates, with the pooled model first and separate estimates for 2021, 2022, and 2023 thereafter. All columns include country fixed effects; the pooled model also includes year fixed effects. The dependent variable is annual inflation from the RAS-calibrated household indices, using expenditure margins and excluding rents. Survey weights remain part of index calibration and national income-quintile construction, but do not weight the regressions. Standard errors are clustered by country. The coefficients are descriptive conditional associations, not causal effects. The complete-case sample contains 138,798 households in 17 countries, observed in each of the three years, for 416,394 household-year observations in the pooled model. Education refers to the household reference person.

Table 6: Household-level inflation regressions: pooled and by year, euro-area sample

| Dependent Variable:<br>Model<br>Model: | Pooled                 | Annual inflation       |                        |                        |
|----------------------------------------|------------------------|------------------------|------------------------|------------------------|
|                                        | (1)                    | 2021<br>(2)            | 2022<br>(3)            | 2023<br>(4)            |
| <i>Variables</i>                       |                        |                        |                        |                        |
| Less than 30 years                     | -0.5943***<br>(0.1048) | -0.0080<br>(0.0911)    | -1.259***<br>(0.2064)  | -0.5159***<br>(0.1387) |
| From 30 to 44 years                    | -0.3599***<br>(0.0790) | 0.1116<br>(0.0834)     | -0.9288***<br>(0.1370) | -0.2624**<br>(0.1110)  |
| From 45 to 59 years                    | -0.2024**<br>(0.0694)  | 0.0882<br>(0.0739)     | -0.4904***<br>(0.1381) | -0.2050**<br>(0.0761)  |
| Education: upper secondary             | -0.0544**<br>(0.0228)  | -0.0821<br>(0.0915)    | -0.0533<br>(0.0509)    | -0.0279<br>(0.1012)    |
| Education: tertiary                    | -0.2287***<br>(0.0291) | -0.2179**<br>(0.0955)  | -0.4128***<br>(0.0614) | -0.0556<br>(0.1113)    |
| Income quintile 2                      | -0.1798***<br>(0.0593) | 0.0855<br>(0.0706)     | -0.4491***<br>(0.1054) | -0.1758<br>(0.1443)    |
| Income quintile 3                      | -0.3066***<br>(0.0865) | 0.0749<br>(0.1022)     | -0.7314***<br>(0.1527) | -0.2632<br>(0.1970)    |
| Income quintile 4                      | -0.4029***<br>(0.1094) | 0.0643<br>(0.1179)     | -0.9866***<br>(0.1940) | -0.2864<br>(0.2442)    |
| Income quintile 5                      | -0.5410***<br>(0.1446) | 0.0427<br>(0.1480)     | -1.314***<br>(0.2731)  | -0.3513<br>(0.2926)    |
| Household size                         | 0.0357<br>(0.0211)     | -0.0081<br>(0.0209)    | -0.0023<br>(0.0324)    | 0.1175**<br>(0.0470)   |
| Intermediate area                      | -0.2081***<br>(0.0321) | -0.2182***<br>(0.0715) | -0.5537***<br>(0.0929) | 0.1476*<br>(0.0790)    |
| Densely populated area                 | -0.4384***<br>(0.0445) | -0.4781***<br>(0.0840) | -1.138***<br>(0.0967)  | 0.3011***<br>(0.1023)  |
| <i>Fixed effects</i>                   |                        |                        |                        |                        |
| country                                | Yes                    | Yes                    | Yes                    | Yes                    |
| year                                   | Yes                    |                        |                        |                        |
| <i>Fit statistics</i>                  |                        |                        |                        |                        |
| Observations                           | 416,394                | 138,798                | 138,798                | 138,798                |
| R <sup>2</sup>                         | 0.61447                | 0.36409                | 0.59431                | 0.60261                |
| Within R <sup>2</sup>                  | 0.01443                | 0.02872                | 0.06969                | 0.01182                |

*Clustered (country) standard errors in parentheses*

*Significance levels: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

*Notes:* Unweighted OLS; fixed 2020 HBS baskets and RAS-calibrated annual inflation excluding rents. Reference groups: quintile 1, age 60 or over, rural areas, ISCED 0–2. Upper secondary and tertiary denote ISCED 3–4 and 5–8. Coefficients are in percentage points; country-clustered standard errors are in parentheses. Fixed effects and sample sizes are reported in the table. The Netherlands is excluded because education is entirely unreported in the available file. Italy is excluded because usable household income is unavailable; Portugal lacks sufficient expenditure detail to isolate rents. Without regression weights, countries contribute in proportion to their sample sizes rather than their represented populations. *Sources:* HBS microdata, Eurostat HICP, authors' calculations.

Table 7 retains the sample and specification of Table 6, replacing observed inflation with the same

household's observed minus counterfactual annual inflation rate. A positive coefficient indicates a more positive, or less negative, inflation difference relative to the reference category, conditional on the other covariates and fixed effects. These coefficients describe conditional differences in the modelled price-policy effect.

Table 7: Household inflation differences: observed minus counterfactual, pooled and by year

| Dependent Variable:        | Observed minus counterfactual inflation |                      |                       |                       |
|----------------------------|-----------------------------------------|----------------------|-----------------------|-----------------------|
| Model                      | Pooled                                  | 2021                 | 2022                  | 2023                  |
| Model:                     | (1)                                     | (2)                  | (3)                   | (4)                   |
| <i>Variables</i>           |                                         |                      |                       |                       |
| Less than 30 years         | 0.2412***<br>(0.0530)                   | 0.0364<br>(0.0216)   | 0.3446**<br>(0.1342)  | 0.3426***<br>(0.0899) |
| From 30 to 44 years        | 0.1662***<br>(0.0289)                   | 0.0296<br>(0.0202)   | 0.2163**<br>(0.0789)  | 0.2526***<br>(0.0552) |
| From 45 to 59 years        | 0.1114***<br>(0.0246)                   | 0.0316<br>(0.0202)   | 0.1402**<br>(0.0529)  | 0.1625***<br>(0.0386) |
| Education: upper secondary | 0.1178***<br>(0.0392)                   | 0.0578**<br>(0.0226) | 0.2537***<br>(0.0775) | 0.0419<br>(0.0707)    |
| Education: tertiary        | 0.1747***<br>(0.0442)                   | 0.0650**<br>(0.0244) | 0.3459***<br>(0.0918) | 0.1131*<br>(0.0607)   |
| Income quintile 2          | 0.1591***<br>(0.0233)                   | 0.0304<br>(0.0211)   | 0.1775***<br>(0.0542) | 0.2694***<br>(0.0717) |
| Income quintile 3          | 0.2404***<br>(0.0311)                   | 0.0433<br>(0.0280)   | 0.2858***<br>(0.0730) | 0.3921***<br>(0.1138) |
| Income quintile 4          | 0.2936***<br>(0.0357)                   | 0.0477<br>(0.0314)   | 0.3612***<br>(0.0872) | 0.4718***<br>(0.1513) |
| Income quintile 5          | 0.3592***<br>(0.0485)                   | 0.0495<br>(0.0348)   | 0.4346***<br>(0.0934) | 0.5935**<br>(0.2193)  |
| Household size             | 0.0301**<br>(0.0137)                    | 0.0102<br>(0.0063)   | 0.0627**<br>(0.0273)  | 0.0172<br>(0.0124)    |
| Intermediate area          | 0.0940***<br>(0.0295)                   | 0.0413*<br>(0.0217)  | 0.2101***<br>(0.0672) | 0.0305<br>(0.0423)    |
| Densely populated area     | 0.1492***<br>(0.0329)                   | 0.0534*<br>(0.0281)  | 0.3585***<br>(0.0954) | 0.0356<br>(0.0551)    |
| <i>Fixed effects</i>       |                                         |                      |                       |                       |
| country                    | Yes                                     | Yes                  | Yes                   | Yes                   |
| year                       | Yes                                     |                      |                       |                       |
| <i>Fit statistics</i>      |                                         |                      |                       |                       |
| Observations               | 416,394                                 | 138,798              | 138,798               | 138,798               |
| R <sup>2</sup>             | 0.24004                                 | 0.54174              | 0.40818               | 0.52802               |
| Within R <sup>2</sup>      | 0.02917                                 | 0.04159              | 0.06237               | 0.07138               |

*Clustered (country) standard errors in parentheses*

*Significance levels: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

*Notes:* Same observations and specification as Table 6. Outcome: the household's observed minus counterfactual annual inflation, in percentage points. Both scenarios use identical baskets and product coverage; the regressions are unweighted. Standard errors are re-estimated for the paired difference and clustered by country.

*Sources:* HBS microdata, Eurostat HICP, authors' calculations.

Table 6 shows that the inflation surge did not affect all household baskets equally. Over 2021–2023, the conditional estimates associate lower income, older reference persons, lower education, and rural residence with higher inflation. In the pooled model, annual inflation is 0.5 percentage

point lower in the fifth income quintile than in the first, 0.6 point lower for households with a reference person below 30 than for those aged 60 or over, and 0.4 point lower in densely populated areas than in rural areas. These differences are all significant at 1 percent. The annual estimates, however, show that this average pattern is concentrated around the 2022 shock rather than equally apparent throughout the period.

In 2021, the clearest differences concern residence and education: inflation is 0.5 point lower in densely populated areas than in rural areas and 0.2 point lower among tertiary-educated reference persons than among those with lower secondary education or less. The income and age coefficients are not significant at 10 percent. By 2022, a broader pattern emerges. The income coefficients decline monotonically across quintiles, reaching  $-1.3$  points for the fifth quintile, while the under-30 and densely populated-area coefficients reach  $-1.3$  and  $-1.1$  points. Tertiary education is associated with 0.4 point lower inflation even after conditioning on income and the other characteristics. All four 2022 estimates are significant at 1 percent. This alignment is consistent with the unequal basket exposure during the energy-price surge documented above, although these regressions do not separately identify the contribution of energy.

The easing of inflation in 2023 does not simply restore the earlier pattern. Income and education coefficients are no longer statistically significant at 10 percent, but the age gradient remains: inflation is 0.5 percentage point lower for households with reference persons below 30 than for those aged 60 or over, significant at 1 percent. Meanwhile, the urban–rural difference changes sign, with inflation 0.3 point higher in densely populated areas, also significant at 1 percent. Larger households now face higher inflation, by 0.1 point per additional member, significant at 5 percent. Thus, the composition of exposure changes as the inflation episode unfolds: some conditional gaps become less precisely estimated, others persist, and the residence gap reverses sign. These comparisons describe the annual estimates; differences in statistical significance alone do not establish statistically significant changes in coefficients across years.

The regression results are consistent with scanner-data evidence, while also clarifying what the present estimates capture. Using US household scanner data, [Kaplan and Schulhofer-Wohl \(2017\)](#) find that lower-income households experience higher inflation on average, but that observable characteristics explain only a small share of the substantial cross-sectional dispersion. For the euro area, [Strasser et al. \(2023\)](#) likewise document small and time-varying differences between income groups, including reversals across countries, and find that product choices and shopping behavior account for considerably more household-level variation than socioeconomic characteristics. Our time-varying income and household-size coefficients indicate that demographic gradients depend

on the composition of each year's price shock. Our basket-based indices do not capture the differences in prices paid within product categories, including variation across brands, retailers, and promotions, that scanner data reveal.

## 4.2 Direct and Indirect Exposure to Extra-EU Import-Price Shocks

We apply the observed extra-EU import-price changes to imported final goods and intermediate inputs. The FIGARO production network propagates the intermediate-input cost impulses across EA20 countries and industries. We then map producer-price responses into consumption products and aggregate them with the household expenditure weights used elsewhere in the paper. Throughout this subsection,  $k \in \{\text{energy, food}\}$  indexes the import-price shock,  $d$  denotes the destination country,  $i$  an imported product,  $j$  a destination industry, and  $g$  a household group. A superscript  $k$  labels the shock scenario; it is not an exponent.

Let  $h_k$  denote the observed percentage change in the annual-average extra-EU import unit-value index for shock  $k$ . For a final product  $i$  absorbed by households in country  $d$ , let  $F_{di}^{\text{extra}}$  denote imports from outside the EU27 and  $F_{di}^{\text{all}}$  total household absorption, including domestic production and all imports. The direct final-demand price impulse is

$$d_{di}^k = h_k \frac{F_{di}^{\text{extra}}}{F_{di}^{\text{all}}}. \quad (12)$$

The ratio therefore scales the common external price shock by the extra-EU share of the product's household supply. The vector  $\mathbf{d}^k$  stacks  $d_{di}^k$  over all country-product pairs.

For intermediate inputs,  $\mathbf{A}$  is the FIGARO matrix of input coefficients for the EA20 country-industry network: each element records the amount of an input required per unit of gross output in the purchasing industry. The vector  $\mathbf{b}^k$  contains the first-round production-cost impulses generated by extra-EU inputs affected by shock  $k$ . Its element for industry  $j$  in country  $d$  is

$$b_{dj}^k = h_k \frac{\sum_{o \notin \text{EU27}} \sum_{i \in \mathcal{S}_k} Z_{oi,dj}}{x_{dj}}, \quad (13)$$

where  $o$  indexes the origin country,  $\mathcal{S}_k$  is the set of products affected by shock  $k$ ,  $Z_{oi,dj}$  is the FIGARO intermediate flow of product  $i$  from origin  $o$  to industry  $j$  in destination country  $d$ , and  $x_{dj}$  is that industry's gross output. Thus,  $b_{dj}^k$  is the first-round increase in unit production costs under complete pass-through of the import-price shock. The vector  $\mathbf{b}^k$  stacks these impulses over all EA20 country-industry pairs.

The corresponding producer-price response is

$$\Delta \mathbf{p}^k = (\mathbf{I} - \mathbf{A}^\top)^{-1} \mathbf{b}^k, \quad (14)$$

where  $\mathbf{I}$  is the identity matrix and  $(\mathbf{I} - \mathbf{A}^\top)^{-1}$  is the Leontief price inverse. The vector  $\Delta \mathbf{p}^k$  includes both the initial imported-input cost impulse and all subsequent rounds of cost propagation through domestic and intra-EA20 input links.

Finally,  $\mathbf{C}$  is the normalized NACE–COICOP concordance matrix: multiplying an industry-level price vector by  $\mathbf{C}$  yields the corresponding vector of COICOP consumption-price changes. The vector  $\mathbf{w}_g$  contains the normalized COICOP expenditure shares of household group  $g$ , so its elements sum to one. The total inflation exposure of group  $g$  is

$$\Delta \pi_g^k = \mathbf{w}_g^\top \mathbf{C} \mathbf{d}^k + \mathbf{w}_g^\top \mathbf{C} \Delta \mathbf{p}^k. \quad (15)$$

The first term is the contribution of directly imported final goods; the second is the contribution of imported intermediate inputs after production-network propagation. Because  $h_k$  and all price responses are expressed as percentage changes,  $\Delta \pi_g^k$  is an inflation exposure in percentage points. The effect on inflation inequality is  $\Delta \pi_{Q1-Q5}^k = \Delta \pi_{Q1}^k - \Delta \pi_{Q5}^k$ , where a positive value means that shock  $k$  raises the inflation exposure of the first income quintile more than that of the fifth.

Panel (a) uses the observed increase in the annual-average extra-EU energy import unit-value index from 2021 to 2022, and panel (b) uses the corresponding food import-price increase. The raw shocks, their conversion into country-industry cost impulses, and the propagation equations are reported in Appendix B. The shocks are common external-price innovations, while bilateral import shares, production structures, and household baskets generate heterogeneous country and household exposure.

Table 8: Product-group contributions of observed extra-EU import-price shocks, euro area, 2022

| Product group                               | Average inflation |            |            | Q1–Q5 gap  |            |            |
|---------------------------------------------|-------------------|------------|------------|------------|------------|------------|
|                                             | Total             | Direct     | Indirect   | Total      | Direct     | Indirect   |
| <i>(a) Energy import-price shock: 83.9%</i> |                   |            |            |            |            |            |
| Food and non-alcoholic beverages            | 0.6               | 0.0        | 0.6        | 0.2        | 0.0        | 0.2        |
| Alcoholic beverages, tobacco and narcotics  | 0.1               | 0.0        | 0.1        | 0.0        | 0.0        | 0.0        |
| Clothing and footwear                       | 0.1               | 0.0        | 0.1        | 0.0        | 0.0        | 0.0        |
| Housing maintenance and furnishings         | 0.2               | 0.0        | 0.2        | -0.1       | 0.0        | -0.1       |
| Water, electricity, gas and other fuels     | 1.7               | 0.5        | 1.2        | 1.7        | 0.5        | 1.2        |
| Operation of personal transport equipment   | 1.8               | 0.3        | 1.5        | -0.5       | -0.1       | -0.5       |
| Transport (excluding vehicle operation)     | 1.2               | 0.2        | 1.0        | 0.0        | 0.0        | 0.0        |
| Other                                       | 0.8               | 0.2        | 0.5        | -0.1       | 0.0        | -0.1       |
| <b>Total</b>                                | <b>6.5</b>        | <b>1.3</b> | <b>5.2</b> | <b>1.1</b> | <b>0.4</b> | <b>0.7</b> |
| <i>(b) Food import-price shock: 21.0%</i>   |                   |            |            |            |            |            |
| Food and non-alcoholic beverages            | 0.5               | 0.3        | 0.2        | 0.2        | 0.1        | 0.1        |
| Alcoholic beverages, tobacco and narcotics  | 0.1               | 0.1        | 0.0        | 0.0        | 0.0        | 0.0        |
| Clothing and footwear                       | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| Housing maintenance and furnishings         | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| Water, electricity, gas and other fuels     | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| Operation of personal transport equipment   | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| Transport (excluding vehicle operation)     | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| Other                                       | 0.0               | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| <b>Total</b>                                | <b>0.7</b>        | <b>0.4</b> | <b>0.2</b> | <b>0.2</b> | <b>0.1</b> | <b>0.1</b> |

Notes. Shocks are changes in the 2022 annual-average extra-EU27 import unit-value indices relative to 2021. Energy is SITC 3; food is SITC 0–1. Direct effects use extra-EU imports purchased by households. Indirect effects use extra-EU intermediate inputs and the full EA20 FIGARO propagation. Direct and indirect effects sum to the total before rounding.

Sources: Eurostat international trade statistics, HBS, FIGARO, authors' calculations.

Table 8 shows that the observed energy import-price shock generates 6.5 percentage points of average EA20 non-rent inflation exposure: 1.3 points through imported final goods and 5.2 points through intermediate inputs and production-network propagation. Household energy and vehicle operation contribute 1.7 and 1.8 points, respectively. The net Q1–Q5 effect is 1.1 points, comprising 0.4 point directly and 0.7 point indirectly. Household energy contributes 1.7 points to the gap, partly offset by vehicle operation (–0.5 point), housing maintenance and other products. The food import-price shock generates 0.7 point of average exposure and a Q1–Q5 gap of 0.2 point. Components and totals are rounded separately.

These are accounting exposures under complete pass-through, not causal estimates or a model fit to observed HICP inflation. Retail margins, taxes, subsidies, regulated prices, substitution, and incomplete pass-through can all attenuate or redistribute the observed consumer-price response.

### 4.3 Inflation Burden

Our indicators measure differences in inflation exposure across household groups, but they do not directly express the cost of inflation relative to household resources. We therefore construct the burden from HBS microdata rather than combining group-level inflation with an external aggregate consumption-to-income ratio. For household  $i$  in country  $c$ , let  $X_{ic}$  denote consumption expenditure on the non-rent products matched to HICP,  $Y_{ic}$  net household income, and  $\bar{\pi}_{ic,y}$  the household's average annual inflation rate in year  $y$ , computed with the RAS-calibrated weights described in Section 1.1. The raw household-level burden is

$$\mathcal{B}_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}, \quad (16)$$

where  $\bar{\pi}_{ic,y}$  is measured in percent, so  $\mathcal{B}_{ic,y}^{\text{raw}}$  is measured in points of net income. Thus, the indicator allows both individual inflation exposure and the consumption-to-income ratio to vary across households. The expenditure numerator and inflation index cover the same matched non-rent products. This fixed-resource exposure indicator is not compensating variation: it holds 2020 expenditure and income fixed and allows neither substitution nor income adjustment.

Very small reported incomes can generate mechanically extreme expenditure-to-income ratios. We therefore winsorize  $100X_{ic}/Y_{ic}$  at the survey-weighted 99th percentile within each country, while retaining the underlying household inflation rate. Denoting the resulting ratio by  $\tilde{\kappa}_{ic}$ , the burden used in the estimates and its aggregation for national income quintile  $q$  are

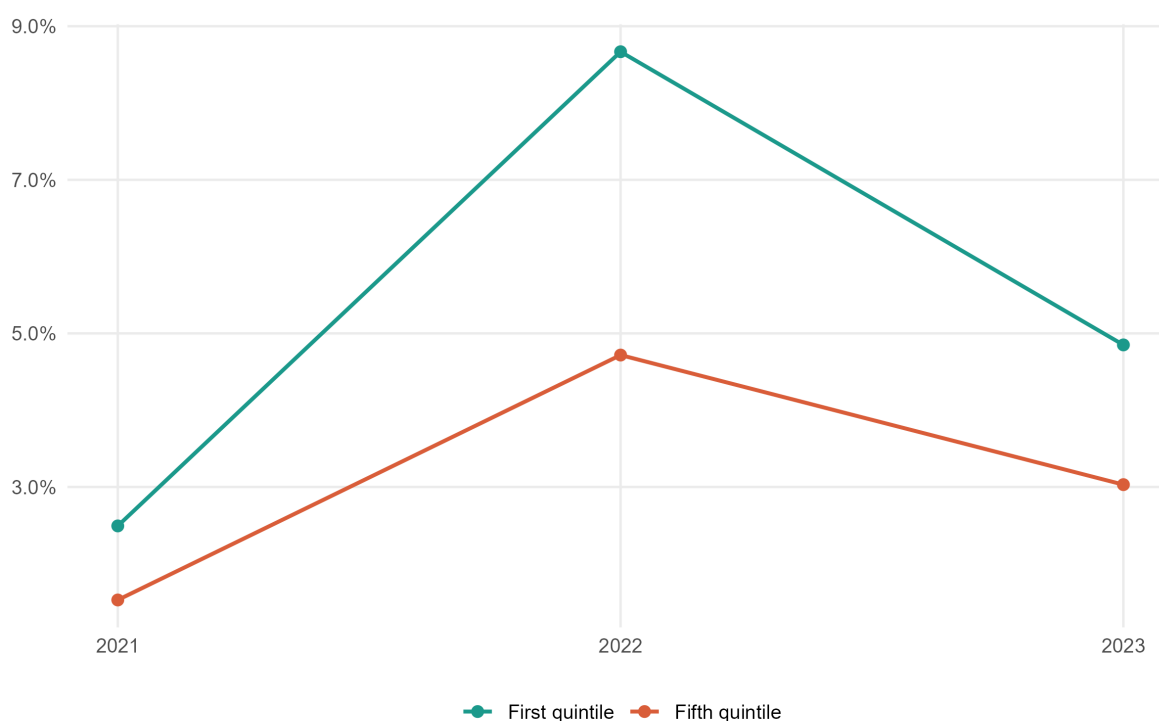
$$\mathcal{B}_{ic,y} = \bar{\pi}_{ic,y} \frac{\tilde{\kappa}_{ic}}{100}, \quad \mathcal{B}_{q,y} = \frac{\sum_c \sum_{i \in q(c)} a_{ic} \mathcal{B}_{ic,y}}{\sum_c \sum_{i \in q(c)} a_{ic}}, \quad (17)$$

where  $a_{ic}$  is the HBS survey weight and  $q(c)$  indicates that total-net-income quintiles are defined separately within each country before households are pooled. The survey weights expand the microdata to national household populations and therefore determine the euro-area aggregation. The 2020 HBS basket, income, and expenditure observations are held fixed, while household-specific inflation is calculated for 2021–2023. For each household-year, the inflation rate is computed over the matched non-rent expenditure categories and the HBS–HICP product weights are renormalized to one. We impose no minimum matched-expenditure threshold.

Higher-income households also generally have greater capacity to absorb a price shock by reducing discretionary expenditure or substituting toward cheaper varieties. Appendix Figure C.2 documents the corresponding difference in basket composition.

Figure 9 uses household indices and matched non-rent expenditure. The first quintile faces a burden of 2.5 percent of net income in 2021, 8.7 percent in 2022, and 4.8 percent in 2023, compared with 1.5, 4.7, and 3.0 percent for the fifth quintile. The Q1–Q5 gap is 1.0, 4.0, and 1.8 points, respectively, calculated before rounding. The income and expenditure filters retain positive-income households; national quintiles are assigned before these burden-specific exclusions.

Figure 9: Household-level inflation burden in the euro area, by income quintile (in % of net income)



Notes: Annual household inflation times matched non-rent expenditure divided by net income; ratios are winsorized at the national survey-weighted 99th percentile. Survey-weighted means within national total-income quintiles, 18 countries excluding Italy and Portugal. Fixed 2020 resources; this is an exposure measure, not a welfare estimate.

Sources: HBS microdata, Eurostat HICP, authors' calculations.

## 5 Conclusion

Household expenditure baskets reveal substantial differences in exposure to the recent inflation surge. Modest income-quintile gaps at the euro-area level conceal larger differences within several countries. Older and rural households also experience higher inflation during the rise in energy prices, although these associations vary over time.

Differences in household resources amplify the burden of inflation. Our fixed-resource exposure measure is higher relative to net income for households in the lowest income quintile than for those in the highest. Similar inflation rates can therefore impose different burdens relative to income.

The price-policy counterfactual indicates that household price measures lower average inflation during the energy crisis and narrow the gap between the lowest and highest income quintiles. Relief on household energy prices accounts for most of the narrowing, while motor-fuel measures partly offset it. These accounting effects depend on the reconstructed and calibrated price wedges and do not measure the policies' full welfare or macroeconomic effects.

This work can be extended in several directions. Transaction data could identify differences in prices paid and substitution within product categories, while linked income and balance-sheet data would permit a broader assessment of welfare effects, including the redistribution between debtors and creditors.

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# Appendix

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## A Aggregation Bias Across COICOP Digits

Table [A.1](#) compares inflation gaps across COICOP digit depths using HBS microdata and reports a digit-3 benchmark from Eurostat national-quintile HBS aggregates. Macro D3 excl. and all microdata columns exclude actual and imputed rents (HBS 041 and 042) and the corresponding HICP actual-rent component (041), and use RAS-calibrated weights. All columns use annual-average chained-index inflation. The macro benchmark is recalculated with RAS; the rent-robustness table retains its relative-expenditure specification.

**Micro–macro consistency.** We first reaggregate the 2020 household records into national income quintiles and compare survey-weighted expenditure shares with Eurostat’s published HBS shares on a common digit-3 support, excluding rents. Ranking households by total net household income substantially improves this match relative to ranking them by equivalised income. Across products and quintiles, the mean absolute difference in budget shares falls from 0.20 to 0.05 percentage points in the Netherlands and from 0.31 to 0.04 in Belgium; the corresponding figures for France are 0.19 and 0.03. This comparison precedes HICP calibration and therefore identifies a difference in the construction of income groups rather than an effect of the inflation-index weighting method. We use total household income for the microdata quintiles throughout this appendix.

This appendix retains its original comparison design: expenditure is rebuilt from the finest available codes and quintile calibration margins come from published tabulations. The main microdata pipeline instead prioritizes directly observed expenditure at the required depth and computes group expenditure margins from the same household sample. The appendix therefore provides a separate digit-depth sensitivity exercise, rather than a replication of the updated baseline.

**Baskets and weights.** Microdata quintiles are defined separately within each country using survey-weighted net household income without division by an equivalence scale (total household income). The sample covers 18 euro-area countries. Fifteen countries provide expenditure variables through digit 5; Germany, France and Spain provide digit 3 in the local microdata files. Italy is excluded because household income is unavailable, and Portugal because its detailed expenditure columns are unpopulated. The 2020 microdata expenditure baskets are reconstructed at each digit depth from the same finest available expenditure variables after removing rents. Nonpositive or missing expenditure entries contribute zero, while eligible households remain in the survey denominators.

For product  $j$  in quintile  $q$ , let  $\bar{x}_{qj}$  be survey-weighted mean HBS expenditure,  $\bar{x}_j$  the national mean,  $s_{q,y}$  the package's consumption share of quintile  $q$ , and  $w_{j,y}^{HICP}$  the HICP product share on matched support. RAS iteratively rescales rows and columns of the positive seed

$$\max\{s_{q,y}w_{j,y}^{HICP}\bar{x}_{qj}/\bar{x}_j, 10^{-12}\}.$$

The calibrated expenditure masses  $M_{qj,y}$  satisfy

$$\sum_j M_{qj,y} = s_{q,y}, \quad \sum_q M_{qj,y} = w_{j,y}^{HICP}, \quad \omega_{qj,y} = M_{qj,y}/s_{q,y}. \quad (18)$$

Both the macro and micro calculations use the package's published consumption shares by income quintile, selecting the latest available wave preceding each weight year, and annual HICP product margins. These quintile margins are not re-estimated from the matched microdata at each digit depth. A shared ECOICOP1 correspondence table defines the D2–D5 mappings for both the household-index function and this appendix. Expenditures and historical HICP prices use the same canonical product codes; rents are removed before D2 aggregation. Micro expenditures are aggregated by quintile before RAS calibration; the table does not average separately chained household indices. Macro baskets use the corresponding published Eurostat HBS expenditures, selecting the most recent available preceding wave for each matched item. Micro prices use the archived ECOICOP1 HICP indices and item weights, which match the HBS classification, and are reaggregated with annual HICP weights. An undivided terminal HICP class is matched to its identical HBS code padded with zeros (for example, electricity 0451/04510 and hospital services 063/06300). Five additional terminal correspondences for hospital services and education are validated against the HBS manual and Eurostat labels and recorded in an explicit mapping table. Each corresponding published price and weight enters once; no distinct subclass receives an imputed parent price. Fuzzy label matching is used to propose candidates, and unvalidated candidates are excluded. Rent components are removed before constructing the broad housing

index. Products without complete monthly prices or annual weights are excluded; prices are not imputed to distinct subclasses. Matched expenditure coverage can vary across digit depths, so the reported differences can combine aggregation and coverage effects. The macro benchmark retains the price source and classification bridge used in the rent-robustness appendix.

**Inflation and aggregation bias.** Let  $I_{q,y,m}^d$  denote the chained Laspeyres index for quintile  $q$  at digit depth  $d$ . Inflation in 2022 is calculated from annual-average indices,

$$\pi_{q,2022}^d = 100 \left( \frac{\frac{1}{12} \sum_{m=1}^{12} I_{q,2022,m}^d}{\frac{1}{12} \sum_{m=1}^{12} I_{q,2021,m}^d} - 1 \right). \quad (19)$$

The gap is  $G^d = \pi_{Q1,2022}^d - \pi_{Q5,2022}^d$  and the microdata aggregation bias is  $B^{d,d'} = G^d - G^{d'}$  for  $d' > d$ . Thus, negative bias indicates a larger gap at finer digit depth.

The micro–macro comparison also reflects differences in source, matched products and income-group construction; it should not be interpreted as aggregation bias. The earlier validation of published HBS expenditure shares uses native digit-3 variables, whereas the inflation calculations aggregate the finest available expenditure variables within each country. In Belgium and the Netherlands, these constructions can differ, notably for insurance expenditure.

Table A.1: Aggregation bias excluding rents: 2022 annual-average inflation

| Country | Inflation gaps (pp) |      |      |      |      | Aggregation bias (pp) |       |       | Coverage (%) |       |       |      |      |
|---------|---------------------|------|------|------|------|-----------------------|-------|-------|--------------|-------|-------|------|------|
|         | Macro D3            | D2   | D3   | D4   | D5   | D2–D3                 | D2–D4 | D2–D5 | Macro D3     | D2    | D3    | D4   | D5   |
| DE      | 1.4                 | 2.0  | 1.5  | –    | –    | 0.6                   | –     | –     | 99.6         | 100.0 | 100.0 | –    | –    |
| FR      | 0.2                 | 0.3  | 0.3  | –    | –    | 0.0                   | –     | –     | 99.6         | 100.0 | 99.9  | –    | –    |
| ES      | 1.0                 | 0.8  | 1.0  | –    | –    | -0.2                  | –     | –     | 99.6         | 100.0 | 99.9  | –    | –    |
| AT      | 0.1                 | 0.5  | 0.0  | 0.0  | 0.0  | 0.5                   | 0.6   | 0.6   | 99.5         | 100.0 | 100.0 | 99.6 | 97.1 |
| BE      | 2.3                 | 2.5  | 2.3  | 2.4  | 2.3  | 0.1                   | 0.0   | 0.1   | 99.3         | 100.0 | 100.0 | 99.3 | 96.9 |
| CY      | 0.8                 | 0.7  | 0.8  | 0.9  | 1.1  | -0.1                  | -0.2  | -0.3  | 99.8         | 100.0 | 100.0 | 99.8 | 98.7 |
| HR      | 0.4                 | 0.1  | 0.3  | 0.2  | 0.3  | -0.2                  | 0.0   | -0.1  | 99.7         | 100.0 | 100.0 | 99.9 | 99.6 |
| EE      | 4.4                 | 3.0  | 4.9  | 4.1  | 3.8  | -1.9                  | -1.2  | -0.8  | 99.3         | 100.0 | 99.9  | 99.5 | 97.2 |
| FI      | -0.9                | -0.6 | -0.8 | -0.6 | -0.8 | 0.2                   | 0.0   | 0.2   | 99.3         | 100.0 | 99.7  | 98.2 | 89.2 |
| EL      | 1.4                 | 1.4  | 0.9  | 0.8  | 0.8  | 0.5                   | 0.7   | 0.6   | 99.2         | 100.0 | 99.5  | 99.3 | 99.0 |
| IE      | 3.0                 | 2.0  | 2.9  | 2.3  | 2.1  | -0.8                  | -0.3  | -0.1  | 99.7         | 100.0 | 100.0 | 99.9 | 97.7 |
| LV      | 4.8                 | 2.9  | 3.7  | 3.3  | 2.0  | -0.8                  | -0.3  | 1.0   | 99.4         | 100.0 | 97.9  | 87.3 | 82.6 |
| LT      | 3.4                 | 2.9  | 3.7  | 4.9  | 4.9  | -0.7                  | -2.0  | -2.0  | 99.2         | 100.0 | 100.0 | 99.8 | 98.6 |
| LU      | 0.2                 | -0.4 | 0.2  | 1.4  | 1.0  | -0.6                  | -1.8  | -1.4  | 99.7         | 100.0 | 100.0 | 99.9 | 96.3 |
| MT      | 0.3                 | 0.6  | 0.4  | 0.3  | -0.5 | 0.3                   | 0.4   | 1.1   | 99.6         | 100.0 | 100.0 | 98.8 | 95.0 |
| NL      | 2.8                 | 2.0  | 3.0  | 2.5  | 1.9  | -1.1                  | -0.6  | 0.1   | 99.3         | 100.0 | 99.9  | 99.7 | 87.8 |
| SK      | 1.2                 | 0.5  | 0.1  | 0.1  | 0.1  | 0.4                   | 0.4   | 0.4   | 100.0        | 100.0 | 100.0 | 99.9 | 92.4 |
| SI      | 0.5                 | 0.5  | 0.6  | 0.6  | 0.7  | -0.1                  | -0.1  | -0.2  | 99.7         | 100.0 | 99.8  | 99.5 | 99.0 |

*Notes:* Inflation gaps and differences are in percentage points; coverage is in percent. Gaps are Q1 minus Q5; D2–D3, D2–D4 and D2–D5 compare micro estimates. Macro D3 uses national-quintile Eurostat HBS aggregates; micro uses survey-weighted total-income quintiles. All columns exclude rents and use RAS on quintile baskets. Construction and coverage are described above. A dash denotes unavailable detail; differences can reflect changes in matched products as well as aggregation.

## B Country Exposure to Extra-EU Import-Price Shocks

For shock family  $k$ , let  $h_k$  denote the observed change in the annual-average extra-EU27 import unit-value index. The energy shock is 83.9 percent and the food shock is 21.0 percent between 2021 and 2022. These are common external-price innovations: country heterogeneity enters through import exposure rather than through different values of  $h_k$ .

For destination country  $d$  and industry  $j$ , the raw import-price shock becomes the imported-input cost impulse

$$b_{dj}^k = h_k \frac{\sum_{o \notin EU27} \sum_{i \in \mathcal{S}_k} Z_{oi,dj}}{x_{dj}}, \quad (20)$$

where  $\mathcal{S}_k$  is the set of shocked products,  $Z_{oi,dj}$  is the intermediate-input flow from origin  $o$  and product  $i$  to industry  $j$  in country  $d$ , and  $x_{dj}$  is that industry's gross output. Thus, an identical raw price shock produces different cost impulses across countries and industries because their extra-EU imported-input shares differ.

The output-weighted country impulse reported in Table B.1 is

$$\bar{b}_d^k = h_k \frac{\sum_j \sum_{o \notin EU27} \sum_{i \in \mathcal{S}_k} Z_{oi,dj}}{\sum_j x_{dj}}. \quad (21)$$

The euro-area row applies this ratio to EA20 sums; it is not a simple average of the country rows.

Table B.1: Import exposure and imported-input cost impulses by euro-area country, 2022

| Country          | Energy products        |             | Food products          |             |
|------------------|------------------------|-------------|------------------------|-------------|
|                  | Raw price shock: 83.9% |             | Raw price shock: 21.0% |             |
|                  | $M/x$                  | $b$         | $M/x$                  | $b$         |
| Austria          | 0.79                   | 0.66        | 0.11                   | 0.02        |
| Belgium          | 2.36                   | 1.98        | 0.24                   | 0.05        |
| Croatia          | 2.62                   | 2.20        | 0.24                   | 0.05        |
| Cyprus           | 0.62                   | 0.52        | 0.14                   | 0.03        |
| Estonia          | 0.87                   | 0.73        | 0.29                   | 0.06        |
| Finland          | 2.27                   | 1.91        | 0.13                   | 0.03        |
| France           | 1.52                   | 1.27        | 0.12                   | 0.02        |
| Germany          | 1.20                   | 1.01        | 0.10                   | 0.02        |
| Greece           | 6.51                   | 5.46        | 0.22                   | 0.05        |
| Ireland          | 0.78                   | 0.65        | 0.30                   | 0.06        |
| Italy            | 2.00                   | 1.68        | 0.29                   | 0.06        |
| Latvia           | 0.73                   | 0.61        | 0.23                   | 0.05        |
| Lithuania        | 6.07                   | 5.09        | 0.23                   | 0.05        |
| Luxembourg       | 0.07                   | 0.06        | 0.05                   | 0.01        |
| Malta            | 0.81                   | 0.68        | 0.10                   | 0.02        |
| Netherlands      | 2.60                   | 2.18        | 0.62                   | 0.13        |
| Portugal         | 2.47                   | 2.07        | 0.42                   | 0.09        |
| Slovakia         | 2.96                   | 2.48        | 0.16                   | 0.03        |
| Slovenia         | 0.96                   | 0.81        | 0.16                   | 0.03        |
| Spain            | 2.70                   | 2.27        | 0.29                   | 0.06        |
| <b>Euro area</b> | <b>1.80</b>            | <b>1.51</b> | <b>0.21</b>            | <b>0.04</b> |

Notes. The raw shocks are the common changes in the 2022 annual-average extra-EU27 import unit-value indices relative to 2021.  $M/x$  is the extra-EU imported-input share of gross output, and  $b = h(M/x)$  is the corresponding cost impulse; both columns are expressed in percent. Energy covers B and C19; food covers A01 and C10–12. The euro-area entries are ratios of EA20 sums. Sources: Eurostat international trade statistics, FIGARO, authors' calculations.

Table B.2 repeats the simulation for each annual energy import-price change from 2021 to 2023 and reports the resulting effects by country. To isolate changes in the external shocks, the simulation holds the 2022 FIGARO network and all 2022 expenditure and country weights fixed across the three columns.

Table B.2: Simulated country effects of the observed extra-EU energy import-price shock, 2021–2023

| Country          | Effect on average inflation |            |             | Effect on inflation inequality |            |             |
|------------------|-----------------------------|------------|-------------|--------------------------------|------------|-------------|
|                  | 2021                        | 2022       | 2023        | 2021                           | 2022       | 2023        |
| Germany          | 4.0                         | 5.6        | -1.8        | 0.9                            | 1.2        | -0.4        |
| France           | 5.1                         | 7.1        | -2.3        | 1.3                            | 1.8        | -0.6        |
| Italy            | 5.8                         | 8.1        | -2.6        | 0.5                            | 0.7        | -0.2        |
| Spain            | 4.3                         | 5.9        | -1.9        | 0.7                            | 0.9        | -0.3        |
| Netherlands      | 5.6                         | 7.9        | -2.5        | 0.9                            | 1.2        | -0.4        |
| Belgium          | 3.8                         | 5.2        | -1.7        | 0.5                            | 0.7        | -0.2        |
| Austria          | 2.6                         | 3.5        | -1.1        | 0.2                            | 0.2        | -0.1        |
| Portugal         | 4.9                         | 6.8        | -2.2        | 0.7                            | 1.0        | -0.3        |
| Greece           | 6.1                         | 8.6        | -2.7        | 0.5                            | 0.7        | -0.2        |
| Finland          | 3.1                         | 4.4        | -1.4        | -0.7                           | -1.0       | 0.3         |
| Ireland          | 3.8                         | 5.3        | -1.7        | 2.5                            | 3.5        | -1.1        |
| Slovakia         | 4.6                         | 6.5        | -2.1        | 1.4                            | 2.0        | -0.6        |
| Croatia          | 6.0                         | 8.4        | -2.7        | 0.5                            | 0.7        | -0.2        |
| Lithuania        | 6.2                         | 8.6        | -2.8        | 0.2                            | 0.3        | -0.1        |
| Slovenia         | 4.0                         | 5.5        | -1.8        | 0.1                            | 0.1        | -0.0        |
| Luxembourg       | 1.6                         | 2.2        | -0.7        | 0.3                            | 0.5        | -0.2        |
| Latvia           | 2.7                         | 3.7        | -1.2        | 0.7                            | 1.0        | -0.3        |
| Estonia          | 3.3                         | 4.6        | -1.5        | 0.3                            | 0.4        | -0.1        |
| Cyprus           | 4.1                         | 5.7        | -1.8        | 1.6                            | 2.2        | -0.7        |
| Malta            | 2.4                         | 3.3        | -1.0        | -0.0                           | -0.0       | 0.0         |
| <b>Euro area</b> | <b>4.6</b>                  | <b>6.5</b> | <b>-2.1</b> | <b>0.8</b>                     | <b>1.1</b> | <b>-0.4</b> |

Notes. Entries are percentage-point effects of the energy import-price shock. Each column uses the change in the annual-average extra-EU27 energy import unit-value index relative to the previous year. The 2022 FIGARO network, product weights, household weights, and country weights are held fixed across years. Countries are ordered by their 2022 HICP country weight. Sources: Eurostat international trade statistics, HBS, FIGARO, authors' calculations.

## C Additional Figures

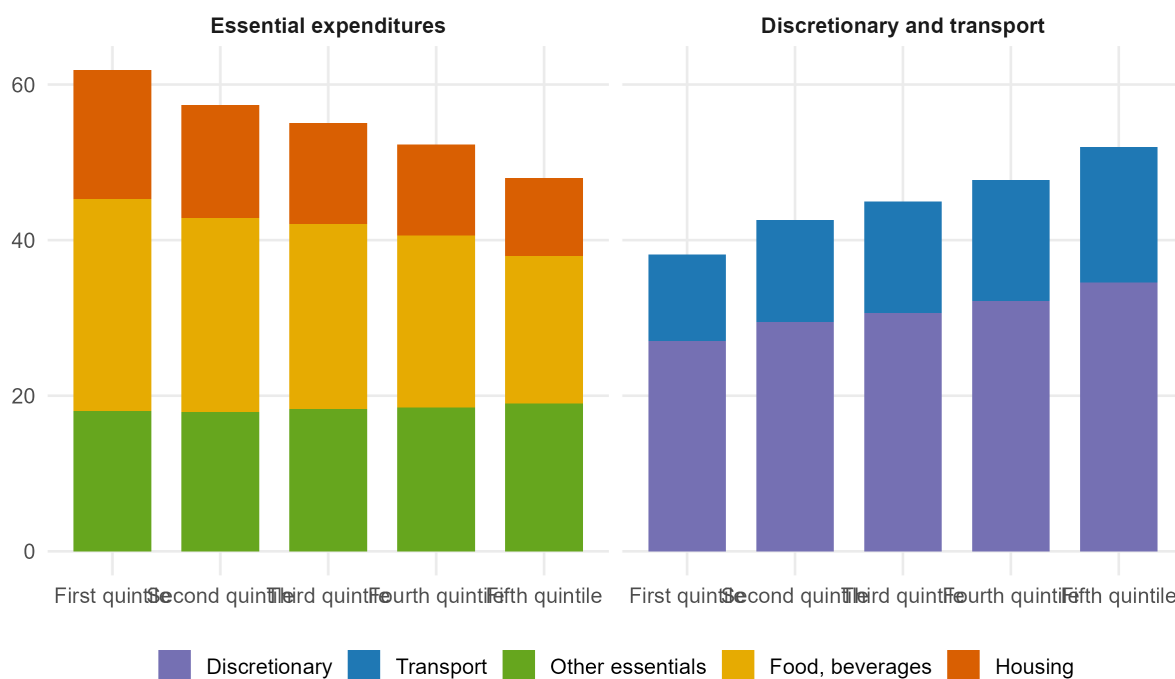
Figure C.1: Main COICOP expenditure weights by income quintile, Spain (in % of total expenditure)



Notes: The figure reports the share of each main COICOP category in total household expenditure for the first and fifth equivalised-income quintiles. Quintiles are computed separately by year using household weights.

Sources: INE Encuesta de Presupuestos Familiares public-use microdata; authors' calculations.

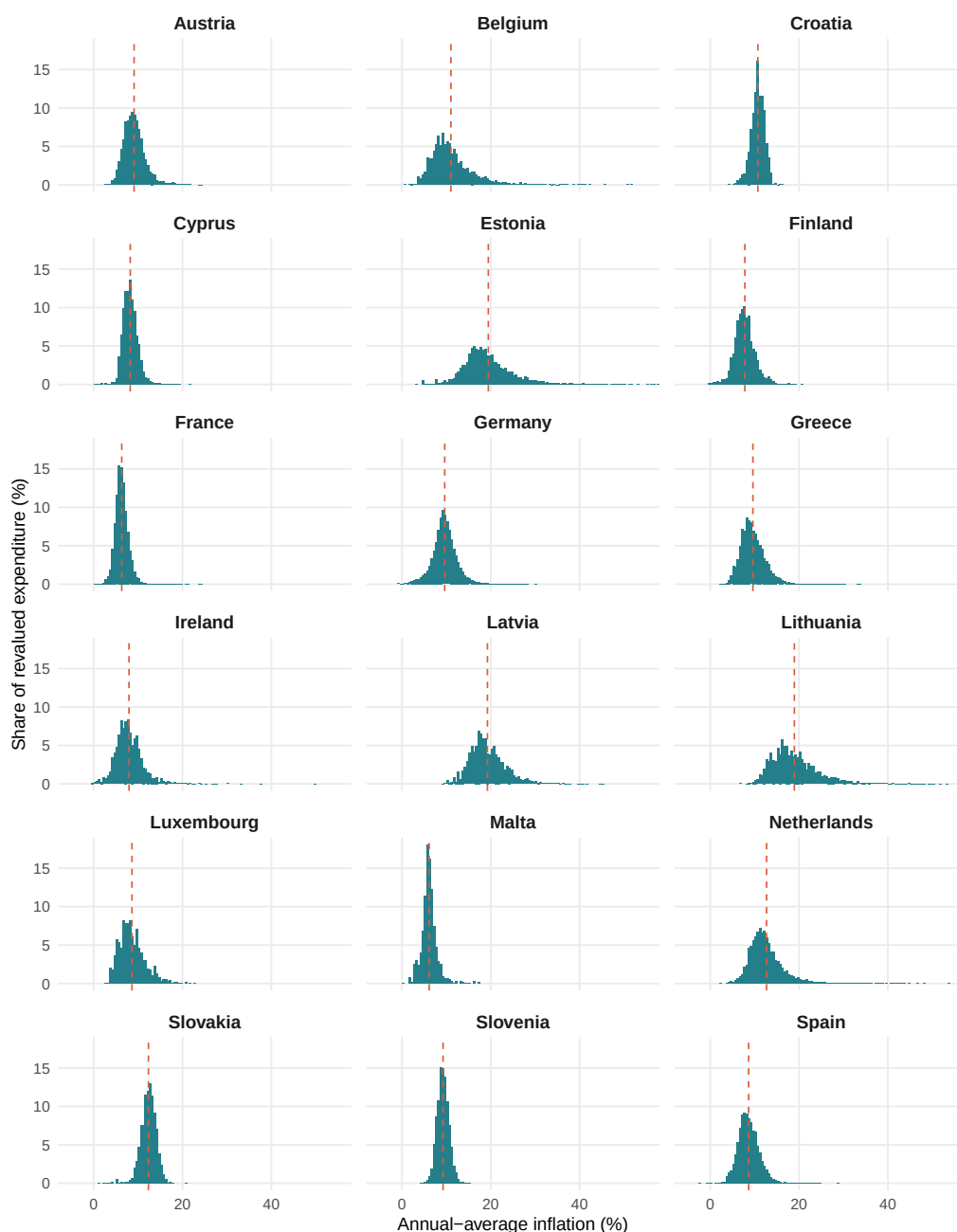
Figure C.2: Essential and discretionary expenditures by income quintile, euro area (in % of total expenditure)



Notes: The figure partitions COICOP expenditure shares into essential expenditures and discretionary/transport expenditures. The euro-area aggregate is computed as the unweighted average of available country-level HBS expenditure shares.

Sources: Eurostat HBS 2015; authors' calculations.

Figure C.3: National distributions of household inflation in 2022: RAS, COICOP digit 3



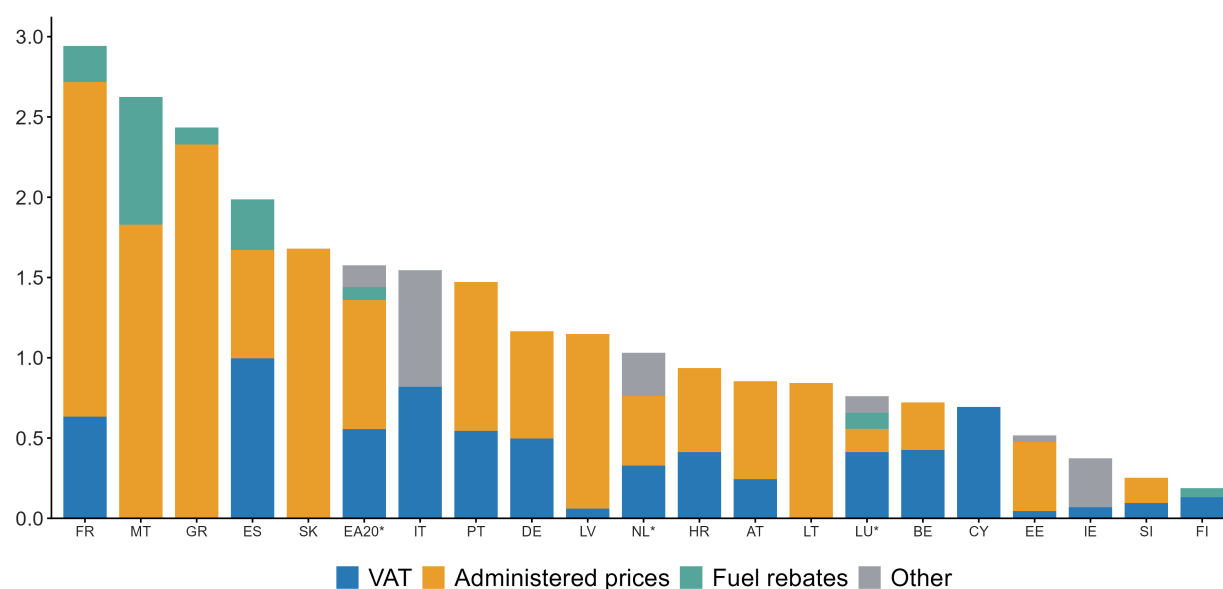
Notes: Annual-average inflation; 2020 HBS baskets, excluding rents (041 and 042). Weights equal survey weights times retained expenditure times the 2021 annual-average household index. Orange lines mark the corresponding weighted means. Bins are 0.5 percentage point wide; the displayed range is  $-5$  to  $55$  percent, without renormalization. The expenditure share outside this range is below  $0.1$  percent in every country. The sample comprises  $152,613$  households in  $18$  countries, excluding Italy and Portugal.

Sources: HBS microdata, Eurostat HICP; authors' calculations.

## D Counterfactuals

This appendix reports country-level observed and counterfactual headline inflation series. Each figure aggregates all simulated energy and food price-policy layers available for the country, using the HICP item weights in the country CPI basket.

Figure D.1: Fiscal Costs of Energy Measures (Percent of 2022 GDP)



Notes: Bars sum documented costs of the 191 measures active in the counterfactual during 2021–2023, counting each shared budget envelope once. VAT includes excise cuts; fuel rebates include fuel subsidies. Other comprises remaining instruments and shared envelopes with no identified instrument split. An asterisk indicates an unidentified instrument split. All active measures have a documented cost, but budget reference periods vary across schemes and do not uniformly cover 2021–2023; costs are not restricted to amounts received by households. EA20 divides summed costs by summed nominal 2022 GDP.

Sources: Bruegel, national official sources, Eurostat; authors' calculations using the fiscal-cost registry underlying the policy tables.

Table D.1: Price measures included in the national counterfactual simulations

| Country | Simplified policy coverage                                                                                                                             | Policy years | Input status                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------|
| Austria | Products: electricity, gas. Instruments: price caps or regulated tariffs, tax and charge relief, 2022–2024 subsidies or rebates.                       |              | 0 reconstructed; 4 calibrated. |
| Belgium | Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge 2022–2023 relief, subsidies or rebates.          |              | 0 reconstructed; 6 calibrated. |
| Croatia | Products: food, electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and 2022–2023 charge relief, subsidies or rebates.        |              | 0 reconstructed; 9 calibrated. |
| Cyprus  | Products: electricity, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and 2021–2023 charge relief, subsidies or rebates.      |              | 0 reconstructed; 5 calibrated. |
| Estonia | Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax 2021–2024 and charge relief, subsidies or rebates. |              | 0 reconstructed; 8 calibrated. |
| Finland | Products: electricity, motor fuels, passenger transport. Instruments: tax and charge relief. 2022–2023                                                 |              | 0 reconstructed; 6 calibrated. |
| France  | Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, subsidies or 2021–2024 rebates.                                 |              | 3 reconstructed; 0 calibrated. |
| Germany | Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax 2022–2024 and charge relief, subsidies or rebates. |              | 0 reconstructed; 7 calibrated. |
| Greece  | Products: electricity, gas, motor fuels. Instruments: subsidies or rebates. 2021–2023                                                                  |              | 0 reconstructed; 3 calibrated. |
| Ireland | Products: electricity, gas, motor fuels. Instruments: tax and charge relief, subsidies or rebates. 2022–2023                                           |              | 0 reconstructed; 5 calibrated. |

*Continued on next page*

| Country     | Simplified policy coverage                                                                                                                                      | Policy years | Input status                    |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------|
| Italy       | Products: electricity, gas, motor fuels. Instruments: tax and charge relief, subsidies or rebates.                                                              | 2021–2023    | 2 reconstructed; 4 calibrated.  |
| Latvia      | Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.                                 | 2022–2023    | 0 reconstructed; 3 calibrated.  |
| Lithuania   | Products: electricity, gas. Instruments: subsidies or rebates.                                                                                                  | 2022–2023    | 0 reconstructed; 2 calibrated.  |
| Luxembourg  | Products: electricity, gas, heating, motor fuels, other consumption. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates. | 2022–2024    | 0 reconstructed; 72 calibrated. |
| Malta       | Products: electricity, motor fuels. Instruments: price caps or regulated tariffs, subsidies or rebates.                                                         | 2022–2023    | 0 reconstructed; 2 calibrated.  |
| Netherlands | Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.                                          | 2022–2023    | 5 reconstructed; 6 calibrated.  |
| Portugal    | Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.                                             | 2021–2026    | 1 reconstructed; 12 calibrated. |
| Slovakia    | Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief.                                                       | 2023         | 0 reconstructed; 3 calibrated.  |
| Slovenia    | Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.                                          | 2021–2023    | 0 reconstructed; 9 calibrated.  |
| Spain       | Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.                       | 2021–2026    | 2 reconstructed; 13 calibrated. |

Notes: Products and instruments are grouped for readability; the underlying workbook retains the measure-level descriptions, dates, and COICOP codes. Reconstructed results use an external monthly path or a direct statutory bill inversion. Calibrated results use documented assumptions or model-based proxies. Sources: Bruegel Dataset (2021), ‘National policies to shield consumers from rising energy prices,’ version of 26 June 2023; authors’ calculations.

## D.1 Austria

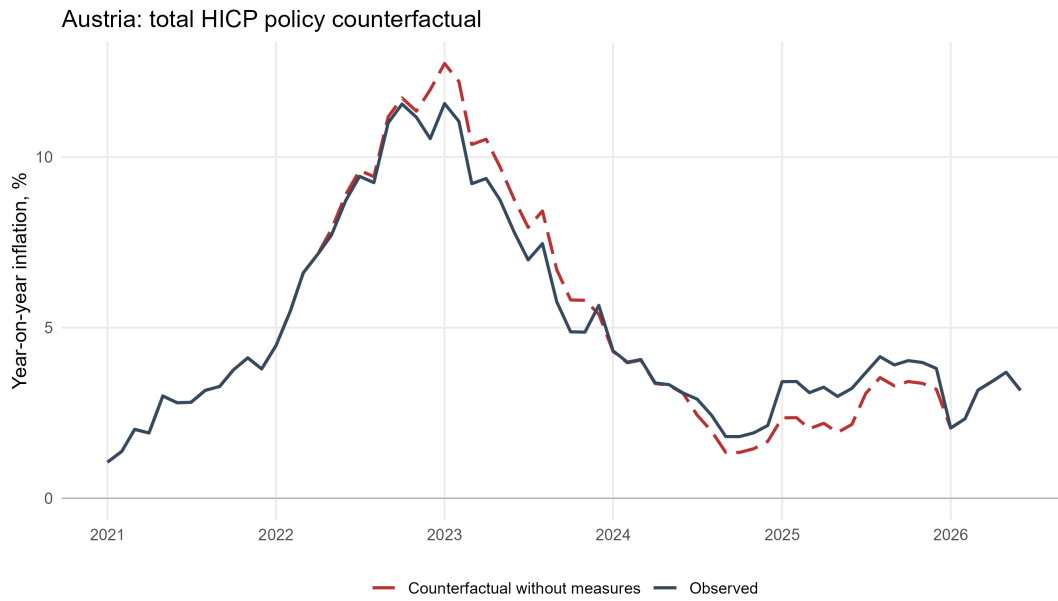


Figure D.2: Austria: observed and counterfactual inflation

## D.2 Belgium

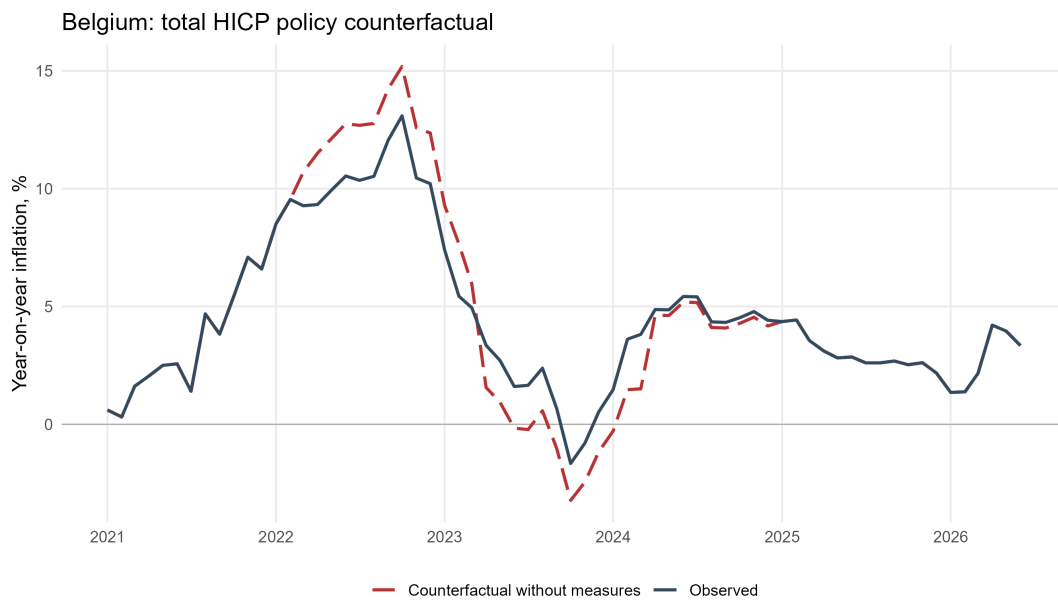


Figure D.3: Belgium: observed and counterfactual inflation

### D.3 Cyprus

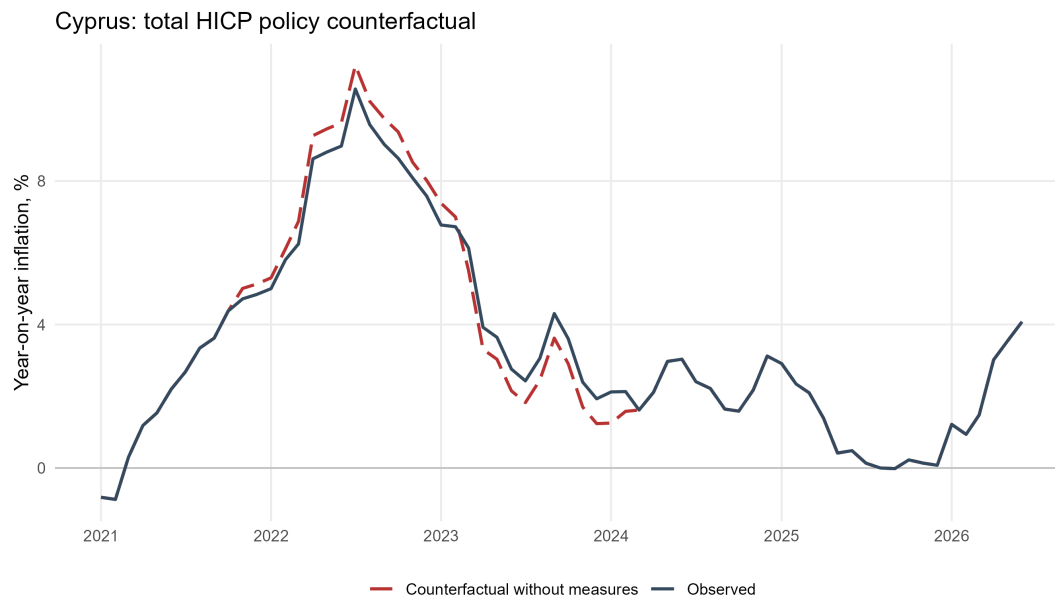


Figure D.4: Cyprus: observed and counterfactual inflation

### D.4 Germany

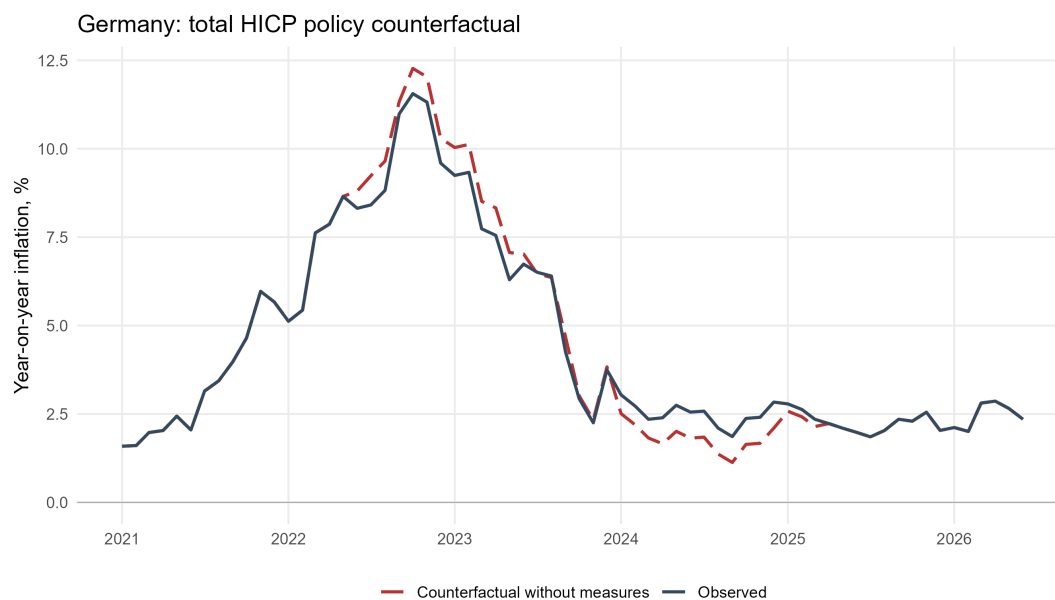


Figure D.5: Germany: observed and counterfactual inflation

## D.5 Estonia

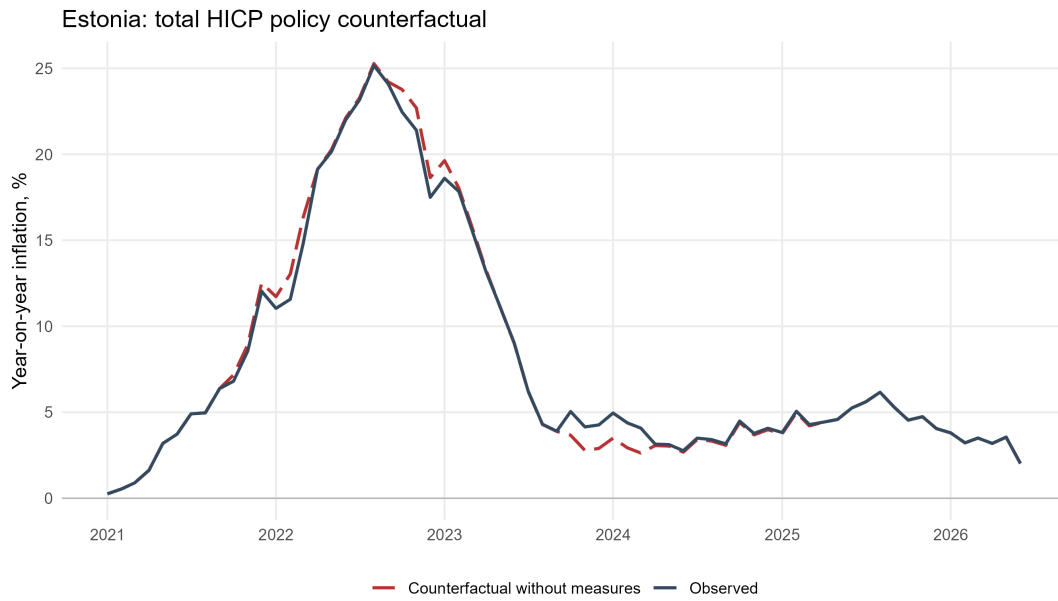


Figure D.6: Estonia: observed and counterfactual inflation

## D.6 Greece

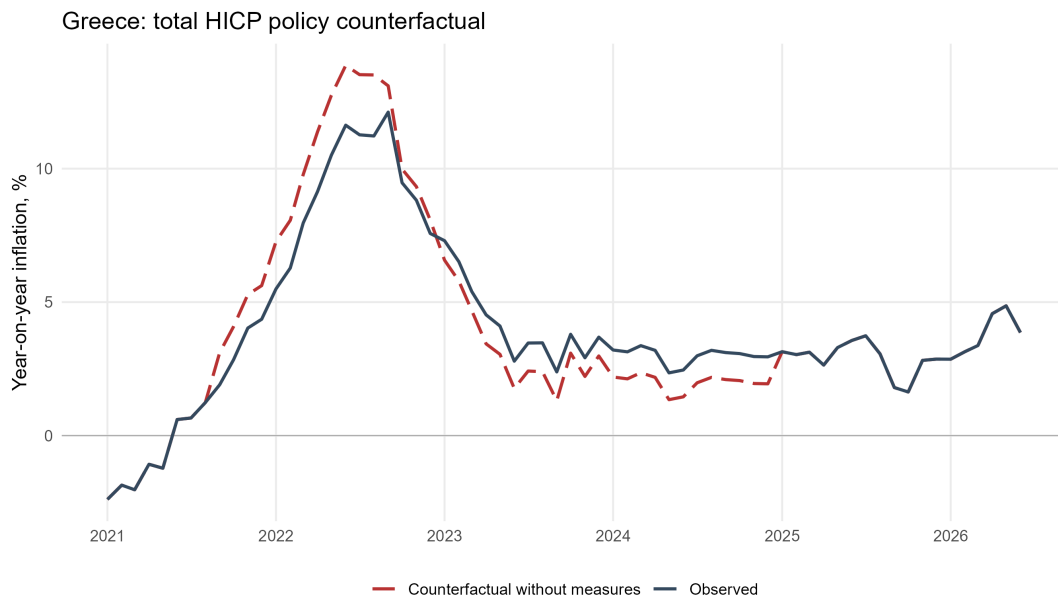


Figure D.7: Greece: observed and counterfactual inflation

## D.7 Spain

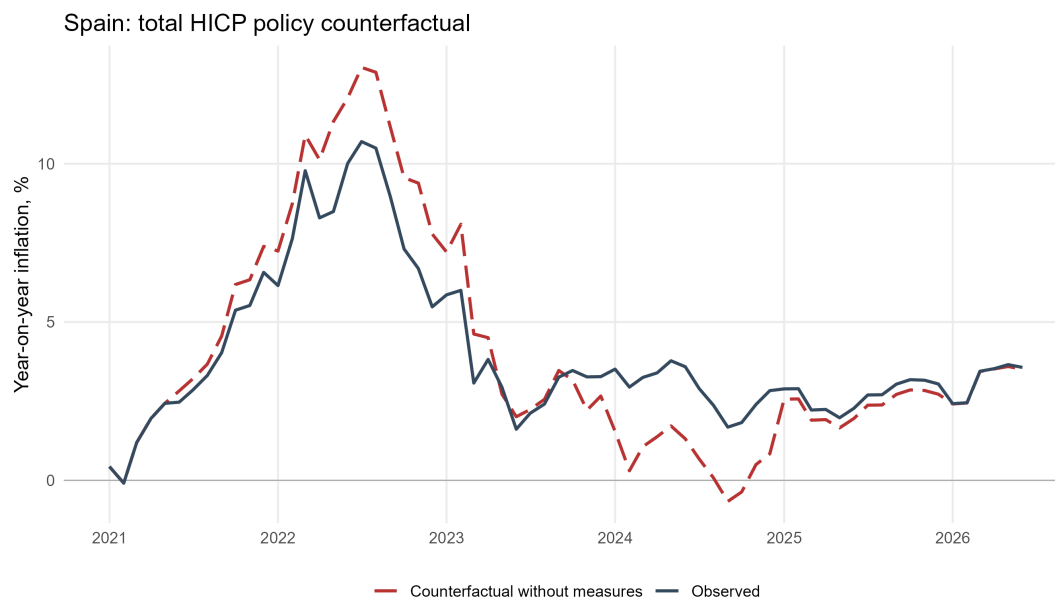


Figure D.8: Spain: observed and counterfactual inflation

## D.8 Finland

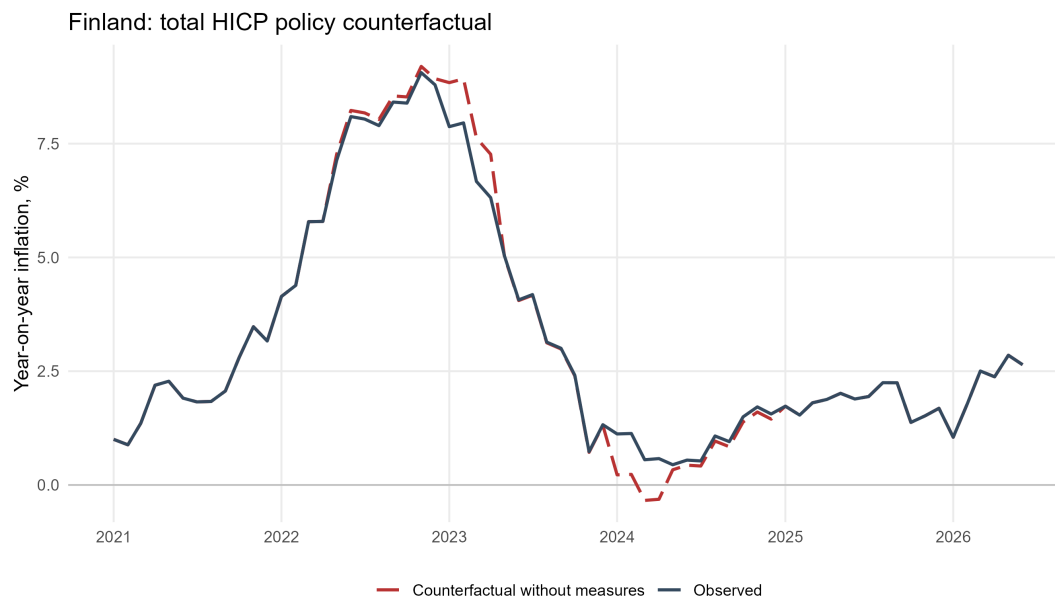


Figure D.9: Finland: observed and counterfactual inflation

## D.9 France

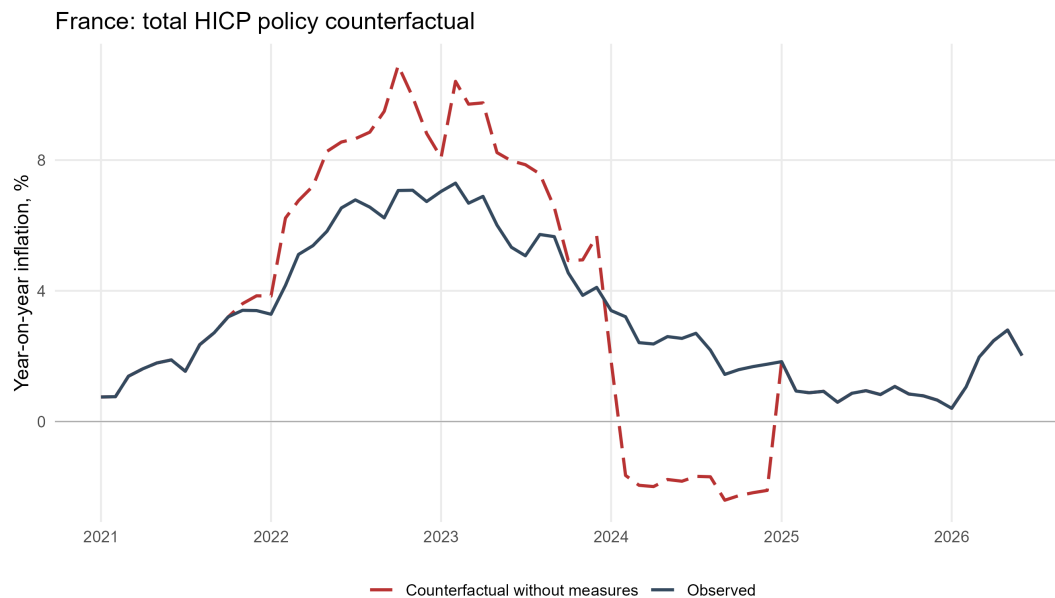


Figure D.10: France: observed and counterfactual inflation

## D.10 Croatia

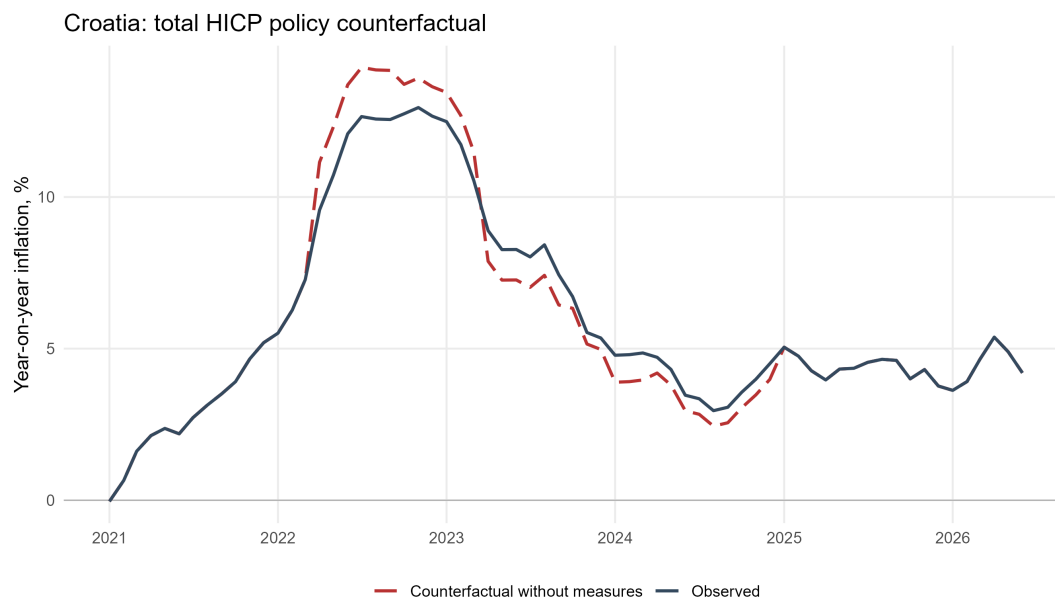


Figure D.11: Croatia: observed and counterfactual inflation

## D.11 Ireland

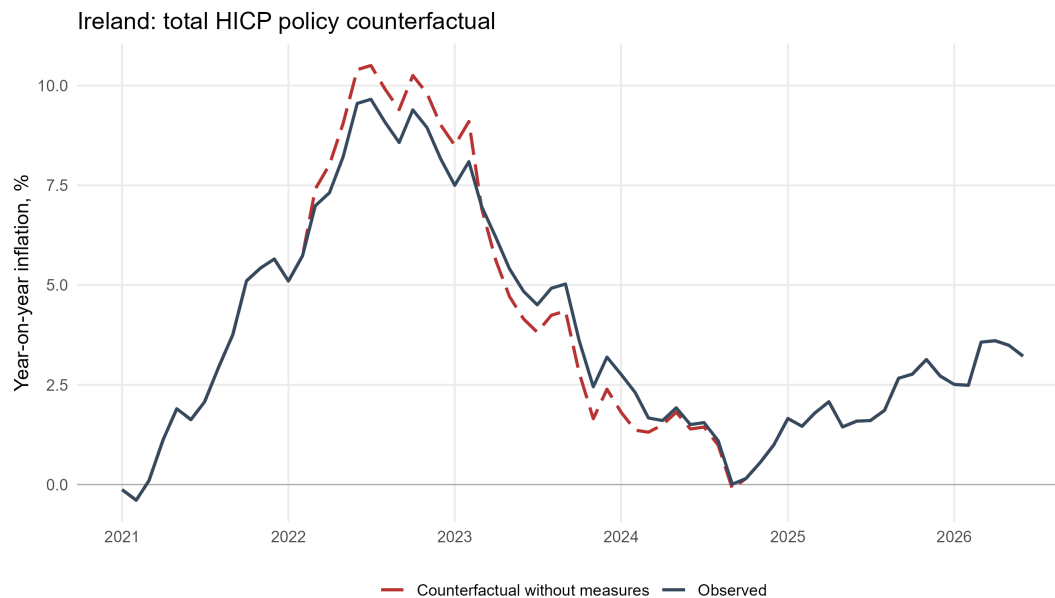


Figure D.12: Ireland: observed and counterfactual inflation

## D.12 Italy

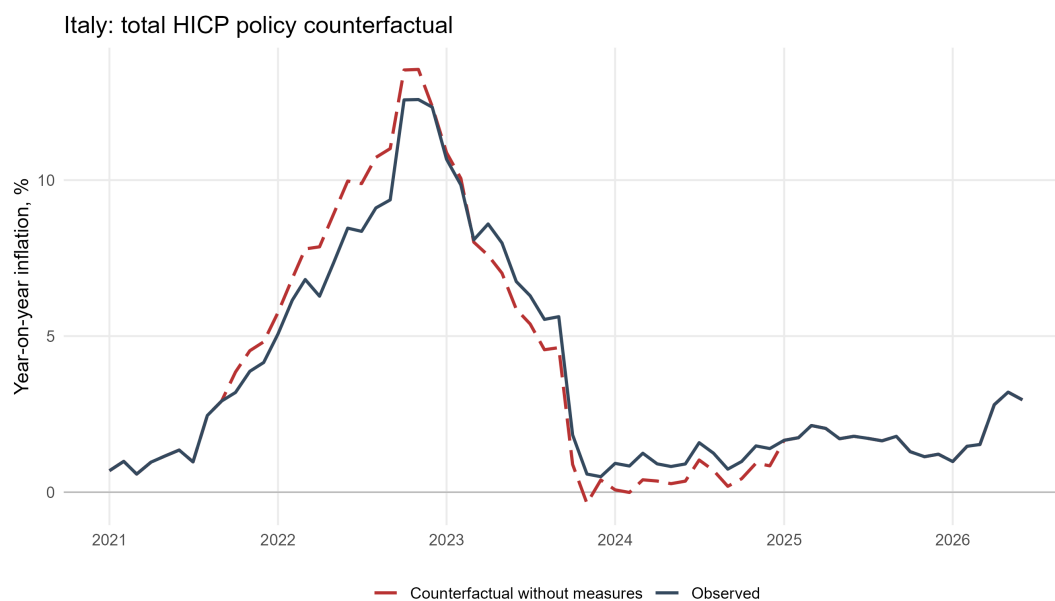


Figure D.13: Italy: observed and counterfactual inflation

## D.13 Lithuania

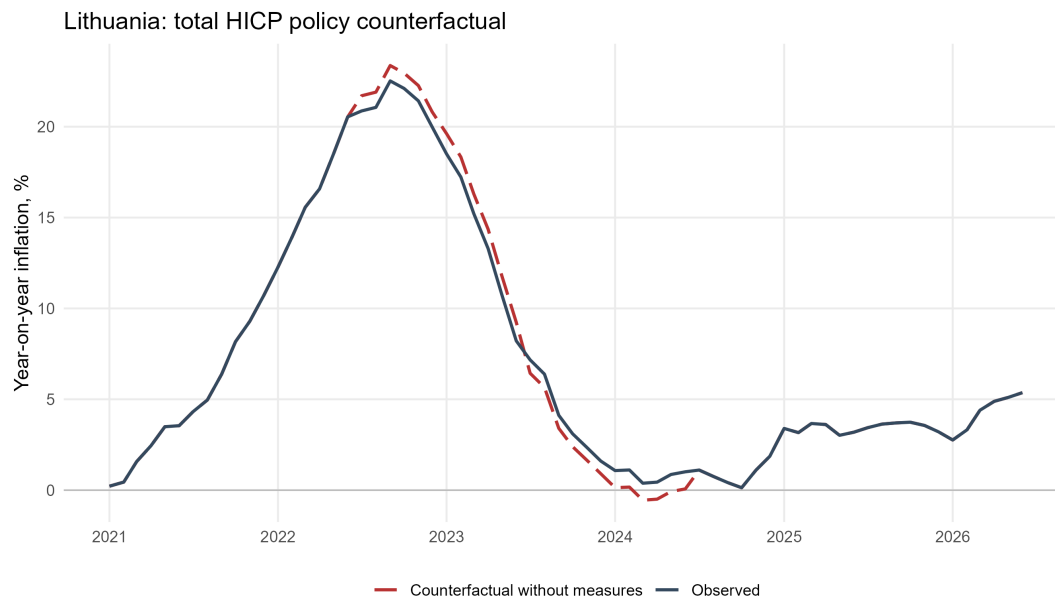


Figure D.14: Lithuania: observed and counterfactual inflation

## D.14 Luxembourg

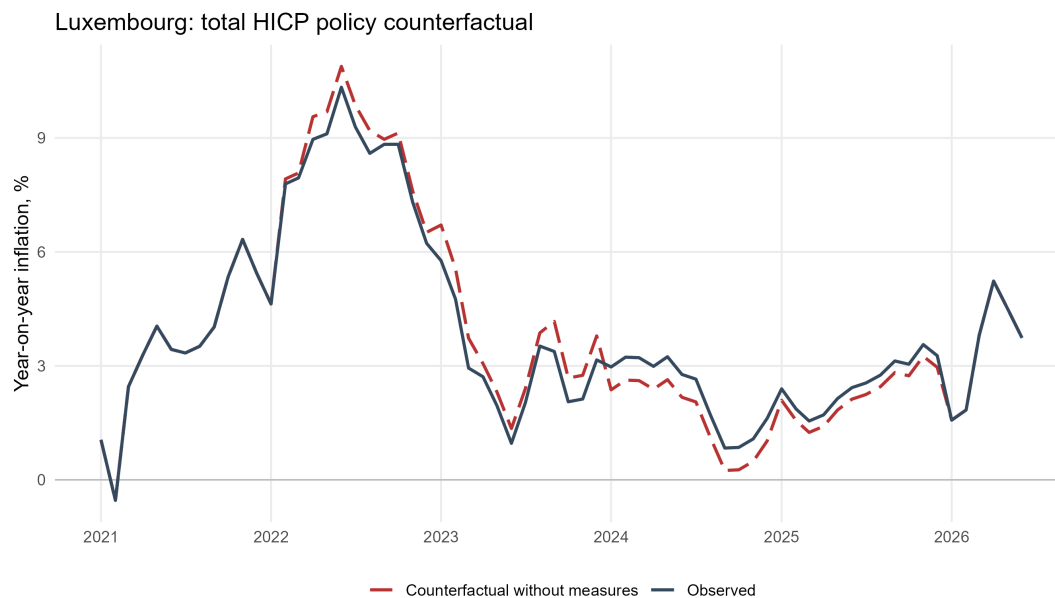


Figure D.15: Luxembourg: observed and counterfactual inflation

## D.15 Latvia

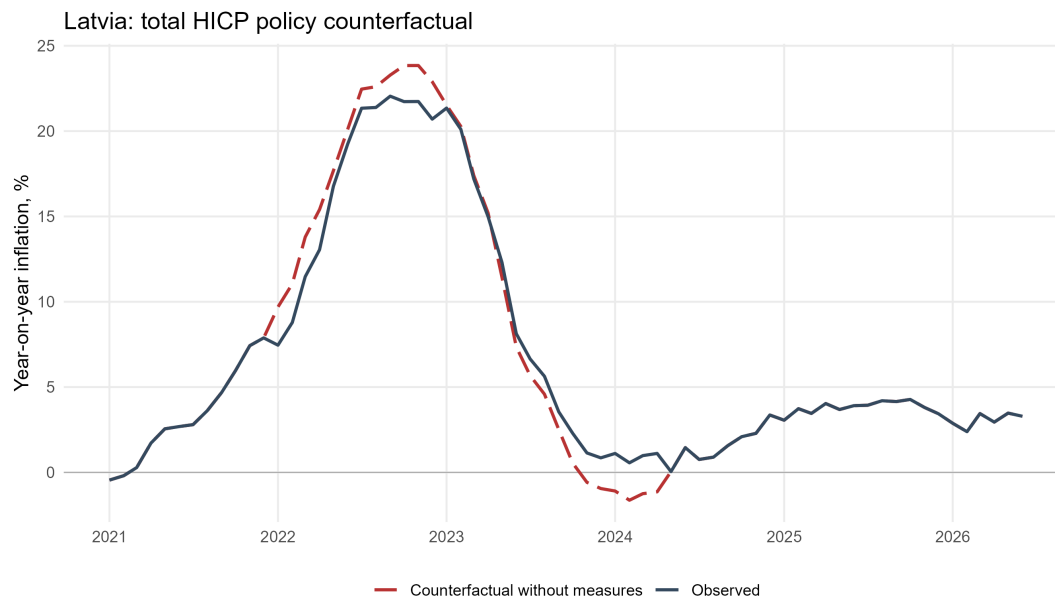


Figure D.16: Latvia: observed and counterfactual inflation

## D.16 Malta

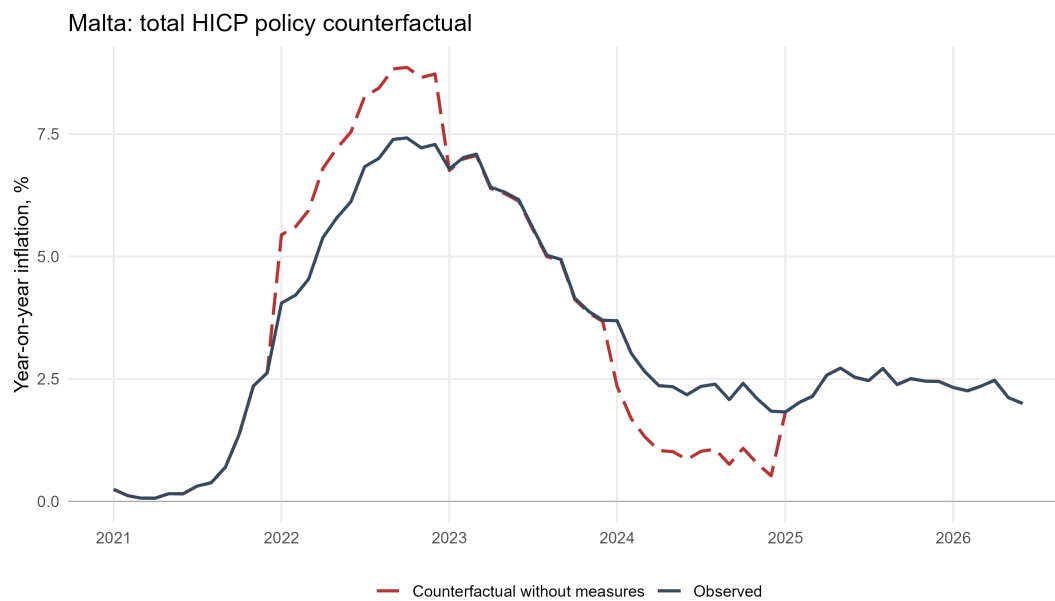


Figure D.17: Malta: observed and counterfactual inflation

## D.17 Netherlands

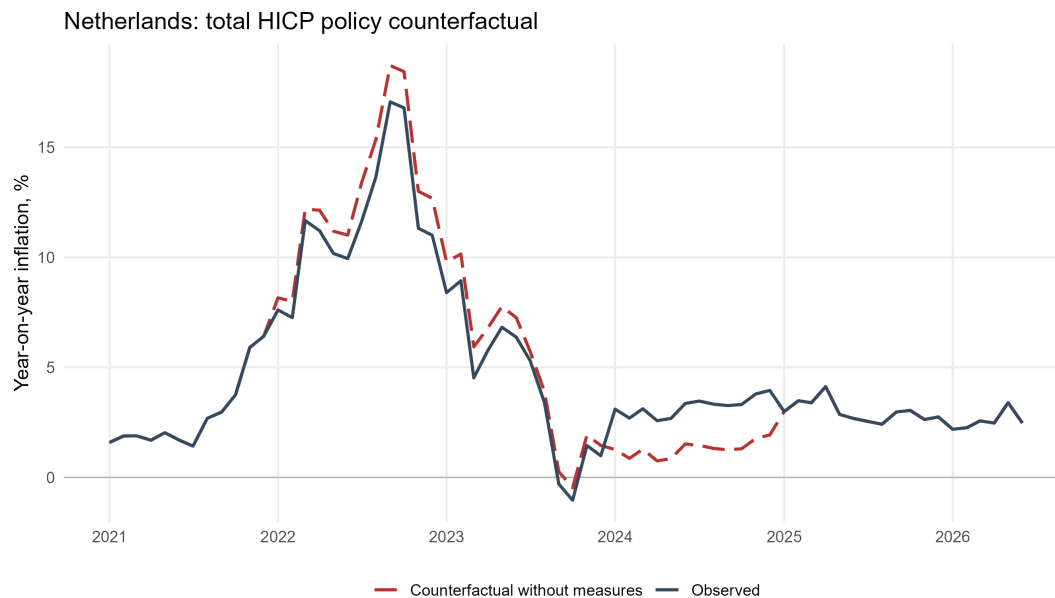


Figure D.18: Netherlands: observed and counterfactual inflation

## D.18 Portugal

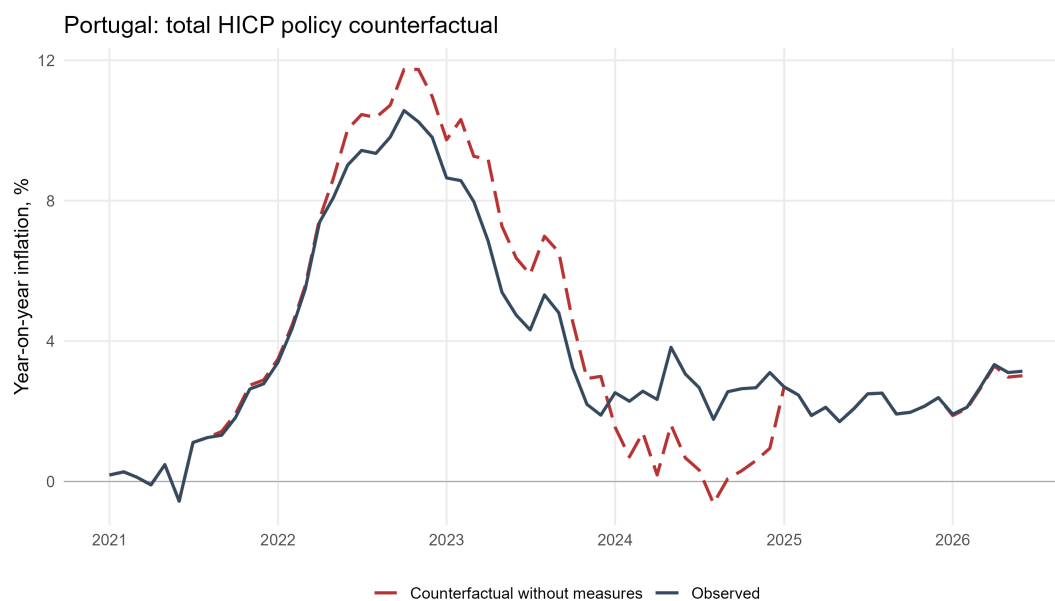


Figure D.19: Portugal: observed and counterfactual inflation

## D.19 Slovakia

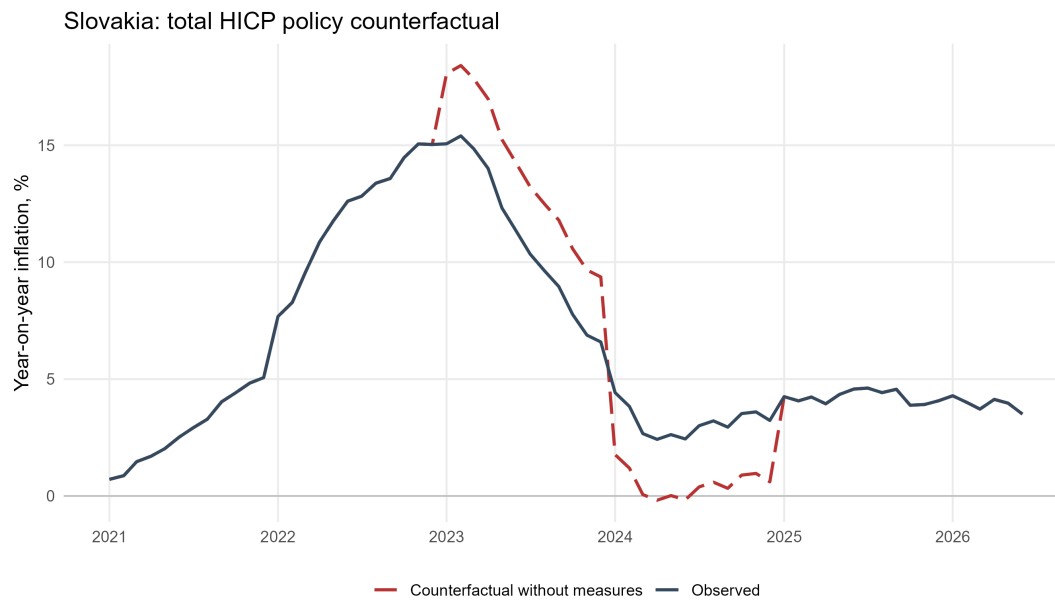


Figure D.20: Slovakia: observed and counterfactual inflation

## D.20 Slovenia

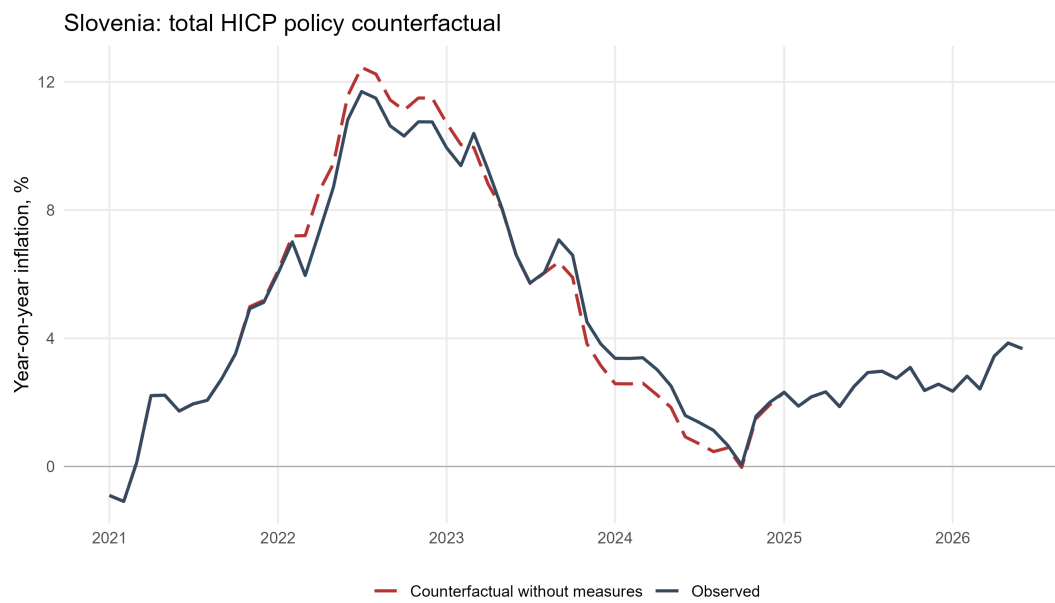


Figure D.21: Slovenia: observed and counterfactual inflation

## E The Household Budget Survey in Italy

The most recent complete harmonized Eurostat HBS entry for Italy is 2005. We therefore combine it with the 2015 and 2020 Istat public-use HBS files to construct the household-group baskets.

Age-group baskets are computed directly from the survey-weighted expenditure aggregates in the Istat files, using the age of the household reference person.

For income quintiles, households are ranked by equivalised total consumption using the Carbonaro scale and the survey weights. Because income quintiles are not observed in the target-year files, we transport the relationship between income- and expenditure-quintile baskets observed in 2005. Let  $E_{jqy}$  denote expenditure on COICOP group  $j$  by expenditure quintile  $q$  in year  $y$ , and let  $s_{jq,2005}^I$  and  $s_{jq,2005}^E$  denote the corresponding 2005 shares for income and expenditure quintiles. The estimated income-quintile consumption structure is

$$\hat{s}_{jqy}^I = \frac{E_{jqy} \left( s_{jq,2005}^I / s_{jq,2005}^E \right)}{\sum_k E_{kqy} \left( s_{kq,2005}^I / s_{kq,2005}^E \right)}. \quad (22)$$

Thus, the 2005 income-to-expenditure ratio adjusts each 2015 or 2020 expenditure-quintile component, after which the basket is renormalized to one. A few broad groups are combined to reconcile the two 2005 classifications. The resulting income-quintile structures are estimated rather than directly observed. The procedure addresses the same absence of jointly observed income and consumption discussed by [Donatiello et al. \(2022\)](#).

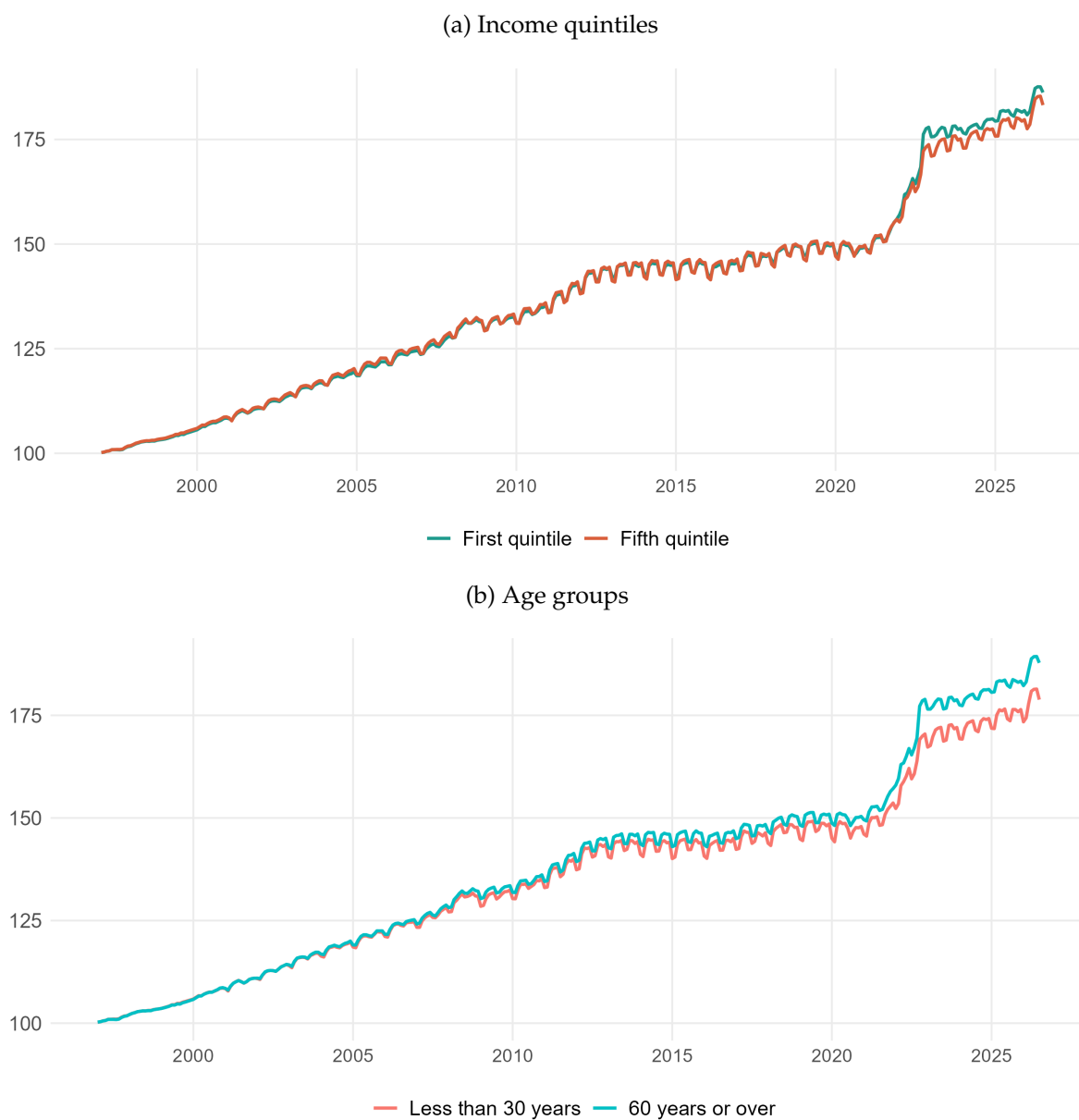
Table [E.1](#) reports the resulting shares of total consumption. Age shares are direct survey estimates; income-quintile shares are bridge estimates. The first and fifth income quintiles account for about 8 and 39 percent of Italian consumption, respectively.

Table E.1: Expenditures by groups (as a share of total expenditures)

| Dimension | Group               | 2015 | 2020 |
|-----------|---------------------|------|------|
| Age       | Less than 30 years  | 2.0  | 2.2  |
| Age       | From 30 to 44 years | 19.7 | 17.9 |
| Age       | From 45 to 59 years | 34.8 | 34.2 |
| Age       | 60 years or over    | 43.5 | 45.7 |
| Income    | First quintile      | 7.8  | 8.1  |
| Income    | Second quintile     | 12.7 | 12.8 |
| Income    | Third quintile      | 17.2 | 17.0 |
| Income    | Fourth quintile     | 23.3 | 22.8 |
| Income    | Fifth quintile      | 39.0 | 39.3 |

Notes. Entries are shares of annual consumption. Age groups follow Eurostat bands; income rows use the Istat–Eurostat quintile bridge. Sources: Istat and Eurostat HBS, authors’ calculations.

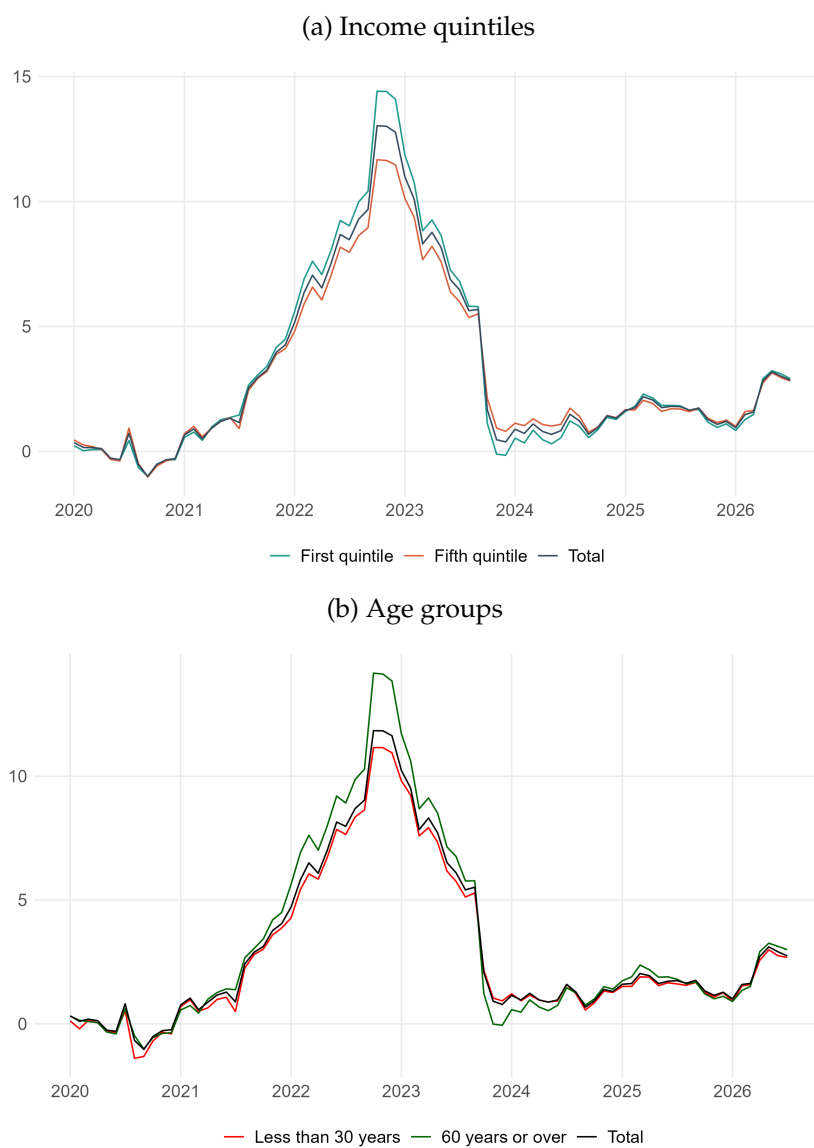
Figure E.1: Price indices by household group, Italy, COICOP digit 3 (index, 1996 = 100)



Notes: The figure reports chained price indices using RAS weighting. Income groups compare the first and fifth quintiles; age groups compare households whose reference person is under 30 with those aged 60 or over.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

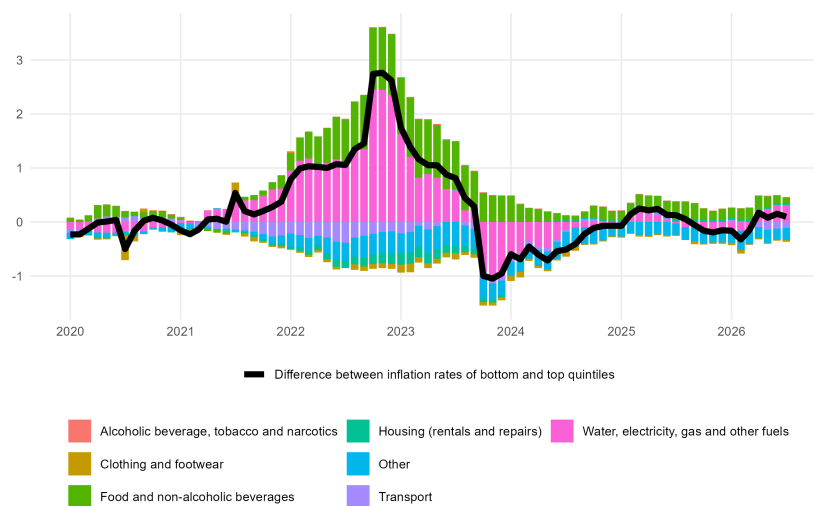
Figure E.2: Inflation by household group, Italy, COICOP digit 3 (in %)



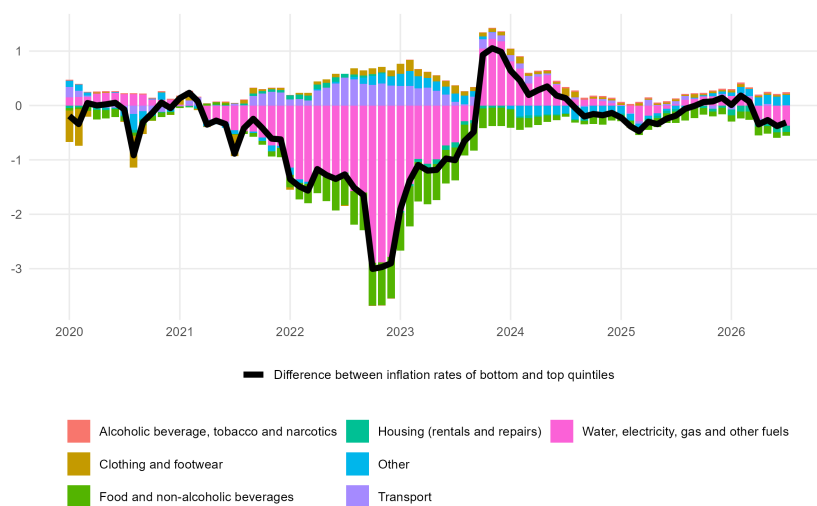
Notes: The figure reports monthly year-on-year inflation rates using RAS weighting for income and age groups.  
Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

Figure E.3: Contributions to inflation gaps, Italy, COICOP digit 3 (in pp)

(a) Income gap: Quintile 1 minus Quintile 5



(b) Age gap: 60 years or over minus under 30



Notes: Bars decompose the monthly inflation gap into broad consumption components; the black line reports the total gap. Positive values indicate higher inflation for the first group named in each panel.

Sources: Eurostat HICP; Italian HBS inputs described in Appendix E; authors' calculations.

## F RAS Weighting

We use the RAS (Row and Column Adjustment Scaling) algorithm. RAS is a bi-proportional matrix scaling method that alternately rescales rows and columns until the required marginal totals are matched ([Bacharach, 1965](#)). In our setting, we combine official HICP item weights with the relative HBS expenditure profile of each household group.

The baseline removes actual and imputed rents (HBS 041 and 042 and corresponding HICP rent components) before constructing matched product baskets. Parent housing aggregates are recalculated without these components. RAS matches group expenditure margins and annual HICP product margins normalized over the remaining matched non-rent products. The resulting indices therefore measure inflation excluding rents; agreement with the all-items HICP is not an identity implied by this calibration.

Rent treatment can affect both the magnitude and the sign of income gaps because housing tenure and rent shares vary across groups ([Geerolf, 2023](#)). Table [F.1](#) retains three alternative conventions as a separate sensitivity comparison: all rents maps HBS 041+042 to HICP 041; effective rents retains HBS 041 only; excluding rents removes 041 and 042 and renormalizes the remaining basket. This reference exercise uses relative-expenditure weights. It is distinct from the RAS-based baseline and the RAS micro–macro comparisons in [Appendix A](#); its excluding-rents column need not reproduce their estimates.

Table F.1: Robustness to rent treatment: inflation gap (in points, 2022)

| Country     | All rents | Effective rents | Excluding rents |
|-------------|-----------|-----------------|-----------------|
| Austria     | -0.2      | -0.7            | 0.1             |
| Belgium     | 2.0       | 0.5             | 2.4             |
| Croatia     | 0.3       | 0.3             | 0.4             |
| Cyprus      | 0.6       | 0.5             | 0.7             |
| Estonia     | 4.5       | 4.6             | 4.7             |
| Finland     | -1.1      | -2.4            | -0.9            |
| France      | 0.1       | -0.7            | 0.2             |
| Germany     | 0.9       | -0.6            | 1.4             |
| Greece      | 1.1       | 0.5             | 1.3             |
| Ireland     | 2.8       | 2.9             | 3.0             |
| Italy       | 1.5       | 1.5             | 1.6             |
| Latvia      | 4.6       | 4.6             | 4.8             |
| Lithuania   | 3.3       | 3.2             | 3.4             |
| Luxembourg  | 0.2       | -0.8            | 0.2             |
| Malta       | 0.2       | 0.2             | 0.3             |
| Netherlands | 1.8       | -0.4            | 2.7             |
| Portugal    | 0.2       | -0.5            | 0.3             |
| Slovakia    | 1.1       | 0.7             | 1.2             |
| Slovenia    | 0.5       | 0.7             | 0.5             |
| Spain       | 0.8       | 0.3             | 1.0             |
| Euro area   | 0.9       | 0.0             | 1.2             |

*Notes:* All scenarios use national-quintile Eurostat HBS baskets, relative-expenditure weights and chained Laspeyres indices. The first pools HBS 041+042 and maps it to HICP 041; the second uses HBS 041 only; the third excludes HBS 041 and 042 and HICP 041 and renormalises. Gap denotes first-quintile minus fifth-quintile inflation, in percentage points. Euro-area indices aggregate national indices with annual HICP country weights. Italy uses the custom national HBS input; Slovakia uses the 2015 HBS wave.

We distinguish three objects: the HICP product shares, the aggregate product-by-group expenditure mass calibrated by RAS, and the resulting within-group baskets. First, let  $s_{g,y}$  denote the share of group  $g$  in total consumption expenditure used for HICP weight year  $y$ :

$$s_{g,y} = \frac{E_{g,y'}^{\text{HBS}}}{E_{\text{all},y'}^{\text{HBS}}}, \quad \sum_g s_{g,y} = 1, \quad (23)$$

where  $y'$  is the HBS wave matched to HICP weight year  $y$ . In the baseline,  $y' = 2020$  and the expenditure margins are computed from the same household sample as the group baskets. Historical applications select the most recent available preceding HBS wave, or the earliest wave when no preceding observation exists. These targets are expenditure shares, not population shares. For Italy, the income targets use the Istat-based HBS inputs described in Appendix E.

Let  $\mathcal{J}_y$  be the set of matched COICOP categories available in the HICP weighting structure in year  $y$ . Let  $E_{j,g,y'}^{\text{HBS}}$  denote expenditure on item  $j$  by group  $g$  in the matched HBS wave  $y'$ , and let  $E_{j,\text{all},y'}^{\text{HBS}}$  denote expenditure on that item by all households. Consistently with the implementation, the HBS correction factor used in the RAS seed is the share of aggregate expenditure on item  $j$  accounted for by group  $g$ :

$$c_{j,g,y'}^{\text{HBS}} = \frac{E_{j,g,y'}^{\text{HBS}}}{E_{j,\text{all},y'}^{\text{HBS}}}. \quad (24)$$

Thus  $c_{j,g,y'}^{\text{HBS}}$  records how expenditure on item  $j$  is distributed across household groups. To avoid mixing index-weight units with expenditure shares, define the HICP weight normalized over the matched product set as

$$\bar{\omega}_{j,y}^{\text{HICP}} = \frac{\omega_{j,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_y} \omega_{k,y}^{\text{HICP}}}, \quad \sum_{j \in \mathcal{J}_y} \bar{\omega}_{j,y}^{\text{HICP}} = 1. \quad (25)$$

The initial product-by-group mass is then

$$Z_{j,g,y} = \max \left\{ s_{g,y} \bar{\omega}_{j,y}^{\text{HICP}} c_{j,g,y'}^{\text{HBS}}, \epsilon \right\}, \quad \epsilon = 10^{-12}. \quad (26)$$

The small positive floor is a numerical safeguard to keep the RAS problem strictly positive after HICP-HBS matching. RAS transforms this seed into a joint expenditure-share matrix  $M_{j,g,y}$ . Its group margins equal the groups' expenditure shares, while its product margins equal the normalized HICP shares:

$$\sum_{j \in \mathcal{J}_y} M_{j,g,y} = s_{g,y} \quad \text{for all } g, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}} \quad \text{for all } j. \quad (27)$$

Both sets of margins sum to one, so the constraints are mutually compatible.

The RAS algorithm starts from  $M_{j,g,y}^{(0)} = Z_{j,g,y}$  and repeats two scaling steps. At iteration  $t$ , the group-margin adjustment is

$$\tilde{M}_{j,g,y}^{(t)} = M_{j,g,y}^{(t-1)} \times \frac{s_{g,y}}{\sum_{\ell \in \mathcal{J}_y} M_{\ell,g,y}^{(t-1)}}, \quad (28)$$

and the product-margin adjustment is

$$M_{j,g,y}^{(t)} = \tilde{M}_{j,g,y}^{(t)} \times \frac{\bar{\omega}_{j,y}^{\text{HICP}}}{\sum_h \tilde{M}_{j,h,y}^{(t)}}. \quad (29)$$

We iterate these two steps by weight year until the maximum absolute row or column marginal error is below  $10^{-10}$ , with a maximum of 1,000 iterations.

The calibrated matrix  $M$  is a joint expenditure distribution. The group-specific HICP basket is the corresponding conditional product distribution within group  $g$ :

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}} = \frac{M_{j,g,y}}{\sum_{k \in \mathcal{J}_y} M_{k,g,y}}. \quad (30)$$

The equality of the two denominators follows directly from the group-margin constraint. It implies that each group basket sums to one. Moreover, the expenditure-share-weighted average of the group baskets reproduces the normalized HICP product weights:

$$\sum_{j \in \mathcal{J}_y} \omega_{j,g,y} = 1, \quad \sum_g s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}. \quad (31)$$

The package implements the same system on a percentage scale. Internally, the calibrated mass is  $M_{j,g,y}^{\text{pct}} = 100M_{j,g,y}$ , so its margins are  $100s_{g,y}$  and  $100\bar{\omega}_{j,y}^{\text{HICP}}$ . The package returns these percentage weights in the R variable `weighted_consumption`. Denoting this variable by  $w_{j,g,y}^{\text{code}}$ , we have

$$w_{j,g,y}^{\text{code}} = \frac{M_{j,g,y}^{\text{pct}}}{s_{g,y}} = 100\omega_{j,g,y}. \quad (32)$$

This quantity therefore sums to 100 within every household group. Thus division by  $s_{g,y}$  in the code is the percentage-scale counterpart of dividing  $M_{j,g,y}$  by its group-row sum in the normalized mathematical representation.

This preserves the relative information contained in the HBS expenditure profiles while making the aggregate of the household-group baskets consistent with the published HICP weighting structure, up to residual differences induced by product coverage and COICOP matching.

The all-items reference exercise in Table F.2 compares the two weights for Germany at COICOP digit 3 over the period beginning in 2019. The RAS calibration tracks the published German HICP more closely than the relative-expenditure weighting, whose implicit aggregate weights are equal across groups.

Table F.2: Germany HICP validation by weighting scheme (differences in pp)

| Method               | Obs. | Mean diff. | Mean abs. diff. | RMSE  | Max abs. diff. |
|----------------------|------|------------|-----------------|-------|----------------|
| RAS                  | 88   | -0.059     | 0.093           | 0.118 | 0.259          |
| Relative expenditure | 88   | -0.070     | 0.143           | 0.166 | 0.449          |

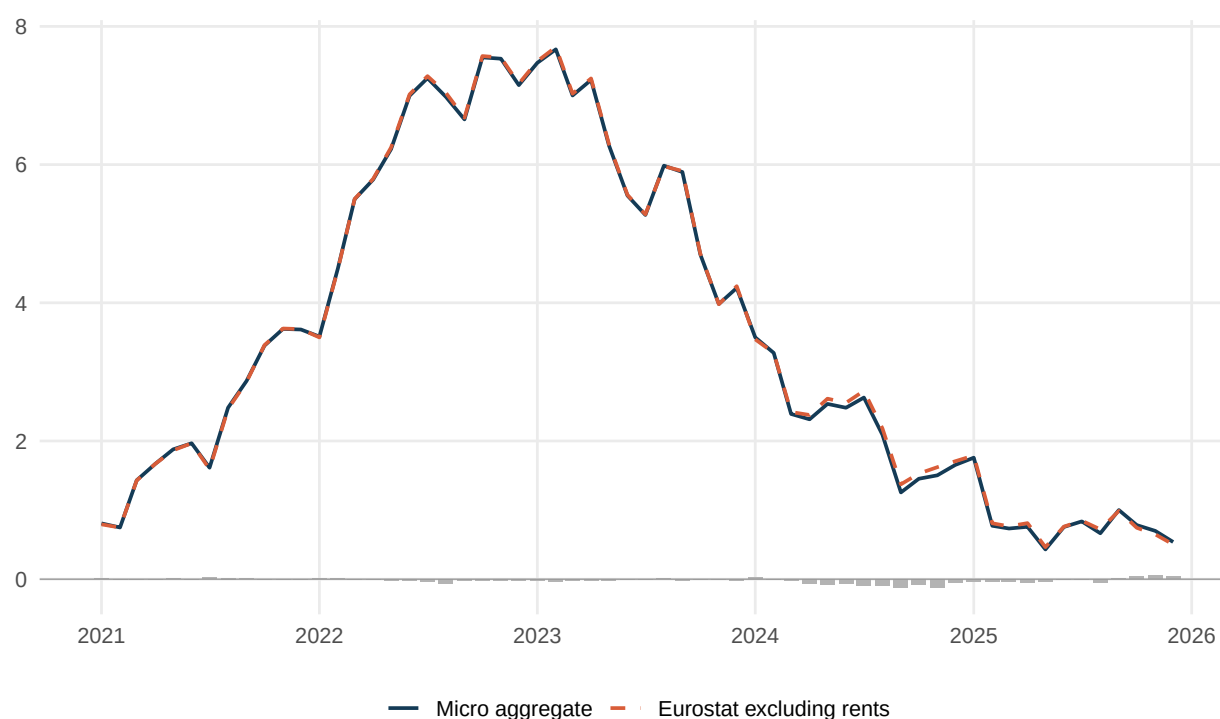
*Notes:* Validation statistics compare the recalculated average of German income-quintile inflation rates with the published all-items HICP inflation rate. The sample is monthly, from January 2019 to April 2026. All differences are recalculated minus published inflation and are reported in percentage points.

## G Validation Tests

Figures G.1 and G.2 compare the microdata indices with Eurostat on matching geographic and non-rent coverage, January 2021–December 2025. The micro aggregate averages December-relative household price links with survey-weighted matched expenditure, then uses annual HICP country weights adjusted for rents before chaining. The household mean instead first averages individual year-on-year inflation with national survey weights, then averages national means with annual all-items HICP country weights normalized over the sample.

### G.1 Internal Consistency

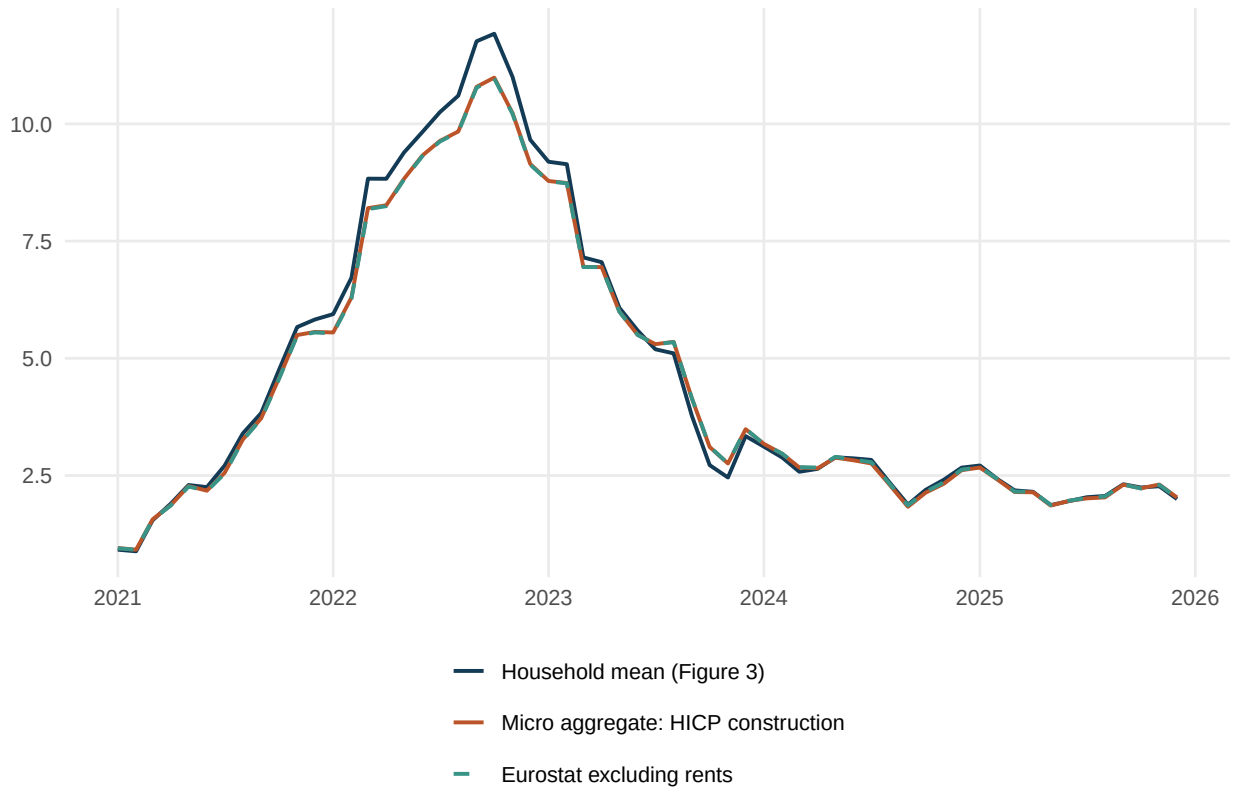
Figure G.1: France: micro aggregate and Eurostat excluding rents (year on year, in %)



Notes: Identical country and time coverage. Grey bars show micro aggregate minus Eurostat, in percentage points. Differences can reflect retained product support and price recoding.

Sources: HBS microdata, Eurostat HICP, authors' calculations.

Figure G.2: Micro–macro reconciliation: 18-country sample excluding rents (year on year, in %)



Notes: All series cover the same 18 countries, excluding Italy and Portugal. The household mean is the Figure 3 series. The two aggregates use identical annual non-rent country expenditure weights and chain price links before calculating inflation.

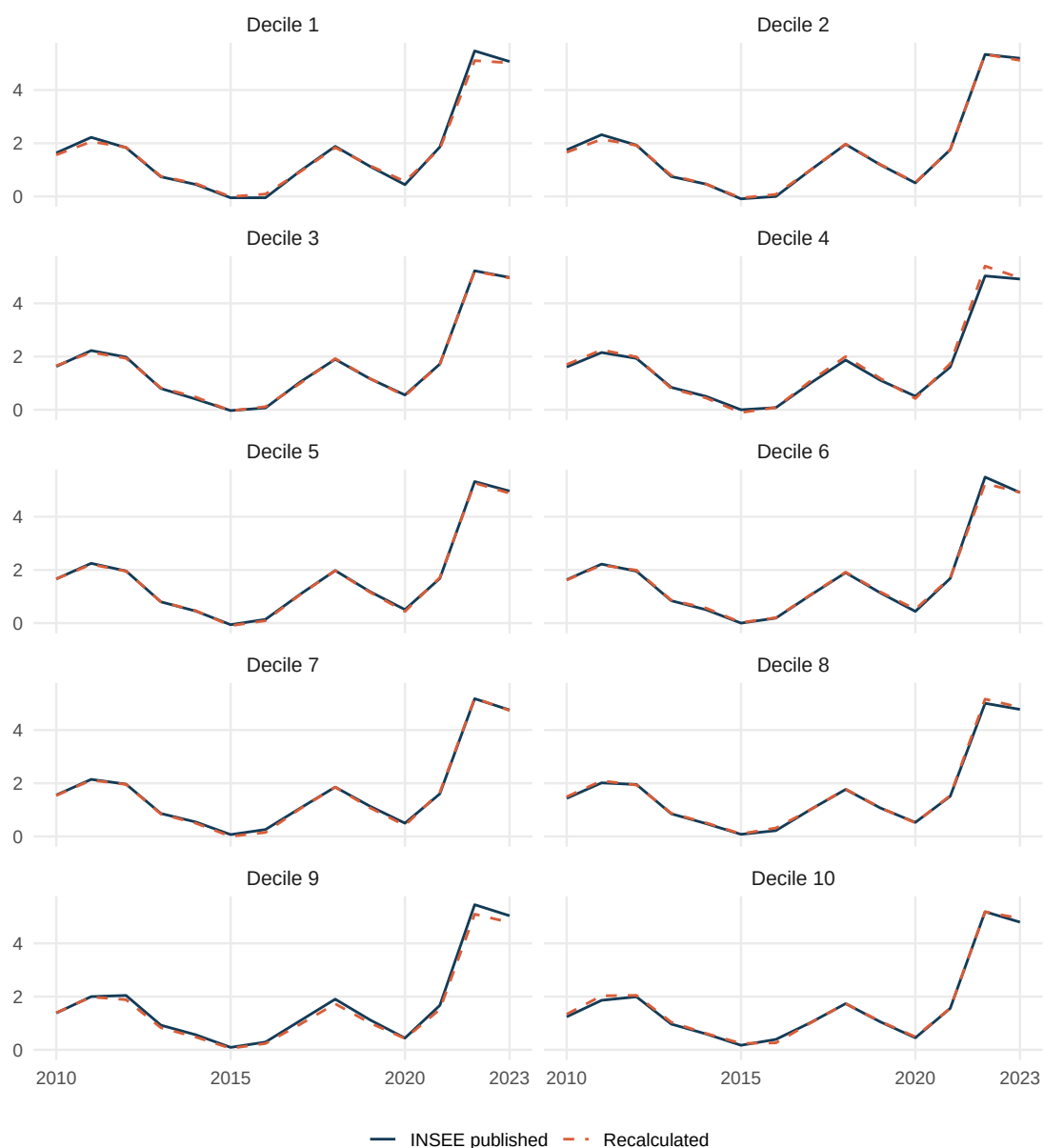
Sources: HBS microdata, Eurostat HICP, authors' calculations.

The mean absolute difference from the Eurostat benchmark is below 0.1 percentage point for the micro aggregate (also below 0.1 at its maximum), compared with 0.2 for the household mean (maximum 1.1). The household mean and expenditure-weighted aggregate measure different objects. The remaining micro-aggregate discrepancy combines product coverage, classification recoding, and price-index construction. Equality on identical product support is partly imposed by RAS calibration and does not independently validate household heterogeneity or policy counterfactuals.

## G.2 External Consistency

Figure G.3 retains the historical French comparison with INSEE annual inflation by standard-of-living decile. It uses its original all-items coverage and equivalised-income groups, separately from the non-rent total-household-income baseline.

Figure G.3: External consistency: annual inflation by standard-of-living decile, France (in %)



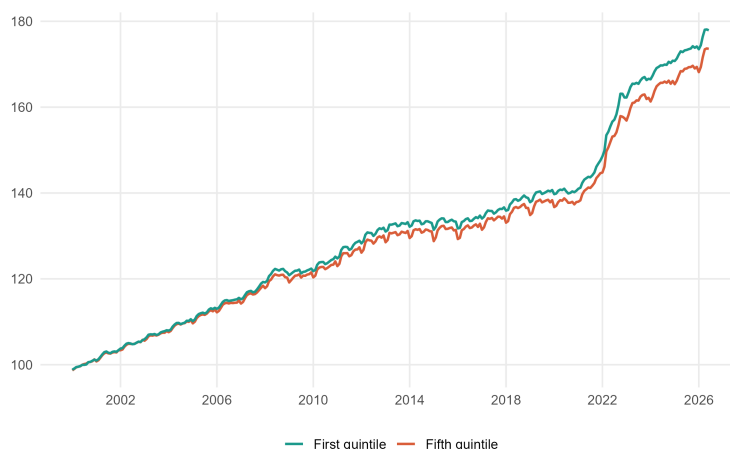
Notes: Annual inflation, 2010–2023. Recalculation uses COICOP digit 4; deciles follow the original INSEE standard-of-living definition. All ten deciles share the same axes.

Sources: INSEE, Eurostat, authors' calculations.

## H Long-Run Inflation Inequality

Figure H.1 shows a modest but persistent divergence between the first- and fifth-quintile price indices after 2008. The post-COVID episode produces the largest visible widening in the series since 2000.

Figure H.1: Long-run inflation inequality in the euro area, by income quintile (index, 2000 = 100)



Notes: This figure shows monthly price indices for the first and fifth income quintiles in the euro area, January 2000 to July 2026. Indices are rebased to 2000 = 100.

Sources: Eurostat HICP, HBS, authors' calculations

This pattern contrasts with the larger long-run gaps documented for the United States in [Jaravel \(2024\)](#). Several methodological differences may account for the contrasting results. First, the euro-area indices reported here aggregate national indices and therefore average across countries whose price dynamics and fiscal responses can offset one another. Second, the euro-area calculation uses harmonized COICOP categories, whereas the U.S. evidence exploits much finer product-level data. Aggregation within broad categories can conceal persistent differences in the prices paid by different income groups, which are captured in the U.S. data.